User Manual

System Administration

This manual provides an overview of the activities which are carried out by a system administrator.

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CCH® Tagetik

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System administration

Contents

This section covers the following topics:

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System administration page 1

System administrator

System administrator's responsibilities

The administrator is a senior-level user responsible for configuring, monitoring and managing the CCH Tagetik environment. Different functions or function modules can be enabled for each user-administrator.

Work flow

The system administrator work flow is unrestricted, and depends on operational needs. It typically follows the phases detailed below.

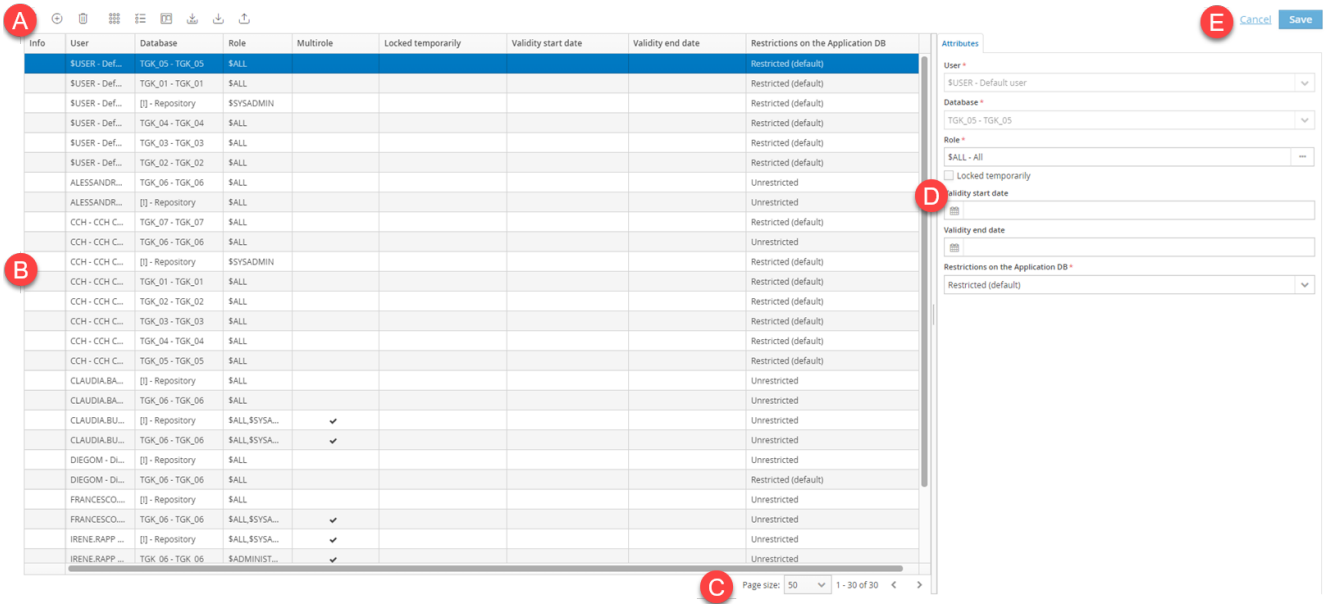
Phase	Activity	See
1	Configure general aspects of CCH Tagetik performance.	"Basic configuration procedures" on page 8 "Audit settings page" on page 3 "Email settings page" on page 4 "System Resources settings page" on page 5 "Security settings page" on page 6
2	Prepare the operatives' database.	"Work on the database" on page 16
3	Load any lists.	"Work on the database" on page 16 "MDM procedures" on page 49
4	Configure the data processing operations.	"How to work with data processing" on page 35
5	Create roles.	"Set roles" on page 28
6	Create users.	"Configure users in the Repository" on page 21
7	Create any endpoints.	"Endpoint management procedures " on page 84
8	Check system performance.	"Server Status page" on page 42 "Running data processing page" on page 41

System administration page

Introduction

Most of the pages on which system administration activities are carried out follow a shared logic. The main features are detailed below.

What it looks like



Element	Description
A	Tool bar with action buttons on the page
B	Table of specific data for the page (e.g. user list, data processing list, etc.)
C	Page size and scroll buttons
D	Configuration tab for data shown in the table (e.g. Attributes, Notifications etc.)
E	Any other action buttons

Buttons on the page

The following buttons are repeated on the system administrator work pages:

Element	Description
	This filters the data shown on the page.
	Adds a row in order to insert a new element.
	Duplicates the selected row and its content, except for the code.
	Deletes the selected row in the table.
	Activates/deactivates the edit function in the table directly and hides the configuration tabs.
	Enables multi-row selection to allow mass changes.
	Selects the columns to be shown/hidden in the table and those to be locked when scrolling the table.
XLS	Exports data in Excel format. This file cannot be re-imported into CCH Tagetik.
	Exports the data in Excel format so they can be edited and reimported into CCH Tagetik.
	Imports data from a previously prepared Excel file.
	Cuts a row so it can be pasted elsewhere.
	Moves a cut row elsewhere.
	Shows a number of actions that can be run on the selected data.

General settings

Contents

This section covers the following topics:

Localization settings page 3
Audit settings page 3
Email settings page 4
System Resources settings page 5
Security settings page 6
Basic configuration procedures 8

Localization settings page

Repository > General Settings > System Localization

Purpose of the page

This page makes it possible to define the languages used by CCH Tagetik.

Page sections

Please find below the most important fields.

Section	Description
Configuration languages	Available languages for metadata descriptions. You can set up to a maximum of four.
Reporting languages	Additional languages for metadata element descriptions in reports. It is possible to add up to 21 languages to the 4 **Configuration Languages**.
Default for new users	Characteristics to be assigned automatically to every new user • **Localization**: User's nationality for the correct display of numbers and dates • **Descriptions language**: Language used to display the descriptions of the metadata elements in the application • **Application language**: Language used on CCH Tagetik pages. • **Fallback application language**: Language used when the user's language is not available

Audit settings page

Repository > General Settings > Audit

Purpose of the page

This page allows the selection of the information required for the audit trail.

Page fields

Please find below the most important fields.

Field	Description
Audit configuration	
Resources monitor	Records instances of memory overload and restoration of normal condition.
Licence	Records the entry of the licence in the audit trail.
Detail of queries run by send e-mail operations	Provides details of email sending operations. This can identify any send errors and is disabled by default.
Detail of queries run by a data process	Records in the audit trail any queries run by data processes for which **Audit SQL data processing** was selected in the **Data Processings** tile.
CPM Operations	
Management of data and metadata by ETL	Records the activities performed during the management of data and tables using the ETL module.
Maximum number of details for modifications by ETL	Tells the system the maximum number of rows to be recorded for operations performed by the ETL engine.
Detail of writeBack value changes	Records all operations to save data from forms using writeBack mode in the audit trail. ***Note**: deselecting the tick-box means that data saves will only be recorded when there is a change to the scenario/period/company triplet.*
Detail of SQL queries run by the writeBack operations	Records queries run by writeBack operations and allows monitoring of errors. ***Note**: selecting the tick-box increases the system workload.* It is useful as a support for monitoring any errors.
Run query from form	Records queries run from forms.
Detail of queries run from forms	Records the queries necessary for running the query included in the form and allows errors to be monitored. ***Note**: selecting the tick-box increases the system workload.*

Email settings page

Repository > General Settings > Email

Purpose of the page

This page makes it possible to configure emails sent by the system, set the notification settings and size of attachments.

Page fields

Please find below the most important fields.

Field	Description
Email configuration	
Method for sending mail	• **None** • **Custom Mail Session**: uses the SMTP session defined in the Repository configuration (**SMTP Server**). • **Mail Session Application Server**: uses the SMTP session configured in the application server (uses the name of the datasource defined in the application server with the subject javax.mail.Session and the email sender as defined in the mail.from name property in the configuration xml file).
Mail From	Only with **Custom Mail Session** selected. Email sender: • **Single**: sender indicated in **Default Mail From** • **User**: sender indicated in the user list (**Users & Roles** tile > **Users**)

Field	Description
Maximum size of attachments in MB from "Report Bursting module"	Maximum size of the .zip file generated by the Report Bursting module 0: no limit
Notifications	
Notification mechanism for scheduled data processing	Type of message that the system sends at the end of scheduled data processing
Automatically notify the user that the validity end date is about to be reached: days remaining	Notifies the user if their **Validity end date** is about to expire.
Notify automatically when about to reach licence limits: number of missing Admin Users	Sends a message that the maximum number of Admin users permitted by the licence is about to be reached.
Email for Cloud Proactive Monitoring Notification	The field is visible for cloud environments only. In the event of infrastructural problems which affect how the application works (Application not available or Partial Outages), a notification will be sent to the email address specified in the field. It is also possible to enter multiple email addresses separated by commas and without spaces (e.g. email1@gmail.com,email2@gmail.com).
In-app notes and attachments	
Attachments: maximum size (in Kbyte)	Maximum size of files that can be attached in CCH Tagetik. 0: no limit

System Resources settings page

Repository > General Settings > System Resources

Purpose of the page

This page makes it possible to adapt the application to the infrastructure on which it is run.

Page fields

Please find below the most important fields.

Field	Description
Main	
Accept lowercase codes	Makes it possible to also accept lowercase identifiers.
Max. number of connected users	Maximum number of users that can log in to CCH Tagetik at any one time. This limit is only applied to users associated with roles not using the function **0000033 Repository - Allows the access without restriction on the max. number of connected users**. 0: no limit
Enable beta features	Enables functions still in development, in order to test them.
Tagetik Data Loader	
Tagetik Data Loader - Write to file: data packet size (Kbytes)	Maximum size of data packets into which to divide the file with data extracted from CCH Tagetik to be transmitted to the Agent
Tagetik Data Loader - write to database: record number for data packet	Maximum number of records that each data packet can contain in order to be transferred to a database managed by the Agent

Field	Description
Time out (min)	Time (in minutes) that CCH Tagetik waits for the Agent's response to a command, at the end of which a timeout error appears. Default: 30 minutes
Endpoint	
FTP endpoint mode	Makes it possible to connect another server using the FTP protocol. • **Active** • **Passive**
OData service: data packet size (rows)	Maximum number of records in each data packet to be loaded to CCH Tagetik via the OData security system. • > 0: packet-based reading enabled • 0: reads all data simultaneously
Data Processing	
Allow manual interruption	Enables the button for manually stopping data processing in progress.
Disable concurrency	Makes it possible to run data processing on several servers with parallel sequencing and scheduling. If not selected, the servers will run several data processes and schedules simultaneously according to the mechanism of the concurrent tasks.
Distribute scheduled processing load	Distributes the scheduled data processing workload among all the active servers depending on their workload.
Distribute queued processing load	Distributes data processing which has been requested but is not yet underway among the active servers, depending on their workload.
Max. scheduled processes per minute/server	Number of data processes per minute that can be run on each server. 0: unlimited
Import/Export Tool: Thread number	Maximum number of threads used by data processing for imports and exports. 1: prevents parallel data processing (default)
Import/Export Tool: batch size	Number of rows to be imported in batch mode (Max batch size). 1000: default ***Note**: we suggest keeping the default value.*

Security settings page

Repository > General Settings > Security

Page purpose

This page makes it possible to manage various security settings, authorising operations performed outside of CCH Tagetik and data processing that does not need CCH Tagetik to be open.

Page fields

Please find below the most important fields.

Field	Description
Security configurations	
Enable Web Services	Enables access to the web services offered by CCH Tagetik. Essential in the presence of the Sharepoint module.
Enable public REST APIs	Enables public APIs without restarting the application server. ***Note**: if disabled, the system will show the **Status 503** error during calls to endpoints.*
Encrypt .NET (Deprecated)	Encodes data exchanged between the .NET client and the application server.
Show server system info.	Sends detailed information to the client, useful in case of errors or issues. The information is also collected when the user is not yet authenticated.

Field	Description
Show last login window	When logging in to CCH Tagetik, a pop-up displays the date of the current user's last login. The information is displayed using the client's time zone.
SSRF Endpoint Check	During creation, it enables validation checks on the URL attribute (or similar) of the Endpoints. ***Note**: only valid for Cloud versions*
Application server: permitted directories	Allows you to indicate the absolute paths of the directories visible from the Application Server for which it is permitted to read or write files with CCH Tagetik tools.
Valid file types	
Attached documents: allowed file types	Types of files that can be attached in CCH Tagetik. ***Note**: metacharacters can be used; use commas to separate the extensions.*
Timeouts	
Client socket (min)	Timeout for the socket in read mode for a certain HTTP client. 0: no timeout
Idle session timeout (min)	Minutes of inactivity after which the system automatically performs the logout. -1: no automatic logout
Data processing (sec)	Timeout for data processing and sub-processing. Once reached, the system stops the current data process or sub-process and sends the message **Data processing <DATA_PROCESSING_CODE> has reached its timeout of <TIMEOUT> seconds**. 0: no timeout ***Note:**: a specific timeout can be set for individual data processes in the data processing table.*
Query 1 (sec)	Indicates after how many seconds any query that is run during data processing must be aborted. ***Note**: If the value of this field is equal to 0, no timeout will be set.*
Query 2 (sec)	Waiting time for long data processes before a current query is stopped.
Thread Dump	
Automatic thread dump in case of lock	Automatically downloads a thread dump locally in case requests are locked out from the application server. If enabled, also specify the time period between two dumps. Default: 60 sec. ***Note**: setting reserved for technical staff.*
Time after which a request is considered blocked (sec)	Time after which a request is considered blocked if not executed. Default: 30 sec.
JVM Heap	
Load % to enter critical state	Maximum percentage of old-generation JVM memory occupied. In addition to this value, the system warns when a critical state is reached.
Action to take when in critical state	Action required when the memory is in a critical state (see "Critical State Memory Manager" on page 43). • **None**: no action to be taken • **Tracking enabled**: no action to be taken. Tracks memory status in application logs. • **Prevent new data processing**: prevents new data processes from being run, and stops processes that are underway. Data processes that use the most memory will be stopped first and then, where memory usage is the same, the most recent processes will be stopped. • **Stop data processing in progress and prevent new data processes using force memory purge of the garbage collector**: prevents new data processing and terminates current processes. Data processes that use the most memory will be stopped first and then, where memory usage is the same, the most recent processes will be stopped. If the status remains critical for more than one minute after data processing has been interrupted, the garbage collector will be activated to force a memory clean-up.

Field	Description
Other	
Number of reruns in case of deadlock	Number of times the system must attempt to rerun a query that has acquired lock status due to an exception, in order to resolve deadlock statuses on the MSSQL Server and Oracle.
Query fetch size	Number of records that must be read by a query during block-based reading

Basic configuration procedures

Enable languages available to users

Repository > General Settings > System Localization

1. In **Configuration Languages**, set the desired languages.
2. In **Default for new users**, set the characteristics to be assigned automatically to every new user.

Customise an individual user's settings

Follow one of the following pathways:

Admin/User | <application database>

Main toolbar > > Settings

Admin | Repository

Repository > Users & Roles > Users

Database management

Contents

This section covers the following topics:

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Manage exported files page 15
Work on the database 16
Query Analyzer 17

The management of CCH Tagetik databases

Introduction

Every CCH Tagetik installation contains a centralised database (Repository) and one or more application databases.

Repository management functions

A list of the main functions is provided below:

Function	Description
General Settings	Sets the general characteristics of the entire application, manages user authentication and the execution settings for all data processing performed by CCH Tagetik.
Users & Roles	Defines users and roles, sets rights, limits the field of action and associates accessible databases.
Database	Displays the linked databases and indicates whether they are accessible.
Data Processing	Defines how CCH Tagetik processes data.
Import/Export	Exports and imports tables from the various databases.
Monitor	Monitors the current data processing , log tracking and server status.
Scheduling	Sets the scheduled data processing and creates processing groups.
Master Data Management	Updates settings and metadata and transfers them from one environment to another (typically from a test environment to a production environment).
Audit	Allows consultation of application logs.

System tables

The CCH Tagetik database contains the following system tables:

Table node	Description
Setup	Tables for setting the metadata, rules and calculation logics
Data	Tables with numerical values
Users	Tables related to user rights over metadata
Deployed	Tables generated by the deployment of CCH Tagetik
Offline data & setup	Tables with data stored offline
Logs	Tables with log information

These tables are displayed by group on the **Import Tool**, **Export Tool** and **Master Data Management** pages to facilitate the administrator's job.

Other tables

The CCH Tagetik database also contains the following tables:

Table node	Description
Service	Table used by the system but not necessary for standard exporting and importing
Custom	Tables created by the user

Characteristics of exported files

The compressed files generated via exporting from the database contain:

- an encrypted file for every exported table containing the table itself
- a log file with information on the performance of the export (rows exported, time taken etc.)
- a file with the metadata of all the tables

The files are exported in compressed format (.zip) and are encrypted.

Example of an exported file

The name of an exported file is structured as follows:

Element	Description	Example
Prefix	Software that generated the code	**tcpm**
User	Code of the user who requested the export	**USER**
Data Processing	Data process that generated the file	**EXP**
Type	Type of database exported: ORA = Oracle SQL = MS SQL Server	**SQL**
Code	Code of the exported database	**TCPM_D0000**
Version	Database version number	**3.1006**
Patch	Database patch number	**0**
Date	Day and month in which the file was created	**2610**
Time	Time (hours and minutes) when the file was created	**1225**
Number	Reason number	**64781**
Part	Files generated by the export as a portion of total files	**part_1_of_2**

As in this example:

tcpm_USER_EXP_SQL_TCPM_D0000_3.1006_0_2610_1225_64781_part_1_of_2.zip

Location of files to be imported

CCH Tagetik allows files to be imported from several locations:

Location	Description
Client	The files are stored locally on a machine accessible by the user.
Remote files	The files are loaded onto the application server in the predefined folder for file uploads.
Application server	The files are loaded onto the application server in any folder. In this case, the file path of the folder must be specified.
Endpoint / Tagetik Agent	The files are uploaded to the location specified as the **Endpoint/Tagetik Agent**.

Import and export of analytical workspaces and analytical dimensions

Analytical workspaces and dimensions are imported and exported by means of the dummy table ANALYTICAL_WORKSPACE_TABLES.

Analytical workspaces and analytical dimensions can be imported under the following conditions:

- the write mode must be “Deletion/Insertion”
- all workspaces and datasets in the environment must be online

The import process comprises the following phases:

Phase	Description
1	The analytical workspaces are placed offline.
2	The data is imported into the analytical workspace and analytical dimensions tables.
3	New tables will be created and the old ones removed.
4	The new analytical workspaces will be placed online again.

In detail, the outcome of the import is as follows:

- The existing settings tables will be overwritten for both the analytical workspaces and the analytical dimensions.
- The existing partitions will be maintained and updated according to the imported settings.
- The list and physical tables of the partitions will not be imported.
- No partitions related to datasets not found in the source environment will be removed. These partitions must be removed manually in the destination database.

Query analyzer

The Query Analyzer function makes it possible to launch script on the database tables. Users for whom this function has been enabled can interact at the basic level with the data present in the database, with the exclusion of data in the audit tables. Specifically, the user can:

- in particular cases, directly control the data
- carry out complex maintenance and restructuring activities on the CCH Tagetik database
- use queries in advanced mode

If the user has the Query Analyzer function enabled in read-only mode, they can only use the *select* function in the query. If editing is also enabled, they can use all the functions (*select, insert, delete, update*).

> ***Note**: the system can only run one query at a time: the first one it finds on the list. The query will be run on the current DB connection.*

Database list page

Repository > Database

Purpose of the page

This page shows a list of all the available databases and allows the system administrator to run management operations.

Buttons and/or links on the page

Element	Description
	**Disconnection message for connected users**: sends a message to users who are about to be disconnected from a specific database.The notification may take a few minutes. If the user is an administrator of repository or of the application DB on which the notification was sent (possesses function 0000008 or 0004132, respectively), he will not receive any notification and will not be logged out **Disconnect users**: disconnects users connected to a specific database. Repository administrators can in any case access that database and carry out activities on it according to their own profile. **Upper/lower case alignment in codes**: makes it possible to correct anomalies in the codes (upper/lower case) of the data tables, using the codes present in the list tables as the master. **Encode data**: encodes the data in a database to prevent unsupported characters from being saved and misinterpreted. Also applicable to individual columns of the database table. **Remove unsupported characters**: removes characters not supported by the application from the database (the char(0) character, for example). ***Note**: this function is only available on application databases connected to the Microsoft SQL Server or Oracle type application.* **Report on data in the database**: generates a report on the quantity of data contained in a database, to assess the dimensions of the tables and the extent to which this affects processing speed. **Benchmarks**: runs a performance test on the database, network, memory (CPU), I/O on disc. **Connection Timeout**: sets the maximum waiting time for the database's response. Identifies any timeouts on the connection between the application server and the database.

Page fields

Please find below the most important fields.

Field	Description
Info	🔒 : the connected user has no rights to edit the database.
Type	Database type: • **Repository**: single centralised database • **Primary**: application database • **Secondary**: accessible external database • **Not Configured**: database which cannot be used
Login Status	**Enabled**: it is possible to connect to the database **Not enabled**: it is not possible to connect to the database

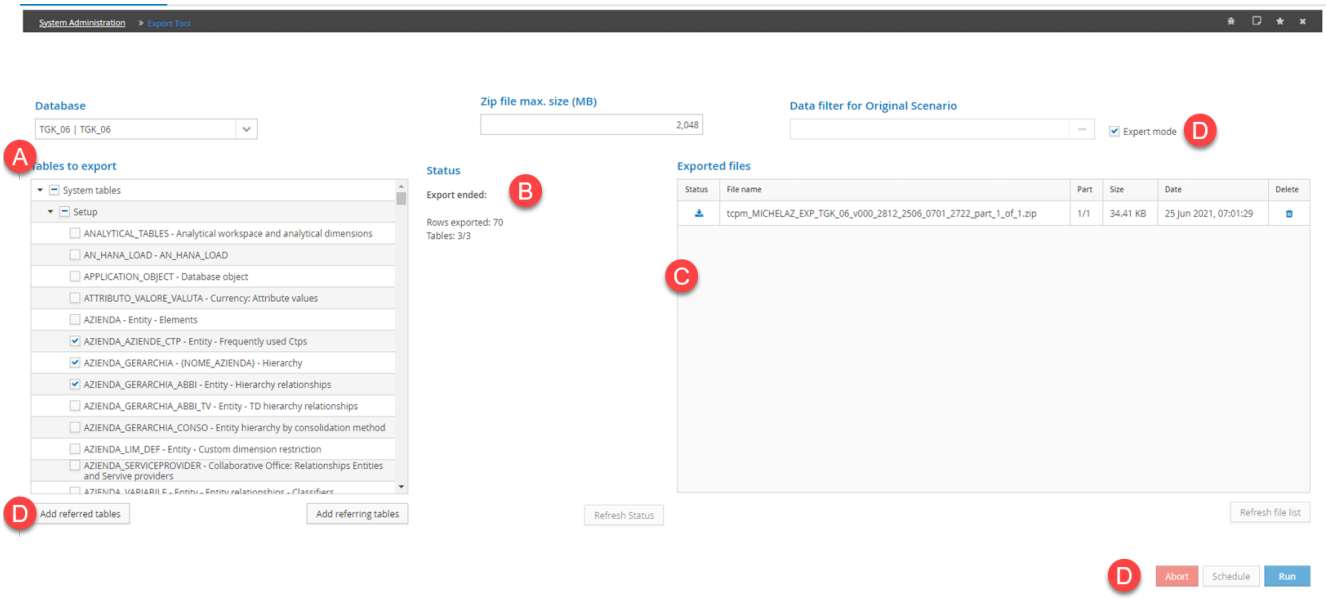
Export tool page

Repository > Import/Export > Export

Purpose of the page

This page makes it possible to export all or part of the CCH Tagetik database.

What it looks like



Element	Description
A	Information on the database to be exported
B	Information on the progress of the export procedure
C	Information on the exported files
D	Action buttons

Buttons and/or links on the page

Element	Description
Expert mode	Displays the additional fields (**Zip file max. size (MB)** and **Data filter for Original scenario**).
Add referred tables	Adds to the list the tables to which the selected tables make reference.
Add referring tables	Adds to the list the tables which make reference to the selected tables.
Refresh Status	Updates the export progress status.
Refresh file list	Updates the list of exported files.
Stop	Stops the export process.
Schedule	Opens the **Scheduling** window to schedule the export.
Run	Exports the selected tables.

Page fields

Please find below the most important fields.

Field	Description
Database	Database from which the data is exported
Tables to export	List of exportable tables contained in the selected database
Status	Export progress
Exported files	List of exported files
Zip file max. size (MB)	Sets the maximum size of the exported compressed file (.zip). If the size of the export exceeds this value, several files will be generated.
Data filter for Original Scenario	Only exports data related to a specific original scenario and potentially the associated consolidation scenarios. Can only be used if the **Tables to export** section contains at least one table of data or offline data.
Personal data protection: export uncoded data values	Decodes the data before exporting them.

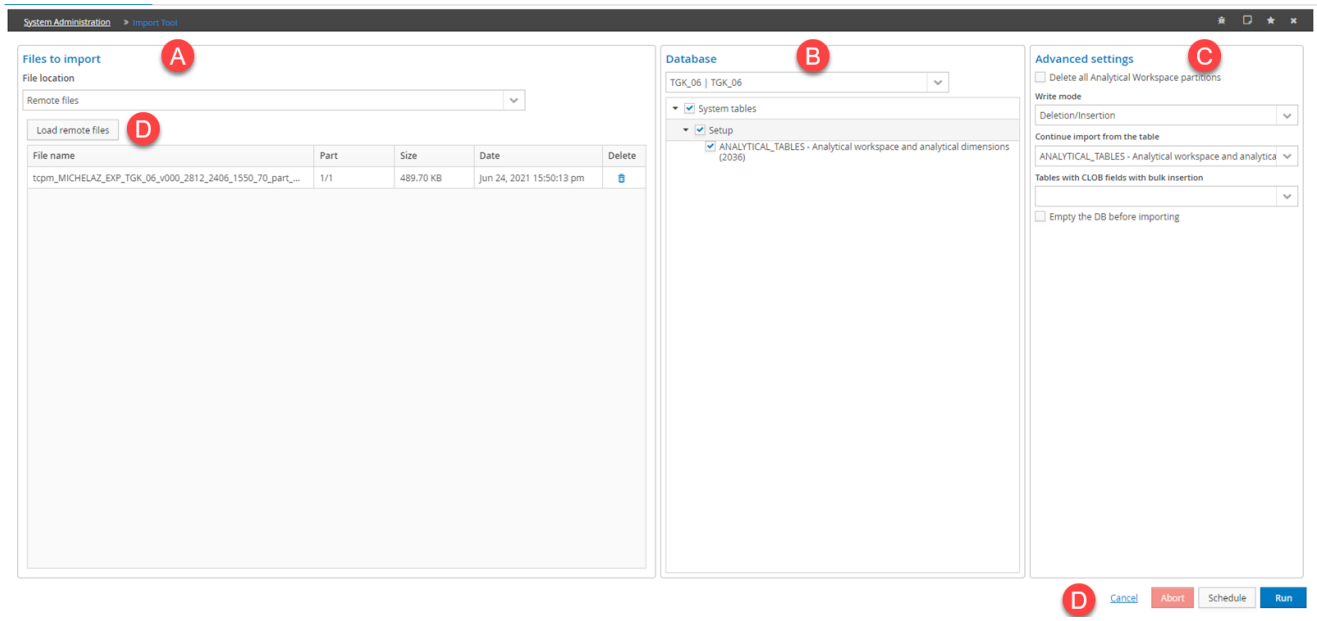
Import Tool page

Repository > Import/Export > Import

Purpose of the page

This page makes it possible to import files from previously created tables into databases.

What it looks like



Element	Description
AA	Information on the files to import
B	Information on the imported tables
C	Advanced import options
D	Action buttons

Buttons and/or links on the page

Element	Description
Load local files	Loads the files selected by a computer accessible by the user.
Load remote files	Loads the selected files into the file upload folder in the application server.
Abort	Stops the import process.
Schedule	Opens the **Scheduling** window to schedule the import.
Run	Run the import.

Advanced import options

Element	Description
Delete all Analytical Workspace partitions	For imports to Analytical workspace tables, delete all partitioning tables before importing the data.
Write mode	Sets the write mode for the imported data: • **Deletion/Insertion**: deletes the data in the tables before inserting the imported data. • **Only insertion**: adds the imported data to the existing data. • **Only deletion**: empties the tables for which data will be imported without inserting new data.

Element	Description
Continue import from the table	Resumes the export from the point at which it was previously interrupted due to an error.
Tables with CLOB fields with bulk insertion	Governs the behaviour of the import for tables with at least one CLOB (character large objects) column. Accelerates data loading during mass insertions. ***Note**: only use when you are certain that the original database contains only a few small columns with these characteristics.*
Empty the DB before importing	Empties the database before importing the new tables. This makes it possible to delete all the tables not involved in the import.

Page fields

Please find below the most important fields.

Field	Description
Files to import	
File location	Location of the file to upload.
Database	
Database	Database into which to import the data and list of tables contained in the imported file

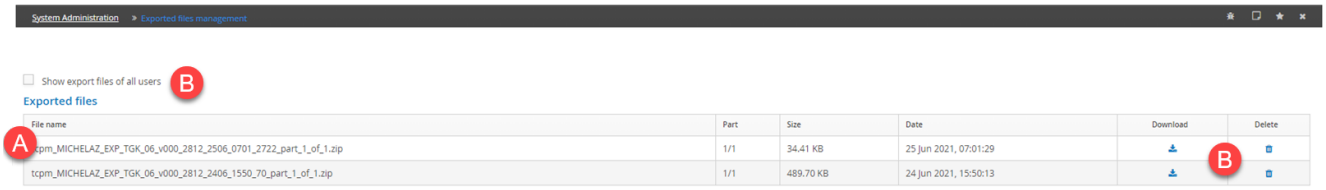
Manage exported files page

Repository > Import/Export > Exported files management

Purpose of the page

This page shows all the export files created and allows them to be downloaded locally or deleted.

What it looks like



Element	Description
A	List of exported files
B	Action buttons

Buttons and/or links on the page

Element	Description
Show export files of all users	Also shows files exported by users other than the one currently connected.

Work on the database

Disconnect users from the database

Repository > Database

To run maintenance operations or tests on the database, it may be necessary to disconnect all the users who are working on it.

- To warn users that they are about to be disconnected from a database, select the database in the list then click ⋮ > **Notify disconnection to connected users** and confirm: users connected to the database will receive a warning message.
- To disconnect users from a database, select the database in the list, click ⋮ > **Disconnect users** and confirm: all connected users will immediately be disconnected.

Export tables from a database

Repository > Import/Export > Export

This procedure creates a compressed file containing all or some of the tables from an individual database.

1. In **Databases**, select the database to be exported.
2. In **Tables to export**, select the desired tables only.
3. To also automatically import all the tables to which the selected tables make reference, click **Add referenced tables**
4. To also automatically import all the tables that refer to the selected tables, click **Add referring tables** .
5. Click **Run** to export the selected tables: the generated file or files will be listed in **Exported files**.

To check any problems during the export, refer to the end-of-process log or "Audit page" on page 69.

Import tables from a database

Repository > Import/Export > Import

1. In **File location**, select the location of the files to be imported.

   ***Note**: if the exported database has been split into several files, they must be located in the same folder in order to be imported.*

2. Select the file and upload it.

   ***Note**: depending on the file location, different identifying information will be required.*

3. In **Databases**, select the database on which to import data.
4. Select the tables to be imported.

   ***IMPORTANT**: only importing some tables could compromise the integrity of the data. For partial imports, create a dedicated export file.*

5. If you also want to import the ANALYTICAL_WORKSPACE_TABLES table, in **Advanced options**, **Write mode** select **Deletion/Insertion**.
6. Set the desired options in **Advanced options** then click **Run**: the tables will be imported immediately.

   ***IMPORTANT**: if the imported database version is less recent than the current one, this may lead to import errors. In such cases, a detailed warning message will appear.*

To check any problems during the import, refer to the "Audit page" on page 69.

Query Analyzer

Repository > Query analyzer tile

Purpose of the page

From this page, you can launch queries on data within the current database
To see the tile and use this feature, the following function needs to be enabled in the user's role:

0002286 - Repository - Tools - Query Analyzer

0002287 - Repository - Tools - Tools - Query Analyzer (Restricted)

for the query analyzer on the Repository

The following functions

0002305 - Query Analyzer

0002306 - Query Analyzer (Restricted)

for the query analyzer on the application database

Users with Query analyzer (Restricted) functions will be able to use in the query the only function: Select, with read-only data

Users with Query analyzer functions, on the other hand, are enabled to edit the database and can use all functions: select, insert, delete, update.

Page fields

Please find below the most important fields.

Field	Description
SQL queries	text of the query to be run
Run	to run the query entered
Number of rows	specifies the number of rows to be returned by the query. By default, the system returns 50 rows;
Result	result returned by the query If the query execution fails due to SQL syntax errors, the system returns an error message.

User management

Contents

This section covers the following topics:

Users 18
Users page 20
User Settings 21
Configure users in the Repository 21
Password rules page 23
Users/Roles relationships page 23

Users

Login management

CCH Tagetik is a highly structured application. For this reason, it is necessary to manage access to data for individual users and groups of users with the same responsibilities. This makes it possible to protect sensitive data and prevents data from being edited by unauthorised users.

Creating users

Users can be created either directly by CCH Tagetik or indirectly by means of external systems integrated with CCH Tagetik. Users' personal details and rights are first added to the Repository database and then to the various application databases they are authorised to access. Further user restrictions on the various dimensions are subsequently detailed in the configuration of working environments on the various application databases.

User lists

The following user lists exist in CCH Tagetik:

Type of lists	Characteristics
Centralised user list	• Saved in the Repository database. • Indicates the accessible databases and the functions available on each of them, for every user.
Application database table	• Saved in the individual application databases. • Only contains data on users authorised to access that specific database. • Indicates the user rights over elements of the data model.

Integration of authentication systems

CCH Tagetik defines the authentication criteria, i.e. the requirements for user access to the system, in the Repository database. However, it can also be integrated with the following external authentication systems to inherit details and credentials:

System	Description
Lightweight Directory Access Protocol (LDAP)	Communicates with Directory services, and particularly with the Windows Active Directory. Tagetik allows more than one LDAP Configuration to be defined, in order to be able to run LDAP authentication for users who are registered on the LDAP service in different domains.
Http Header Customizable (HHC)	This allows automatic user access to the application. The authentication data are read directly in the headers of the HTTP request. (Proprietary Tagetik authentication mechanism)

System	Description
Remote Authentication Dial-In User Service (RADIUS)	This allows access with a username and password, and also the enabling of a token-based two-factor authentication system.
Internet X.509 Public Key Infrastructure (IPKI) Client Certificate	If there is more than one HTTPS connection, this allows access by requesting a client certificate to authenticate the user. Normally in HTTPS connections, only the server provides a certificate. When the server also requests a certificate from the client, there is two-way SSL authentication.
Kerberos	This allows automatic user authentication for users already authenticated in a Windows domain.
SAML	This allows automatic access for users already authenticated in another secure domain.
OpenID	This allows access for users with an ID represented by an URL provided by an identity provider. The user must already be registered on this site.

For more details on interactions between CCH Tagetik and the LDAP system, see the documentation on that specific topic.

The **Default authentication** mechanism also automatically manages Sharepoint authentication. This is a proprietary CCH Tagetik mechanism and is based on symmetrical key encryption which allows users connected to the Sharepoint portal to access CCH Tagetik as well.

Reference users

To facilitate and speed up the creation and configuration of new users, pre-existing *reference users* can be used as a template. The functions and authorisations of the reference users will be added to the functions attributed to the user.

Area profiles

Area profiles are dummy users to whom roles and rights are attributed. Administrators can associate area profiles with users and thus restrict the number of functions that can be enabled for those users. The area profile function therefore gives a maximum limit of possible authorisations for the associated user. For example, users associated with the “Italy” area profile can be configured to have access to different functions and data compared with users associated with the “USA” area profile.

Users associated with an area profile in an application environment will inherit the rights of the area profile in the same environment for each individual dimension.

Operationally, the area profiles are a way of delegating user management. They can be used by peripheral administrators to control authorisations for specific groups of users.

User alignment

After the users have been created in the Repository database, they must also be copied into the application databases in which they are authorised to work. This operation is called *alignment*.

Rights over the data model, or the rights over the various dimensions and functions accessible to the user, will be defined subsequently.

If the user running the alignment is associated with an area profile, any users not associated with the same area profile will be excluded from the alignment.

The alignment operation also eliminates users not present in the Repository and locks users who are temporarily locked in the Repository.

User configuration process

The user configuration process typically follows these phases:

Phase	Description	See
1	Create users in the Repository database or import users already registered on other authentication system	"Add a user manually" on page 22 "Import the user details from the LDAP system " on page 34
2	Create any custom roles	"Create a custom roles" on page 28

Phase	Description	See
3	Associate a role to each user	"Associate one or several roles with a user" on page 28
4	Align users on the working application database	"Authorise a user to work on an application database" on page 22
5	Set rights: assign specific rights in the context of the data model.	-

User Creation and Deletion Log

To verify user creation or deletion operations, you can use the Metadata Log Report (see "Metadata log report page" on page 73), available at:

Repository > Audit > Report > Metadata Log Report.

The report displays a list of recorded operations and includes, for each of them:

The user who performed the action.

The date and time when the operation was executed.

The identifier of the object involved (for example, the user code that was created or deleted).

This information makes it possible to accurately monitor and verify all relevant administrative activities, ensuring full traceability of the changes applied.

Password security requirements

The system administrator defines the characteristics and requirements for passwords that can be set by the user. Those characteristics and requirements are listed in the table called "Password rules page" on page 23. CCH Tagetik also allows for more restrictive password management, using rules that require the following characteristics in order for the password to be considered secure:

- length must be at least eight characters
- must contain at least one number
- must contain at least one special character
- must never have been used before
- valid for a maximum of three months

To enable this type of management, activate the **it.grupposervizi.easy.pwd.rule.security.enabled** system property on the application server. The property also acts on passwords already in use. Once activated, the pre-existing passwords will be changed so that they meet the requirements defined by these rules.

Users page

Repository > Users & Roles > Users

Purpose of the page

This page makes it possible to manually add new users and define their characteristics. It also makes it possible to associate every user with one or more roles.

Buttons and/or links on the page

Element	Description
	**User type on licence**: generates a report with a list of all the configured users. For every user, it indicates whether they are active and what type of user they appear to be, in relation to the licences purchased. Makes it possible to check whether the licence has been infringed.
	**Users/Roles mapping**: opens the page to associate roles with the selected user. **Change Password**: makes it possible to set a new password for the selected user. **Copy preferences from**: makes it possible to copy another user's preferences to the selected user. **Apply the right to be forgotten**: deletes all personal and sensitive data for the selected user. **Mark users as deleted**: lock the selected user without deleting their personal data. This information remains available for auditing. **Recode users**: changes the code associated with the selected user.

User Settings

Repository > Users & Roles > Users

Attributes tab

Field	Description
Code	User name for login
User type	**Local User**: user added directly within CCH Tagetik. **External user**: user created within an external system and imported into CCH Tagetik.

Security tab

Please find below the most important fields.

Field	Description
Locked temporarily	Prevents the user from accessing the application.
Password rule	Rule applied to the password.
Manage password rules	Opens the page to set restrictions on passwords and password management.
Password to be changed	Forces the user to change their password upon next login. Once changed, this field is automatically deselected.
Reference users and area profile	
Reference user for other users	User who can be used as a reference user for other users.
Inherit roles from Reference user	Enables the role and accessible databases to be inherited from a reference user.
Reference Users	Users from whom the selected user inherits their roles and accessible databases.
Area profile	The selected user is a dummy user with an area profile function.
Inherit Area profile from another user	The user is associated with an area profile which limits their authorisations.
Max number of area users	Maximum number of users that can be associated with the area profile.
Area	User group. If the user represents an area profile, this field contains the user code of that user.

Configure users in the Repository

Add a user manually

There are several ways that user lists can be added to CCH Tagetik. This procedure explains how to add them manually to the user list in the Repository.

Repository > Users & Roles > Users

1. Click : a new row will appear in the table.
2. In the **Attributes** tab, enter the user's personal details.
3. In the **Security** tab, if necessary, restrict the period for which the login credentials are valid and any reference and area profile users to be associated.

Authorise a user to work on an application database

Repository > Users & Roles > Users

Repository > Users & Roles > Users/Roles mapping

1. Select the desired user.
2. Click > **Associate Users/Roles**: the relationship page opens for the selected user.
3. Click : a new row will appear in the table.
4. In the **Attributes** tab, select the database for which access is to be authorised, and the role that the user can perform on it.
5. If useful, set a period of validity for this authorisation.
6. Set the type of access restrictions for the selected database.

Create an area profile user

Repository > Users & Roles > Users

1. Click : a new row will appear in the table.
2. In the **Attributes** tab, enter the dummy user details.
3. In the **Security** tab, in **Reference users and area profile**, select **Area profile**.
4. In **Max number of area users**, set the maximum number of users that can be associated with this profile. If set to zero, there are no limits on the number of users that can belong to the area.

View the roles associated with a user

Repository > Users & Roles > Users

Repository > Users & Roles > Users/Roles mapping

1. Select the desired user.
2. Click > **Users/Roles mapping**: the roles associated with each database to which the user has access are indicated in **Role**.

Disconnect the users from the database.

Repository > Monitor > Connected Users

1. Select the user to disconnect.
2. Click > **Disconnect the selected users**: the user is immediately disconnected from the database.

Allocate a temporary password to a user

Repository > Users & Roles > Users

1. Select the desired user.
2. Click > **Change Password** and enter a password: this is the user's temporary password.
3. In the **Security** tab, select the **Password to change**: on next login, the selected user must choose a new password.

Change a user's password

Repository > Users & Roles > Users

1. Select the user whose password is to be changed.
2. Click ⋮ > **Change Password** and enter the new password.

Unlock access for a user

This procedure makes it possible to unlock access for users who have been locked manually by the administrator or automatically by CCH Tagetik.

Repository > Users & Roles > Users

1. Select the locked user.
2. In the **Security** tab, deselect **Locked temporarily** and click **Save**. The user can now access all databases on which they are authorised to work.

Password rules page

Repository > Users & Roles > Password rules

Purpose of the page

This page makes it possible to define rules related to the following aspects:

- password validity over time
- conditions for the length and format of accepted passwords
- preventing reuse of the same password on expiry
- maximum number of times a user can enter an incorrect password before being locked

Page fields

Please find below the most important fields.

Field	Description
Validity and Expiration notifications	
Validity days	Password duration 0 = no expiry
Days left before password expires	Determines how many days in advance a user should be notified of password expiry. 0 = no warning email
Days user locked after password expiration	Indicates the number of days for which the user is locked after password expiry. ***IMPORTANT**: Only the system administrator can remove this lock.*

Users/Roles relationships page

Repository > Users & Roles > Users > ⋮ > Users/Roles mapping

Repository > Users & Roles > Users/Roles mapping

Purpose of the page

This page makes it possible to associate, with each user, one or more roles in the database to which they have access.

Page fields

The characteristics of the role associated with the user in the specific database can be defined in the **Attributes** tab.

Please find below the most important fields.

Field	Description
Database	Name of the database on which the user is authorized to operate
Role	Code of the roles associated with the user when operating on the specified database
Locked temporarily	Temporarily prevents the user from accessing the application.
Restrictions on the Application DB	Defines the user's rights over the dimensions of the selected database.

Roles management

Contents

This section covers the following topics:

Roles 25
System roles page 27
Custom roles page 27
Set roles 28
Functions & Modules page 29
Users/Roles relationships page 29

Roles

What are roles?

Roles identify types or groups of users who perform similar tasks and therefore need to access specific functions or data. Each role contains a list of functions or operations that users associated with that role can perform. For example, a role can be associated with running specific data processing.

Types of role

There are the following types of role:

Roles	Characteristics
System roles	• Predefined, cannot be changed or deleted. • Can be used as reference for creating custom roles. • Are identified by the $ symbol.
Custom roles	Created for specific needs, can be changed and deleted.

System roles

The available system roles depend on the type of licence installed. A list is included below:

Name	Description	Licence requested
$ALL	Can access all functions	Administrator
$ADMINISTRATOR	Has administrator privileges for an application database and can: • add new users • define Cash Flow Planning, Allocation and Closing, consolidation, ETL rules • apply the data entry and report generation settings • define the reporting forms and templates • configure the data collection process • prepare preliminary data prior to data entry (e.g. insertion of exchange rates and tax rates) Original data cannot be added.	Administrator
$AUDIT	Can view data and lists without editing them.	Data Entry & Reporting o Viewer

Name	Description	Licence requested
$CONSOLIDATOR	Has consolidator privileges and can: • enter data into the consolidated financial statements (e.g. consolidation adjustments, consolidation data processing) • enter data on shareholdings • manage intercompany entries • monitor the activities of individual contributors	Administrator or Power User or Contributor
$SYSADMIN	Has system administrator privileges, works only on the Repository and can: • create custom roles • add users and define their roles • manage user visibility on the application databases • manage data processing run modes and scheduling • manage database imports and exports • manage logs	Administrator
$USER	Has contributor privileges and, for the entities for which they are responsible, can: • enter data on balances and preconsolidation adjustments • enter data from forms • enter data on intercompany entries • reconcile intercompany entries • run data processing operations for individual companies • run the diagnostic check, by company • certify data	Power User or Contributor
$DEREPORT	Can generate reports, enter data from forms and run data processing operations entered on data entry forms or configured in a process workflow.	Data Entry & Reporting or Contributor
$ANALYTICS	Can see reports published on the Tagetik Analytics web portal according to their own rights. Cannot access the web interface.	Tagetik Analytics

Functions and function modules

Specific functions are associated with every role, acting as authorisations to perform specific operations. Users associated with a role are authorised to perform all the functions that are in turn enabled for that role. Several related functions can be grouped into a module. Therefore, every function allows data to be worked on in a specific, controlled and restricted way.

Type of change to data

Below is a list of the possible data work permitted by the various functions:

Type	Description
Don't edit data	Prevents data from being edited.
Consolidated amount	Allows editiing of tables of consolidated data, both for balances and for adjustments.
Adjustment	Allows editing of adjustment data tables, both gross and intercompany.
IC amount	Allows editing of tables of original intercompany data.
Gross amount	Allows editing of tables of original gross data.
All	Allows editing of data in more than one of the tables specified above. This function does not necessarily fall within one of the above classifications/

Association of roles with users

Associating a role with a user determines which menus the user can see and which functions they can perform. Users who have access to various application databases may have different roles for each database. When these associations are created, the activities to which the user will have access in the application database will also be defined. More specific restrictions are defined in the application database itself.

Example of association: different roles for different databases

USER1 needs to work on application databases DB1 and DB2. On DB1 he performs the function associated with role RA, while on DB2 he performs the functions associated with role RB. Roles and databases must therefore be associated with the user as follows:

User	Application database	Role
USER1	DB1	RA
USER1	DB2	RB

Example of association: different roles for the same database

USER2 needs to perform functions on DB2. These functions are associated with roles RC, RD and RE. The user must therefore be associated with all the necessary roles.

User	Application database	Role
USER2	DB2	RC
USER2	DB2	RD
USER2	DB2	RE

Restrictions on application databases

The types of restrictions that can be associated with a user in relation to specific roles that they perform on specific databases are outlined below. The relative user rights will consequently be configured by the system in the application environment.

Restriction	Description
Unrestricted	Unrestricted user
Restricted on all dimensions	User restricted on all dimensions
Restricted	User only restricted for the entity, category and consolidation scenario dimensions

System roles page

Repository > Users & Roles > Roles > System roles

Purpose of the page

This page makes it possible to view a list of system roles with the functions associated with them.

Buttons and/or links on the page

Element	Description
	Enabled functions: opens the page with the list of functions enabled for the selected role.

Custom roles page

Repository > Users & Roles > Roles > Custom roles

Purpose of the page

This page makes it possible to create customised roles in addition to the predefined system roles and view a list of custom roles already created.

Buttons and/or links on the page

Element	Description
⋮	**Enable functions**: opens the page for associating functions with a custom role.

Attributes tab

This tab allows you to define the characteristics of the selected custom role.

Please find below the most important fields.

Field	Description
Reference role	Custom or system role from which the custom role inherits all characteristics. ***Note**: defining a role as the sum of other roles, or using reference roles, is particularly useful when overrides need to be applied. Otherwise, the "sum" of the functions of roles can be obtained by associating several roles to individual users, without defining further "sum" roles.*

Set roles

Create a custom roles

Repository > Users & Roles > Roles > Custom roles

1. Click ⊕: a new row will appear in the table.
2. In the **Attributes** tab, assign a code and meaningful description to the new custom role.
3. In **Reference Roles**, if necessary, select one or several roles from which to inherit enabled functions.
4. Click **Save**: the new custom role will appear in the custom roles table and can be customised.

Customise a custom role

Repository > Users & Roles > Roles > Custom roles

1. Select the role to be modified and click ⋮ > **Enable functions**: the page for configuring the functions associated with the selected role will open.
2. Click ⊕: a new row will appear in the table.
3. In the **Attributes** tab, from **Module/Function**, select a function or module to be enabled for the role.
4. In **Action**, set the type of enabling.
5. Click **Save**: the enabled function appears in the custom roles table.
6. Click **Definition result** to see details of all the functions enabled for the selected role.
7. To enable several functions, repeat the process from step 2.

Associate one or several roles with a user

Repository > Users & Roles > Users

Repository > Users & Roles > Users/Roles mapping

1. Select the desired user.
2. Click ⋮ > **Associate Users/Roles**: the relationship page opens for the selected user.
3. Click ⊕: a new row will appear in the table.
4. In the **Attributes** tab, select the desired database and the desired role.
5. If useful, define a period for which the association with the role will be valid.
6. Set the type of access restrictions for the selected database.

Functions & Modules page

Repository > Users & Roles > Roles > Functions & Modules

Purpose of the page

This page shows a list of all the functions, grouped into modules, available to the connected user.

Page fields

Please find below the most important fields.

Field	Description
Module	Code and description of the functions module to which the selected function belongs
Function	Code and description of the function ◦ Consolidated data: functions that modify consolidated data tables both balances and adjustments ◦ Don't modify data: functions that do not modify data ◦ Adjustment: functions that modify the tables of adjustment original data both gross and intercompany ◦ Intercompany balance: functions that modify the tables of original data intercompany ◦ Gross Balance: functions that modify the original gross data tables ◦ All: functions that modify more than one of the above tables and therefore do not fall uniquely into one of the above classifications.
Licence type	Type of licence for which the function is available
Edit type	Type of data edit that the function allows

Users/Roles relationships page

Repository > Users & Roles > Users > ⋮ > Users/Roles mapping

Repository > Users & Roles > Users/Roles mapping

Purpose of the page

This page makes it possible to associate, with each user, one or more roles in the database to which they have access.

Page fields

The characteristics of the role associated with the user in the specific database can be defined in the **Attributes** tab.

Please find below the most important fields.

Field	Description
Database	Name of the database on which the user is authorized to operate
Role	Code of the roles associated with the user when operating on the specified database
Locked temporarily	Temporarily prevents the user from accessing the application.
Restrictions on the Application DB	Defines the user's rights over the dimensions of the selected database.

External Authentication

Contents

This section covers the following topics:

Configuration of External Authentication Systems page 30
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LDAP search for roles page 32
LDAP search for rights page 33
LDAP alignment window 33
Manage users from the LDAP system 34

Configuration of External Authentication Systems page

Repository > ⋮ > External Authentication

Purpose of the page

This page makes it possible to set up communication with external authorisation systems such as the LDAP system, and determine how user data is imported from the various systems.

Page fields

The fields displayed for communication with the external LDAP system are shown below.

Field	Description
Use Third Party Authentication	Enables the use and configuration of communication with the various external systems and displays the necessary fields.
Available authentication types	Makes it possible to select the external system with which to configure communication.
General Settings	
Fail Safe authentication enabled	The user can access CCH Tagetik using their own credentials even if the external system is not responding.
Enabled Login for all users of external authentication mechanism	All users authenticated on the external system will be added to the Repository and can therefore access CCH Tagetik.
Password rule for open an external user	Password rule that a user from an external system must follow if they become a local user.
Customized URL for redirect after Logout	URL to which users are redirected after disconnecting from CCH Tagetik.
Specific Settings	
Update User data on Login	Every time a user connects, the relative data in the Repository will be updated with the data on the LDAP system.
Read profiles from LDAP	The user's roles and rights will be inherited directly from the LDAP system.
Filter: User name	Script for recovering the user name based on the user name entered at login.
Domain rule	Script for recovering the domain name based on the domain name entered at login.
Browse LDAP with specific user	Makes it possible to set the user who by default runs the user search on LDAP.

Field	Description
Configuration sets	
	Opens the window of requirements to be configured for the LDAP servers. = configuration complete = configuration incomplete, required data are missing. For more details on configuring these requirements, see the documentation on integration with the LDAP system.

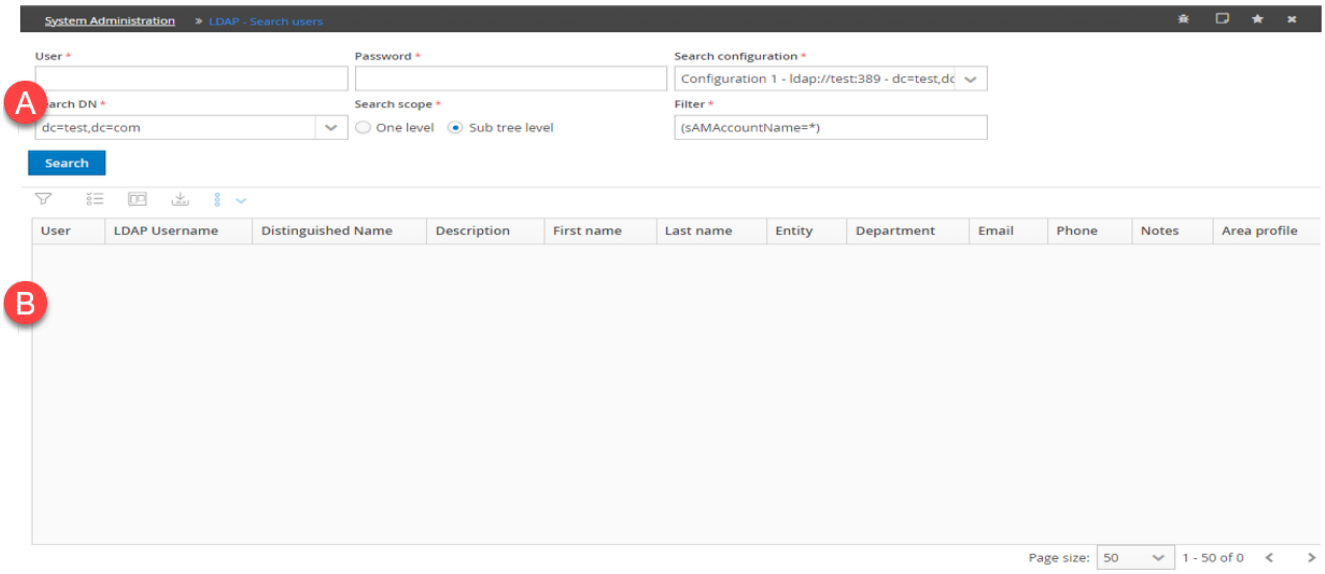
LDAP search page

Repository > Users & Roles > LDAP > Search users

Purpose of the page

This page makes it possible to search users registered in the LDAP system and import the relative tables into CCH Tagetik.

What it looks like



Element	Description
A	Search filters
B	List of users found on the LDAP system

Buttons and/or links on the page

Element	Description
Search	Runs the search according to the filters that have been set.
	Enable Users: imports the list of selected users into the repository. **Show info**: shows the details of the selected user.

Data on the page

Please find below the most important fields.

Field	Description
Search configuration	Combination of pre-defined search parameters
Search DN	Distinguished Name suffix in LDAP used as the starting point for the search
Search scope	**One level**: the search is only performed in the sub-levels immediately below the starting point of the search specified in the **Search DN** field. **Sub tree**: a recursive search is performed on all sub-levels below the starting point of the search specified in the **Search DN** field.
Filter	Rule with which the search is run. The initial value will be taken from the **Attribute Mapping - Username** field defined in the selected configuration.

For more details on user searches in the LDAP system, see the documentation on that specific topic.

LDAP search for roles page

Repository > Users & Roles > LDAP > Search reference users for roles

Purpose of the page

This page makes it possible to search users registered in the LDAP system so they can be used as reference users for roles. It also allows the users that have been found to be imported, associating them with the roles on CCH Tagetik.

Buttons and/or links on the page

Element	Description
Search	Runs the search according to the filters that have been set.
⋮	**Enable Users**: imports the list of selected users into the repository. **Show info**: shows the details of the selected user.

Data on the page

Please find below the most important fields.

Field	Description
Search configuration	Combination of pre-defined search parameters
Search DN	Distinguished Name suffix in LDAP used as the starting point for the search
Search scope	**One level**: the search is only performed in the sub-levels immediately below the starting point of the search specified in the **Search DN** field. **Sub tree**: a recursive search is performed on all sub-levels below the starting point of the search specified in the **Search DN** field.
Filter	Rule with which the search is run. The initial value will be taken from the **Attribute Mapping - Username** field defined in the selected configuration.
Attribute - Reference user for roles	Name of attribute containing the profile to be used for inheriting roles. The initial value is recovered from the **Attribute mapping - Reference users for roles** field in the selected configuration.
Rule - Filter: Reference users for roles	Rule to be applied to identify the profile name, starting with the attribute. The initial value will be taken from the **Recover rules and filters - Filter: Reference users for roles** field in the selected configuration.

LDAP search for rights page

Repository > Repository > Users & Roles > LDAP > Search reference users for rights

Purpose of the page

This page makes it possible to search users registered in the LDAP system so they can be used as reference users for the user rights. It also allows the users that have been found to be imported, associating them with the roles on CCH Tagetik.

Buttons and/or links on the page

Element	Description
Search	Runs the search according to the filters that have been set.
	Enable Users: imports the list of selected users into the repository. **Show info**: shows the details of the selected user.

Data on the page

Please find below the most important fields.

Field	Description
Search configuration	Combination of pre-defined search parameters
Search DN	Distinguished Name suffix in LDAP used as the starting point for the search
Search scope	**One level**: the search is only performed in the sub-levels immediately below the starting point of the search specified in the **Search DN** field. **Sub tree**: a recursive search is performed on all sub-levels below the starting point of the search specified in the **Search DN** field.
Filter	Rule with which the search is run. The initial value will be taken from the **Attribute Mapping - Username** field defined in the selected configuration.
Attribute - Reference user for rights	Name of attribute containing the profile to be used for inheriting rights. The initial value will be taken from the **Attribute mapping - Reference users for rights** field in the selected configuration.
Rule - Filter: Reference users for rights	Rule to be applied to identify the profile name, starting with the attribute. The initial value will be taken from the **Recover rules and filters - Filter: Reference users for rights** field in the selected configuration.

LDAP alignment window

Repository > Users & Roles > LDAP > Refresh users

Purpose of window

This window makes it possible to align configurations in CCH Tagetik, for a single user, with those present in the LDAP system.

Fields in window

Please find below the most important fields.

Field	Description
Username	LDAP service login credentials
Password	

Field	Description
Configuration	Specific configuration from which to read settings for communication with the LDAP service
Users	List of users for which to align the table
Synchronize	Rule according to which user data is aligned: **Rights & Roles**: personal data and data on the roles and rights associated with users are aligned. **Only Rights**: only personal data and data on the rights associated with users are aligned. **Only Roles**: personal data and data on the roles associated with users are aligned.

Buttons and/or links in the window

Element	Description
Schedule	Schedules the running of a user alignment.
Run	Runs the user alignment immediately.

Manage users from the LDAP system

Enable communication with the LDAP system

Repository > ⋮ > External Authentication

1. Select the **Use Third Party Authentication** option: all the necessary settings for configuring communication with the external system will be displayed.
2. From **Available authentication types**, select LDAP.
3. Set all the desired fields.

Import the user details from the LDAP system

This procedure describes how to import the user details already registered in the LDAP system. For more details, see the documentation on that specific topic.

Repository > Users & Roles > LDAP > Search users

Repository > Users & Roles > LDAP > Search reference users for roles

Repository > Repository > Users & Roles > LDAP > Search reference users for rights

1. Enter the search parameters, then click **Search**: a list of all the users that fulfil the search criteria will be displayed, together with their user details.
2. Select the users to be imported into CCH Tagetik, then click ⋮ > **Enable Users**.

Data processing management

Contents

This section covers the following topics:

Data Processing page 35
How to work with data processing 35

Data Processing page

Repository > Data Processing

Purpose of the page

The processing characteristics can be set on this page.

Page fields

Please find below the most important fields.

Field	Description
Type	Processing performance mode. • **Synchronous**: processing performed in real time by the user's client. Temporarily locks the CCH Tagetik client awaiting the result. • **Asynchronous**: processing performed by the server. Makes it possible to continue working with the CCH Tagetik client. ***Note**: scheduled data processing is always performed in asynchronous mode.*
Audit SQL data processing	Activates a query audit which the system will perform on the individual data processes. Useful during maintenance.
Max Concurrent tasks	Maximum number of parallel tasks that the available servers can run for each individual data process
Timeout (sec)	Seconds of inactivity after which the selected data process and related subprocesses will be suspended

How to work with data processing

Set the processing characteristics

Repository > Data Processing

1. Do the following:

To edit the processing characteristics...	Then...
in the table	1. Click ⁘: the table will expand. 2. In the table, click the desired data process. 3. Edit the desired fields.
in the configuration tab	1. Run the desired procedure: **To edit...** / **Then...** a single data process → click the desired data process several processes simultaneously → 1. Click ☰. 2. Select the desired data processes. 2. In **Attributes**, edit the desired fields.

2. Click **Save**.

Scheduling of data processing

Contents

This section covers the following topics:

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Scheduling 38
Schedule data processing 39

Scheduling page

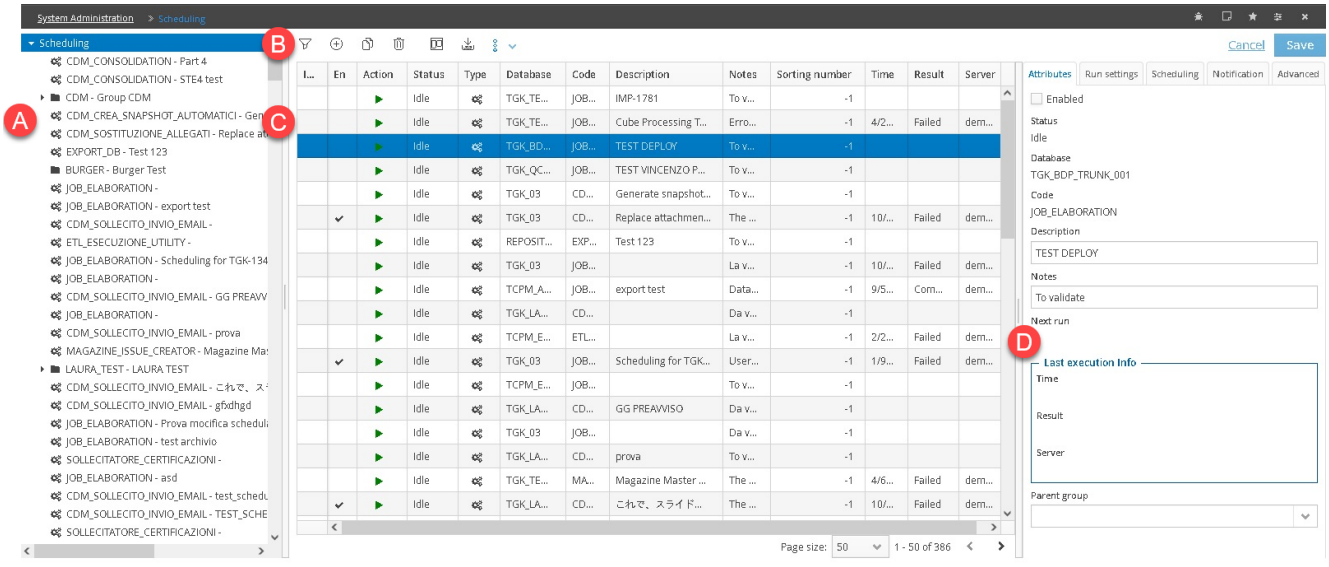
Repository > Scheduling

Purpose of the page

This page makes it possible to:

- Approve the data processing operations scheduled by the user.
- Schedule new data processing operations.
- Manage the existing scheduled data processing.
- Create data processing groups to be scheduled.

What it looks like



Element	Description
A	Tree with scheduled data processing operations and scheduled data processing groups
B	Tool bar
C	Table with details of the items selected in the tree
D	Configuration tab for the data processing selected in the list

Buttons and/or links on the page

Element	Description
	Duplicates the selected row in the table, customising the following fields: • (**Run settings** tab) **Running as**: user who duplicated the row. • (**Scheduling** tab) **Minutes** and**Hours**: time at which the row was duplicated. • (**Scheduling** tab) **Day of the month**, **Month** and **Day of the week**: take the value *.
	Deletes the selected row in the table. The deletion of a group entails the deletion of all operations scheduled within it.
	Shows the selected scheduled data processing graph (code **JOB_ELABORATION**).
	Immediately runs the selected scheduled data processing, if inactive.
	Stops the selected scheduled data processing and the data processing of its group (if any), if underway or running. ***Note**: The data processing will resume automatically at the scheduled time, if it has not passed already.*

Scheduling elements

Element	Description
	Scheduled data processing group
	Individual scheduled data process

Configuration tab

Element	Description
Attributes	Details on the data processing
Run settings	Data processing run details
Scheduling	Scheduling of data processing
Notification	Details of data processing notifications
Advanced	Insertion of any SQL scripts

Scheduling

Repository > Scheduling

Attributes tab

Please find below the most important fields.

Field	Description
Enabled	Indicates that the data processing has been approved and enables it to be run.
Parent group	Indicates the group to which the data processing belongs.
Note	Reports the reason why the scheduled data processing is not valid and may contain comments added by the user.
Last execution Info	Reports the date, time and result of the last data processing and the name of the server that performed it. If the system property *it.grupposervizi.easy.server.scheduling.allowed* (selected by default) is deselected in the application server configuration file, the system will prevent pending or booked data processing from being run on the server. ***Note**: scheduled data processing is always performed in asynchronous mode.*

Run settings tab

Please find below the most important fields.

Field	Description
Enable sorting	Only with **Parent group** specified. Makes it possible to specify the run order in relation to other data processing operations scheduled in the same group.
Permitted servers	Indicates the servers on which the scheduled data processing can be performed.
Booking Maximum Waiting Time	Amount of time for which the system will try to run the scheduled data processing in the queue. The system assigns **Booked** status to the data processing and automatically attempts to run it every minute until it succeeds or until the time limit set by the user is reached. • -1: no time limit • 0: prevents the data processing from acquiring **Booked** status and schedules it for the next Crontab cycle. • > 0: minutes in queue. Once expired, the data processing is considered to have failed and will be scheduled for the next Crontab cycle.
Errors management	This makes it possible to configure the launch of failed scheduled data processing.
Action in case of Error	Only with **Parent group** specified. Indicates the action that the system must perform if a data process in the group fails. • **Continue**: the system moves on to the next data process in the group. • **Exit**: the system stops data processing in the group.

Scheduling tab

Please find below the most important fields.

Field	Description
Days of the month	Data processing run date or dates. *: all possible values
Month	Data processing run month indicated in figures. *: all possible values

Crontab syntax

The fields *Minutes*, *Hours*, *Days of the Month*, *Month* and *Days of the Week* can contain a number, a range of values, multiple values, particular strings, following the Crontab syntax. The operators that can be used are:

- comma (,): specifies a list of values e.g. 1,3,4,7,8
- hyphen (-): specifies a range of values, e.g. 1-6 which equals 1,2,3,4,5,6
- asterisk (*): specifies all possible values of a field. For example, an asterisk in the time field means 'every hour';
- bar (/): is used to skip a certain number of values.

Schedule data processing

Approve a data processing operation

Repository > Scheduling

1. Select the operation to be approved, first in the tree and then in the table.
2. In **Attributes**, select **Enabled**: the approval tick will appear in the **Enabled** column of the table.

Create a scheduled data processing group

Repository > Scheduling

1. Click : a new row will appear in the table.
2. In the **Attributes** tab, enter the group's ID code and description.
3. Set the desired parameters in the remaining configuration tabs.
4. Click **Save**: the group schedule will appear in the tree.

Add data processing operations to a group

Repository > Scheduling

1. Run the desired process:

To act...	Then...
in the tree	Select and drag the desired operation to the group schedule.
in the table	1. Select the desired data processing operation. 2. In the **Attributes** tab > **Parent group** select the desired group.

2. Click **Save**.

Manage a scheduled data processing group

This procedure makes it possible to set valid parameters for all scheduled data processing operations in the group.

Repository > Scheduling

1. Select the group of interest in the tree and then in the table.
2. Change the parameters of the configuration tabs as desired.
3. Click **Save**.

Enable/disable scheduling on a specific server

Repository > Monitor > Server Status

1. Select the row for the desired server.
2. Click in the **Allowed schedules** field: select/deselect the tick box to enable/disable the data processing schedule on the server.

Associate a scheduled operation with a specific server

Repository > Scheduling

1. Select the desired scheduled data processing operation in the tree and then in the table.
2. In **Run settings**, select **Permitted servers**.
3. In **Specific servers** indicate the server to which to associate the selected data processing.

Monitor

Contents

This section covers the following topics:

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Running data processing page 41
Log levels page 42
Server Status page 42
Critical State Memory Manager 43

Connected users page

Repository > Monitor > Connected Users

Purpose of the page

This page makes it possible to view all the users currently connected to CCH Tagetik and, if necessary, disconnect them.

Buttons and/or links on the page

Element	Description
⋮	**Disconnect the selected users**: disconnects the selected users from CCH Tagetik without advance warning but with a notification.
Refresh	Updates the list of connected users.

Running data processing page

Repository > Monitor > Running data processings

Purpose of the page

This page shows a list of all the data processing operations currently being run, indicates their status and allows the interruption of any incomplete processes. It also specifies the processes that have been completed and reported to the user.

Buttons and/or links on the page

Element	Description
⋮	**Stop**: stops the selected data processing operation if its status is **BOOKED** or **RUNNING**. **Details**: shows the details of the selected data processing operation, if still running (**RUNNING**). If the data processing status is **PARTIALLY_COMPLETED**, **COMPLETED**, **ABORTED** or **FAILED**, it shows the relative Audit page.

Data processing status

Status	Description
PARTIALLY_COMPLETED	Data processing operation structured into several parts and partially completed
COMPLETED	Data processing completed successfully
RUNNING	Data processing in progress
FAILED	Data processing stopped with error
ABORTED	Data processing stopped by user

Log levels page

Repository > Monitor > Log Levels

Purpose of the page

This page makes it possible to define the level of detail according to which significant events in CCH Tagetik are monitored. You can also create several types of log and set more or less detailed log levels for specific events occuring on CCH Tagetik.

Changes made on this page will remain valid until the application server is next restarted. For every log category, the level specified in the system log configuration file will be restored.

Buttons and/or links on the page

Element	Description
	This adds a row in order to insert a new log type.
	Duplicates a log type with the same characteristics as the one selected.

Log levels

Level	Description
DEBUG	Most detailed log level. Tracks debug and information messages, warnings and critical errors.
INFO	Medium log level. Tracks information messages, warnings and critical errors.
WARN	Log level that tracks warning messages and critical errors.
ERROR	Log level which only tracks critical errors and unexpected conditions.

Server Status page

Repository > Monitor > Server Status

Purpose of the page

This page shows the status of all the servers currently connected to the Repository database.

Buttons and/or links on the page

Element	Description
	Updates the displayed data.
	**Download Thread Dump from the current server**: downloads a text file with the code for all active processes on the current server.
	**Download Server Details**: downloads information on the status of the selected server in a .json file.

***Note**: if the data are not available within two minutes from the request, a warning message will be displayed and the operation may be repeated.*

Server status

Status	Description
Shutdown	Server shut down
On	Server on
Verified	Server turned off. Another server has verified that the database contains no suspended records related to this server.
Broken	Server no longer correctly connected.

Attributes

Allowed schedules Status For each server, you can use the checkbox to change the server selected when running scheduled or booked data processing.

Critical State Memory Manager

What the critical state memory manager is for

The amount of memory used by an application server depends on CCH Tagetikusage and varies a lot based on the number of user sessions, the number and type of data processes running, the application setup and the size of the database.If the available memory is exhausted, the application may remain in an inconsistent state or no longer respond to user requests.

The critical state memory manager is used to avoid approaching the threshold past which the available memory will be exhausted.

Depending on the configuration that has been set, it makes it possible to block the launch of new data processing and potentially also terminate processes that are running, where they are defined as “interruptible”, if the application server is using more memory than the threshold for entering a critical state.

To manage the memory when it is in a critical state, it is necessary to define the following elements:

- Action required when in critical state
- JVM heap load threshold before entering a critical state (%)
- JVM heap load threshold for exiting a critical state (%)

The default action to be taken in a critical state is to block the launch of new data processing, thus avoiding further overloading a system under stress.

If the application server terminates due to a lack of memory, identify the origin of the problem and check whether the size of the application server's memory is correct.In this case, enable the critical state memory manager and block data processing found to be in a critical situation.

Determining critical states

Every time the system needs to increase the amount of memory being used, the manager checks whether or not it is in a critical situation.

If the system is in a critical situation and the memory usage trend is showing an increase compared with the previous measurement, then the critical state memory manager requests the termination of most recently started data processing that is defined as “uninterruptible” and potentially uses the most memory.

The same thing occurs when the system is in a critical situation and the memory usage trend is showing a reduction, but there is no pending request to terminate data processing.

Blocking the launch of new data processing

When the application’s status is critical in terms of memory usage, the critical state manager may refuse to launch new data processing, if this behaviour is enabled in the settings.This applies both for requests made by users on the user interface and for scheduled data processing.

If the data processing was launched manually from the web interface, the following behaviours may take place:

If the user chooses to...	Then the system...
cancel the data processing	does not perform any action.
relaunch the data processing	attempts to launch the data processing again. If there is still insufficient available memory, it blocks the data processing immediately and displays and error message.
book the data processing	the data processing is added to the queue. In single-server environments, the data processing is launched as soon as the critical state has been exited.in multi-server environments, the data processing can be performed on a server subject to a lower load.

Termination of data processing

When the critical state memory manager terminates data processing, it does so to ensure that other data processing underway is able to be completed successfully.Data processing completed by the critical state memory manager ends with the message “The data processing was stopped by the system”.

For data processing launched manually from the web interface, the user is given the options described in "Critical State Memory Manager" on the previous page.

Audit

If at least the *Tracking enabled* is set in the *Action required when in critical state* configuration, then the Audit registers every time the system enters or exits a critical on the *Resources Monitor* operation.

The two types of event registered by the *Resources Monitor* operation are:

- *Critical memory usage detected* - indicates that the system has entered a critical state
- *Healthy memory come back* - indicates that the system has exited the critical state

Four pieces of information are registered in the details of the Audit operation:

- the garbage collector counter - useful information for recovering the same event in the logs
- the percentage of memory used
- the memory trend, which may be INCREASING or DECREASING
- the memory status, which may be RED (in a critical state) or GREEN (not in a critical state)

The audit also registers, in the *Operations on data processing in progress* category, any individual termination requests for each of the data processes in progress that the memory manager has requested to stop.

All audit registrations made by the memory manager are made in the name of the system ser *TagetikSystemAdministrator*.

Master Data Management (MDM)

Contents

This section covers the following topics:

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Master Data Management

What is it for

Master Data Management (MDM) makes it possible to manage metadata and rules that need to be shared among several databases, such as a test database and a production database. The purpose is to update and synchronise databases quickly and easily, avoiding the need to export and import the entire database. Update and synchronisation take place by means of routines.

What are routines?

Routines are packets of instructions containing all the attributes and information related to a data synchronisation. The following types of routine exist:

Routine	Description
From database to database (DB to DB)	The data are transmitted from one CCH Tagetik database to another. These databases share the same Repository from which the routine is performed. Syncing is direct and does not require the file containing the data to be transmitted to be exported first.
From database to file (DB to file)	The desired data will be exported from a database and saved in an external .zip file encrypted within the chosen path. The encrypted data remain encrypted.
From file to database (file to DB)	An previously-created encrypted .zip file is imported into the desired database.

Routines can also be grouped.

Analytical and Custom Dimensions

The data processing operates on the analytical or custom dimension tables, synchronising the dimension elements in the source and target environments, including custom attributes. If new dimensions are open in the source environment, the data processing automatically opens those dimensions on the target and synchronises their content. For dimensions that are closed in the source environment, the work must be done manually, repeating the same operation on the target environment dimensions homepage.

Run in encrypted environments

If the environment requires the encryption of sensitive information, such information will be kept encoded in the created file and will be illegible when imported, unless the data is imported to an environment with an encryption configuration identical to the source environment.

MDM page - Routines list

Repository > Master Data Management > Routines > List

Purpose of the page

This page makes it possible to create routines for importing and exporting data.

Buttons and/or links on the page

Element	Description
	• **Details**: makes it possible to define the content of the routine. • **Copy routine details**: creates a copy of the selected routine.

Attributes tab

Please find below the most important fields.

Field	Description
Info	: routine locked; can only be edited and run by authorised users. the user is not authorised to access the Source DB specified in the routine, or the role associated with the user or the role of their reference user does not have the 0003568 - Master data management: Export DB with MDM function enabled; the user is not authorised to access the Target DB specified in the routine, or the role associated with the user or the role of their reference user does not have the 0003569 - Master data management: Import DB with MDM function enabled;
Code	Routine ID
Target DB	Database into which the selected routine will update the tables or import the setups
Sorting	Order in which the selected routines will be performed
Disable detail audit	Determines the level of detail at which the performance of the routine is audited. If selected, it ignores changes to the synced tables and only registers the performance of the routine. ***Note**: Please only select the tick-box to speed up operations for en masse synchronisations.*
Automatically include the required tables	Also syncs the tables linked to the selected table to avoid primary key violations and partial synchronisations of metadata. Synchronisation of linked tables is limited to the elements called up by the tables chosen by the user. ***Note**: please use the tick-box for all routines.*
Run in steps only	Runs the routine in steps, requiring synchronisation to be manually confirmed for each one. ***Note**: if not selected, the system shows the changes at the end of the operation without allowing the user to restore the previous situation.*
Do not run deployment of the application after termination of the routine	Prevents the MDM engine from automatically launching the necessary deployments, in relation to the recently updated definition tables. ***Note**: if the tick-box is selected, deployments must be launched manually.*

Details attributes tab

Repository > Master Data Management > Routines > Elements > Details

On this page, it is possible to define the content of the routine.

Please find below the most important fields.

Field	Description
Action	Include or exclude the selected table.
Table	Specify the table or logical groupings of the tables to be synchronised. • **$ | ANALYTICAL_DIMENSION - Analytical Dimension**: contains the tables relating to the type of routine, including custom attributes. For routines from databases to files without tables, the section appears empty to allow new analytical dimensions included in the import files to be imported. • **$| DESTINATIONS - Custom dimensions**: contains tables relating to the type of routine, custom attributes included. For routines from databases to files without tables, the section appears empty to allow new analytical dimensions included in the import files to be imported.
Include the child tables automatically	Includes all tables linked to the selected table.
Deletion method	Manage data in databases before synchronising. • **Delete**: automatically deletes absent records from the source database, only if the table on which the routine acts has no links to other tables in the database. *Note: if there are links to other tables, CCH Tagetik highlights the records that should be deleted manually from the target database in the performance audit ( ).* • **Delete only the rows produced by MDM**: automatically deletes from the source database the absent records that were inserted during the previous running of Master Data Management. *IMPORTANT: The origin of the record taken into account by the MDM does not correspond to the 'origin' field available in the metadata list. While the latter may change even on records entered via MDM as a result of manual changes, the origin of the records in this field will always be preserved. So Tagetik always knows if a record has been entered through previous synchronisations run by MDM.* • **Do not delete**: syncs the existing records or new insertions without deleting or highlighting anything else.
Import SQL filter **Export SQL filter**	Only for **From database to database** and **From file to database** routines . Filters the metadata tables to be synchronised. If applied to a node, it acts on all tables linked to that node.
Fields exceptions	Selects the fields of the table not to be synchronised. • From database to file routine: the exclusion of columns is applied to the source environment and the excluded fields are included in the produced file. (Information is stored offline within the file's metadata) • From file to database routine: the exclusion only applies to rows added, for which the destination values will be kept. For rows being inserted, the target values of the excluded fields are left empty or set to the default values in the database. • From database to database routine: the exclusion applies to the rows added and rows updated, but the target values of the excluded fields will be kept for the latter. For rows being inserted, the target values of the excluded fields are left empty or set to the default values in the database. *IMPORTANT: the exclusions are will only be applied if the tables to be synchronised were defined individually, not using a grouping.*

MDM page - Routines - Elements - Details

Repository > Master Data Management > Routines > Elements > Details

Purpose of the page

On this page, it is possible to define the content of the routine.

Buttons and/or links on the page

Element	Description
	This filters the data shown on the page.
	This adds a row in order to insert a new table.
	This duplicates the selected row.
	Deletes the element contained in the selected row.
	Enables/Disables the grid modify function. This allows you to change the characteristics of the selected elements directly in the table, and to hide the configuration tabs.
	Enables multi-row selection to allow mass changes.
	Selects the columns to be shown/hidden in the table and those to be locked when scrolling the table.
	Exports data in Excel format. This file cannot be re-imported into CCH Tagetik.
	Exports the data in Excel format so they can be edited and reimported into CCH Tagetik.
	Imports data from a previously prepared Excel file.
	Cuts a row so it can be pasted elsewhere.
	Moves a cut row elsewhere.

MDM page - Routines - Grouping

Repository > Master Data Management > Routines > Grouping

Purpose of the page

This page makes it possible to manage routine groupings.

Buttons and/or links on the page

Element	Description
Switch view	This displays the hierarchy in two different ways. • **Element/node relationships**: displays the routines in alphabetical order with the reference node. Non-related routines are not shown. • **Structure**: displays the routines in alphabetical order, specifying the parent node.
	Adds a routine to the grouping.
	Adds a node to the grouping.

Attributes tab

Field	Description
Info	🔒: routine locked; can only be edited and run by authorised users. the user is not authorised to access the Source DB specified in the routine, or the role associated with the user or the role of their reference user does not have the 0003568 - Master data management: Export DB with MDM function enabled; the user is not authorised to access the Target DB specified in the routine, or the role associated with the user or the role of their reference user does not have the 0003569 - Master data management: Import DB with MDM function enabled;
Code	Routine ID
Target DB	Database into which the selected routine will update the tables or import the setups
Sorting	Order in which the selected routines will be performed
Disable detail audit	Determines the level of detail at which the performance of the routine is audited. If selected, it ignores changes to the synced tables and only registers the performance of the routine. ***Note**: Please only select the tick-box to speed up operations for en masse synchronisations.*
Automatically include the required tables	Also syncs the tables linked to the selected table to avoid primary key violations and partial synchronisations of metadata. Synchronisation of linked tables is limited to the elements called up by the tables chosen by the user. ***Note**: please use the tick-box for all routines.*
Run in steps only	Runs the routine in steps, requiring synchronisation to be manually confirmed for each one. ***Note**: if not selected, the system shows the changes at the end of the operation without allowing the user to restore the previous situation.*
Do not run deployment of the application after termination of the routine	Prevents the MDM engine from automatically launching the necessary deployments, in relation to the recently updated definition tables. ***Note**: if the tick-box is selected, deployments must be launched manually.*

Relationships tab

This tab is only visible for routines.

Field	Description
Nodes	Specifies the name of the grouping node.

MDM procedures

Create a routine

Repository > Master Data Management > Routines > Elements

1. Click ⊕: a new row will appear in the table.
2. In the **Attributes** tab, set the parameters as desired.
3. To assign the routine to a node, in the **Relationships** tab, select the desired node.
4. Click **Save**.

Define the content of a routine

Repository > Master Data Management > Routines > Elements > Details

1. Click : a new row will appear in the table.
2. In the **Attributes** tab, set the parameters as desired.
3. Click **Save**.

Manage routine groupings

Repository > Master Data Management > Routines > Grouping

1. Click to create a new element and on for a new node.
2. In **Attributes** and **Relationships**, set the desired parameters to define the grouping logics.
3. Click **Save**.

Export data

This procedure makes it possible to export data manually into a file, via a routine defined previously.

Repository > Master Data Management > Execution > Export

1. Select the routine to be used to export the data.
2. Click **Run**.

Import data

This procedure makes it possible to import data manually, via a routine, from a previously-exported file.

Repository > Master Data Management > Execution > Import

1. Select the routine to be used to import the data.
2. To report any errors found in real time or to stop the routine if necessary, select **Manage errors**.
3. To synchronise without checking the changes again, select **Apply routines without checking**.

   ***Note**: after synchronisation, it will not be possible to restore the previous situation.*

   ***Note**: if the selected routines include at least one routine with the option **Run only in steps** selected, the **Apply routines without checking** option is disabled.*
4. Click **Run**: the changes will appear.

Schedule data imports

Repository > Master Data Management > Execution > Import

1. Select the desired routine and options.
2. Click **Schedule**.
3. For **File to DB** routines, consider the following:

If the File location is ...	Then...
Client	the process cannot be scheduled.
Application server	in **File name** indicate the name of the file of interest.
Endpoint/Tagetik agent	In **Endpoint** and **File name** indicate the endpoint and the name of the file of interest, respectively.

***Note**: **From database to database** routines can be scheduled without restriction.*

Schedule data exports

Repository > Master Data Management > Execution > Export

1. Select the desired routine.
2. Click **Schedule**.

Manage exported files

Repository > Master Data Management > Exported files management

1. Select the desired routine.
2. To work on routines created by other users, select **Show export files of all users**.
3. Download or delete the desired file.

Configuration Deployment Tool (CDT)

Introduction

The **Configuration Deployment Tool** (**CDT**) is an innovative solution designed to simplify and speed up the transport of metadata and configurations between different environments (e.g. from development to test or production environments). This tool eliminates the need to export and import the entire database, while ensuring consistency, synchronisation and data integrity across environments.

Supported features and functionality

The tool features a completely revamped and intuitive user interface that enables system administrators to work efficiently. It allows them to directly select the functional objects of interest, without requiring any specific knowledge of tables, fields, or the underlying database structure. This approach reduces operational complexity and significantly improves the user experience, making it simpler, smoother and more rewarding.

The tool currently supports the management of the**Analytical Information Hub** objects and, in future releases, will be progressively extended to also include the main functional objects of the **Financial Workspace**, offering increasingly comprehensive support for cross-environmental configuration needs.

Homepage Configuration Deployment tool (CDT)

Repository > Configuration Deployment Tool

Page purpose

The Configuration Deployment tool (CDT) homepage guides the user through the configuration and start-up of the **data transfer** in a simple and intuitive way.

What it looks like

Below are the main elements of the page:

Letter	Icon	Element	Function
To		Filter Feature	allows the selection of a subset of objects to be transferred between environments source and target. The filter button, associated with the selected object category, opens the list of items of the selected category, allowing the users to filter by code, description or other attributes.
B		Deletion Policy	Indicates the deletion policy. See Policy Update
C		Transfer run button	Starts the transfer procedure
D		Transfer processes	Opens the transfer notification section. See Notification Panel
E		Transfer Logs	Opens the Transfer logs page showing all transfers made to all accessible targets. See Transfer Logs Page
F		Export page	Opens the Export page, which allows you to export data by writing to a file.
G		Import page	Opens the Import page, which allows you to import data by reading from files.

This page includes two main cards that allow you to select the environments involved in the transfer:

- **Source Environment:** where the data to be transferred are located.
- **Target Environment:** the destination to which the data will be transferred.

After selecting the desired environments, the user can start the transfer by clicking the **Run** button. This action initiates the process and automatically opens the transfer page.

For further details on how to set up and start the transfer, see the procedures section.

Compatibility between different CPM versions

From **SP35**, the CDT supports the transfer of configurations between Tagetik environments with **different source and target** versions, as long as the difference between the two does not exceed two releases (e.g. from source SP33 to target SP35).

When performing a transfer between non-aligned versions:

- the tool operates in backward-compatible mode, using the older version between the source and the target for metadata and configurations
- if the source is on a newer version than the target, some resources may not be transferred, because they are not yet supported by the target.

***IMPORTANT**: if the difference between source and target exceeds two releases, the transfer can still be started, but may fail. In this case, the system signals the condition with a warning message and the operation may generate errors.*

Deletion Policy

Introduction

This feature manages the update between the Source and Target environments via three main methods:

- **None**
- **Full**
- **Partial**

***IMPORTANT**: Each method defines a specific behaviour for updating and handling objects in the two environments.*

***Note**: This feature is accessible via the Deletion Policy button, which allows you to select the preferred update method.*

Update methods

Below is a description of the available update methods.

Method	Description
None	It updates the Target only with insertions and updates, without deleting any data in the latter. ◦ Objects that exist on both Source and Target are updated on Target; ◦ Objects that exist in the Source but are missing in the Target are added to the Target; ◦ Objects that exist on Target but are missing in Source are kept on Target (not deleted).

Full	It performs insertions, updates and deletions based on the difference between Source and Target. ◦ Objects that exist on both Source and Target are updated on Target. ◦ Objects that exist in the Source but are missing in the Target are added to the Target. ◦ Objects that exist on Target but are missing in Source are deleted from Target.
Partial	Deletions are limited to objects originating from the use of CDT/MDM ◦ Objects that exist on both Source and Target are updated on Target. ◦ Objects that exist in the Source but are missing in the Target are added to the Target. ◦ Objects that exist on Target but are missing in Source are only deleted if they originate from the use of CDT/MDM.

***IMPORTANT**: Any changes at the node level will automatically affect settings at a lower level. The settings on the lower elements of the hierarchy can later be changed individually by selecting the specific function.*

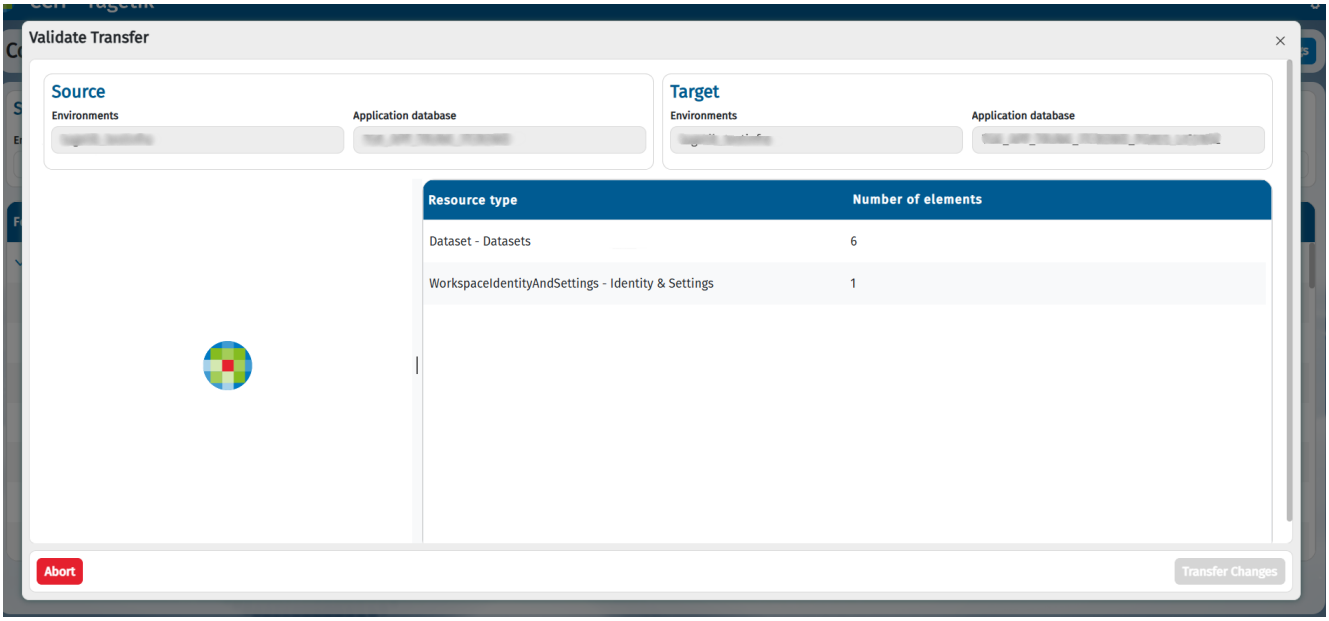
Transfer page

Repository > Configuration Deployment Tool > Run

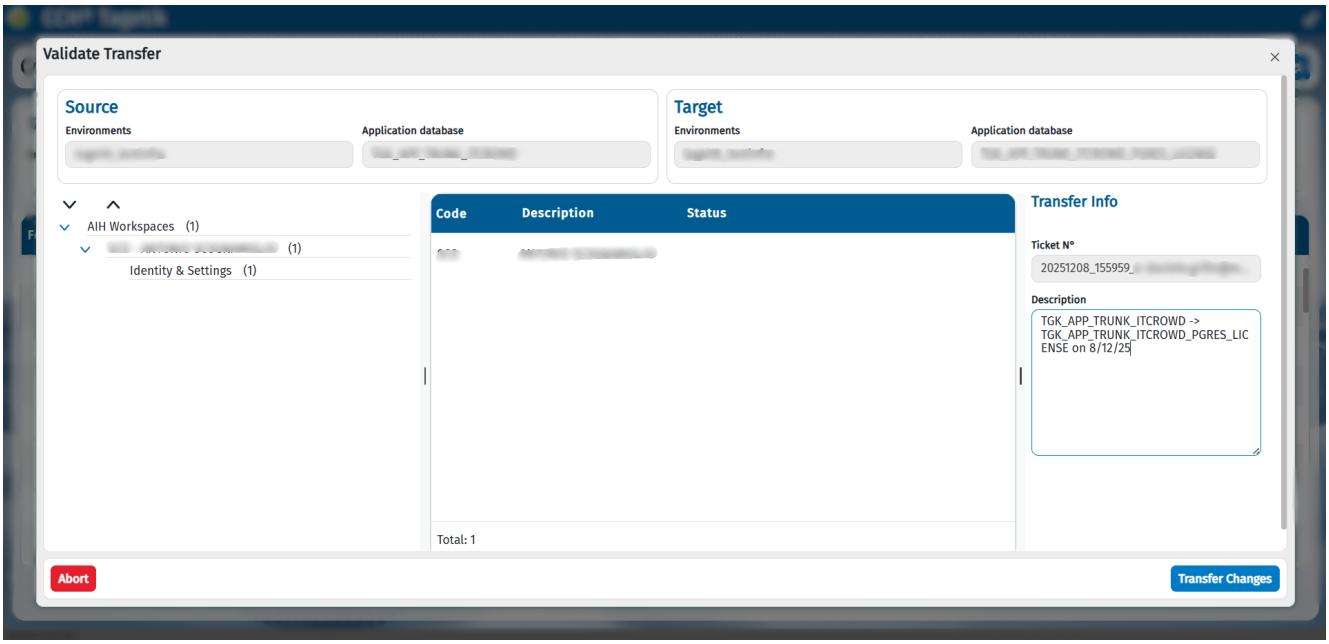
Purpose of the page

This detail window, available during the execution of the transfer, allows you to monitor the status and progress of the process.

What it looks like



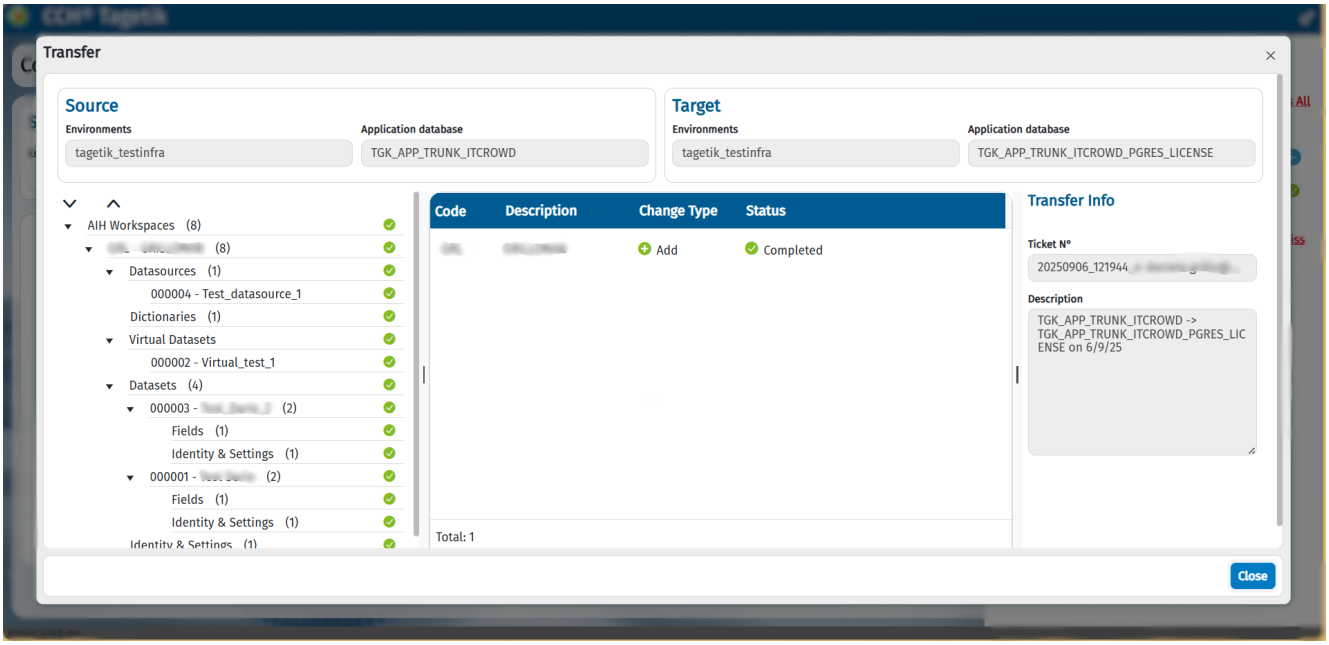
At the end of pre-processing, the window changes as shown:



***IMPORTANT**: Within the window, it is possible to request an abort via the Abort button.*

For further details, see *Abort*.
Once validation is complete, the user can run a transfer operation and start the data processing.

When the transfer is complete, the window will update automatically.

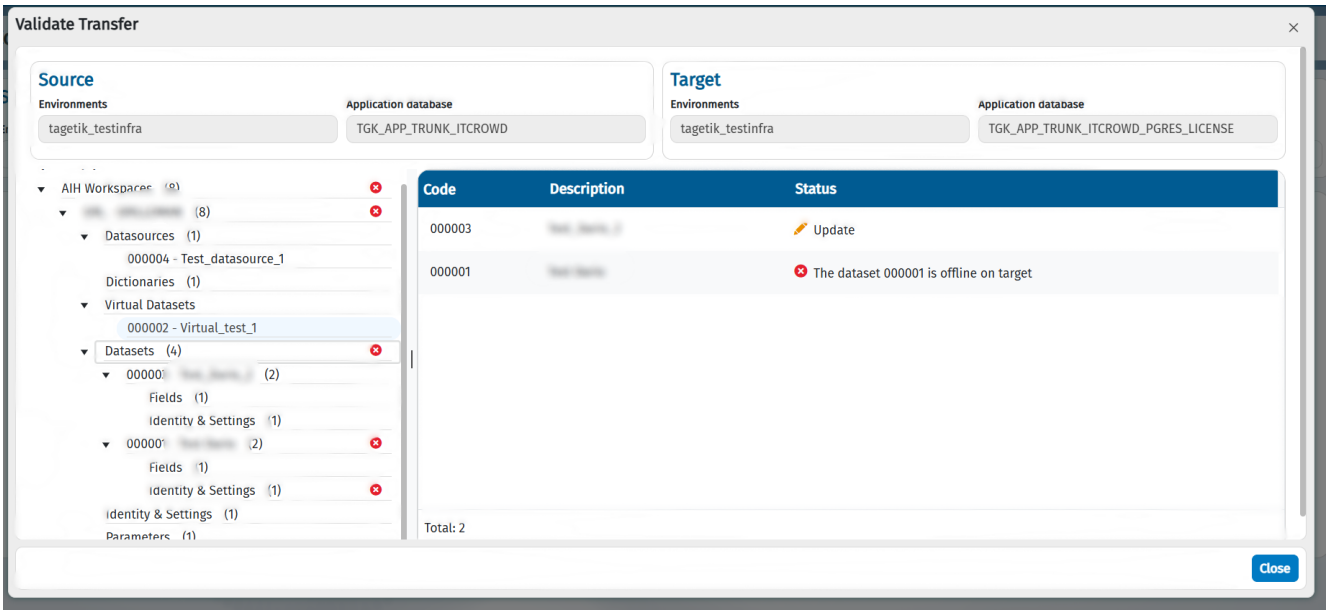


The system shows three main sections:

- **Left section - Tree of elements:**
  Displays the list of objects involved in the transfer, organised in a tree structure. The list updates in real time during data processing, dynamically displaying the elements that are progressively handled.
- **Central Section - Detail Table:**
  It shows a table with information on each item processed, including:
  - Code
  - Description

- Type of change made to the target database
- Data Processing Status

This section is useful for detecting errors and understanding their causes, as in the example below:



- **Right Section - Identifying Information**
  It includes two areas:
  - **Ticket Number:** is a unique code identifying the transfer in progress.
  - **Description:** is a customisable field where the user can enter a short description of the transfer before starting it. The field is automatically pre-filled with a value in a predefined format: ambienteSource - ambienteTarget on data.

CDT: Import Page

Repository > Configuration Deployment Tool > Import

Page purpose

The Import page of the Configuration Deployment tool (CDT) guides the user through the process of **importing data from files**. From the **Target** section, it is possible to select the target database for the data from files.

What it looks like

Letter	Element	Function
To	Tab switch button	allows you to change the CDT usage mode
B	Select file from PC	Allows you to select a file from the internal memory of type .cdt
C	Remove uploaded file button	Removes the file to be imported
D	Button for importing the data contained in the file	Starts the import procedure

Fields/Actions on the page

Target Section	
Field	**Description**
Environments	Pre-filled field showing the application environment of the purchased CPM version.
Application database	Drop-down field for selecting the import target database.
Other fields/actions	**Description**
Select a File/drag & drop	Mode for selecting the file from which the CDT will read the data to be imported: • **Select a File**: opens a window to select a .cdt file • ***Drag & drop* action: *allows*** you to upload the file by dragging it into the area
Import	Starts the process of importing data from the uploaded file and automatically opens the transfer page.

CDT: Export Page

Repository > Configuration Deployment Tool > Export

Page purpose

The **Export** page of the Configuration Deployment tool (CDT) guides the user through the process of **exporting data to a file**. It features the same buttons as the home page and includes the Source section, which is used to select the source environment from which to export data.

What it looks like

Letter	Element	Function
To	Tab switch button	allows you to change the CDT usage mode
B	Run export button	Starts the data export procedure in a .cdt file

Fields/buttons on the page

Source Section	
Field	**Description**
Environments	Pre-filled field showing the application environment of the purchased CPM version.
Application database	Drop-down field for selecting the source database where the data to be transferred is located.
Export	It starts the export process by automatically opening the export page. Once the operation is complete, the **Download** button allows you to download the file containing the exported selected resources.

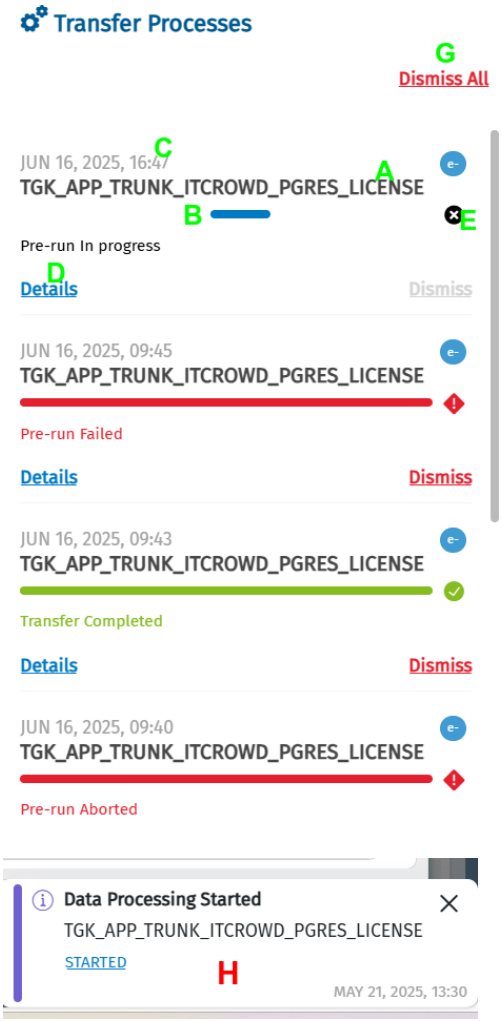
Notification Panel

Repository > Configuration Deployment Tool > Transfer Process

Purpose of the page

This section allows you to view the real-time updated status of transfers run by you.

What it looks like :



Components of the Notification Panel tab

Below is a description of the main components:

Letter	Element	Function
A	Target Database	Shows the database to which the transfer is being made
B	Status bar	Indicates the status of the transfer in real time
C	Date of transfer	Shows date and time of the last notification update
D	'Details' button	Opens the section with full details of the transfer See Transfer Page
E	'X' icon (Abort)	Starts the procedure to abort the transfer See MDM20 Abort
F	'Dismiss' button	Removes the notification of the selected transfer
G	'Dismiss all' button	Removes all notifications of completed transfers
H	Information pop-ups	Pop-up messages indicating changes in data processing status

For further information on the operation of the notification panel, see Notification Management.

Transfer procedures

Repository > Configuration Deployment Tool

Select Source and Target environments

1. Select **Environment** and Source **Application Database**
2. Select **Environment** and Target **Application Database**

Selecting Functional Objects and Deletion Policy

1. Select categories and functional objects from the source environment
2. Set the deletion policy via the appropriate command(cancellation policy)
3. Set up filtering to determine a subset of objects to be transferred between environments
   Click on the filter button associated with the selected object category. A window will open listing all the items in the category. From there, you can apply filters to the available columns (such as code, description, or other attributes) to select only the desired objects.

   Note: Filtering allows only a subset of objects to be transferred from the source environment to the target environment

Launching the objects pre-calculation

1. select the start calculation command
2. wait for the calculation of the selected functional objects

Note: You can interrupt the ongoing calculation using the Abort option.

Checking the pre-calculation result

1. Check the result of the pre-calculation through the detail in the table
   a. Added: the system will add the object to the Target environment
   b. Updated: the system will update the object on the Target environment
   c. Deleted: the system will delete the object on the Target environment

Running the data processing

1. run the transfer data processing. Once started, the tool will open a window that displays the progress of changes on the Target environment in real time.
2. wait until the transfer of objects is complete


Note: You can interrupt the data processing using the Abort option.

Checking the data processing result

1. check the data processing on the Target environment through the detail in the table
   1. Added: The system has added the object to the Target environment
   2. Edited: the system has updated the object on the Target environment
   3. Deleted: the system has removed the object from the Target environment.

Transfer Notification Management

The notification side panel allows the user to monitor the progress of transfers in real time, through a compact and intuitive view.

Each transfer is displayed through a dedicated tab, which provides essential information on the status of the process. The tab includes quick commands and real-time updates, allowing the user to intervene promptly if needed.

For a summary of the notification components, refer to the page about the Notification Panel Components

Main features

- Real-time monitoring of the status of transfers.
- Ability to **access details**, **stop** or **hide** notifications of individual transfers.
- Option to remove all completed notifications, useful for keeping the panel tidy.
- Dynamic status updates (via progress bar and pop-up information messages).

Possible transfer statuses

During the procedure, each transfer can go through the following statuses:

- Data Processing Started
- Data processing in progress
- Data processing completed with success
- Data processing warning
- Data Processing error
- Aborting data processing
- Data Processing was stopped

These statuses are signalled both within the notification itself and by pop-ups showing the change.

Abort feature

This feature allows the user to manually interrupt a pre-calculation or processing operation in progress.

***IMPORTANT**: Objects processed before the interruption will remain saved in the Target environment.*

Objects that have not yet been processed will not be transferred to the Target environment.

The Transfer Logs functionality

Basic concepts

The **Transfer Logs** functionality allows the user to view the list of transfers completed or in progress for the tenants and databases to which the user has access. The aim is to provide a clear and up-to-date overview of transfer processes, enabling the following actions:

- status check
- if necessary, execution of interruption, single and multi-row deletion , rollbackoperations.

***IMPORTANT**: Only delete or rollback operations can be performed in the Transfer Logs view.*

Available operations

- Single and multi-row **deletion (Delete)**
- **Rollback**

***IMPORTANT**:Deletion is only available for transfers that are not in progress and not subject to activerollback processing .*

For further information on rollbackoperations see Rollback

Transfer Logs page

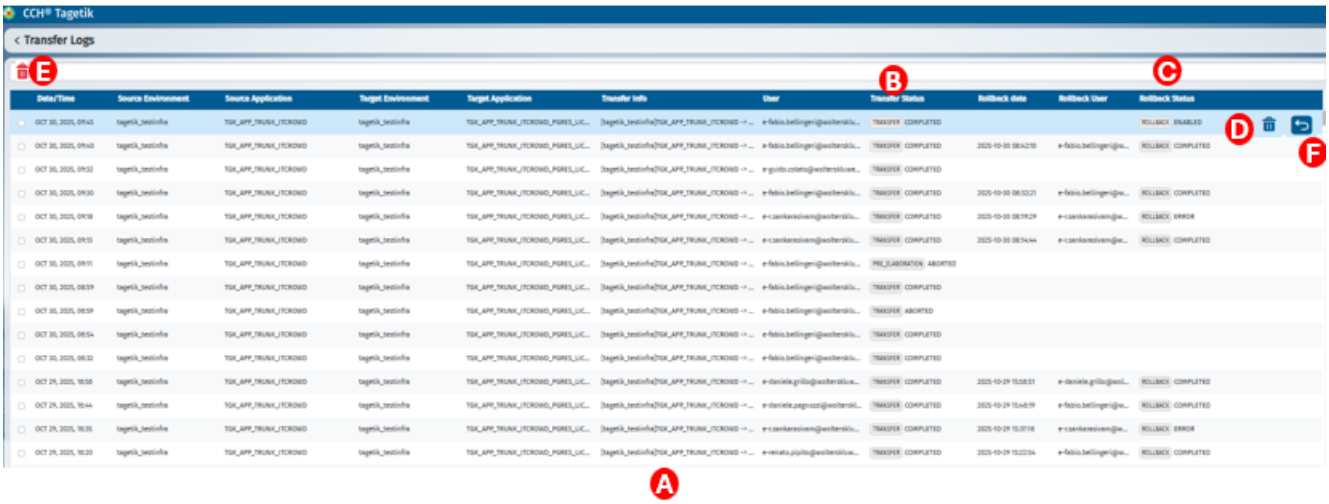
Repository > Configuration Deployment Tool > Transfer Logs

Purpose of the page

This page shows a summary grid of the transfers, in chronological order. For each transfer, the following data is shown:

- Date/time of transfer
- Source Environment
- Source Application
- Target Environment
- Target Application
- Transfer Info
- User who performed the transfer
- Transfer Status
- Date of any rollback
- User who run the rollback
- Rollback status
- Available Actions

What it looks like



Elements on the page

Below are the functions of the page elements:

LETTER	ELEMENT	FUNCTION
A	**Transfer Logs Grid**	Shows the list of transfers performed or in progress. Each row represents a transfer.
B	**'Transfer Status' column**	It displays the current status of each transfer and provides the option to view its details (see Transfer Page
C	**'Rollback Status' Column**	Shows the outcome of any rollbacks associated with the transfer and allows you to open the relevant details(see Transfer Page)
D	Single Delete button	Enables deletion of the selected record. It is not possible to delete processes that are currently running, or the last transfer performed for the same target.

E	**Rollback** (Play) button	**Rollback** is enabled only for the most recent transfers performed with the same target and with **no deletion policy set.**
F	**Filter** button	Applies filters for a more precise search within the table
G	**Refresh** button	Updates the table
H	**Multiple Deletion** Button (DELETE ALL)	Allows the deletion of selected records. It is not possible to delete processes that are currently running, or the last transfer performed for the same target.
I	**Multiple selection** button	Enables the selection of multiple rows, required to activate the **H** button

Rollback

A rollback is an operation that allows you to undo the last transfer performed on a given target environment, restoring the metadataand configuration to the state prior to the transfer.

The rollback option is available, on specific target , only for the last transfer in chronological order with deletionpolicy = "none"(see Deletion policy)and only if no other processes are currently active

Data processing logic

When the rollback data processing starts , the system displays a view showing the detail of the changes to be restored.

The system automatically pre-fills the Ticket Numberand Transfer Info fields with their original values, prefixed with 'RB_', in order to make the rollback operation immediately identifiable.

During processing, the system displays the rollback progress and, once completed, updates the Transfer Logs view with the corresponding completion entry in rollbackstatus.

In the event of an error, the operation is marked as ERROR, while still keeping a record of the event in the log.

Note: It is not possible to relaunch a rollbackthat has failed or has been aborted. However, you can relaunch a rollback that failed during initialization phase (pre-initialization operations).

Rollback Procedures

Homepage > Transferlogs

Procedure

Run Rollback

1. Identify the transfer for which the procedure will be carried out
2. Check that the status of the rollbackis ENABLED
3. Click the **rollback** button
4. Verify the data displayed in the modal
5. Use the Rollback Changesbutton **to confirm**
6. Click **confirmation** on the pop-up

Notes: you can interrupt the running rollback (see Interruption)

Data processing Auditing

To monitor the activities performed by the **Configuration Deployment Tool** on the target environment, you can access the relevant Audit page (Metadata Log Report).
This section records the complete history of the data processing activities, allowing you to trace each individual execution of the transfer process and its details (date, user, description, etc.).
By selecting the row corresponding to the desired data processing, you can access a detailed report of the transfer process, including the outcome and any relevant events.

Different types of detailed view are available: **DETAILS , DETAILS OF ALL PARTS and DETAILS AGGREGATED**. All views follow a tree structure, where the main node is the **Configuration Deployment Tool** branch. This branch includes a sub-branch called **Part 1,** which contains the ticket number associated with the data processing. See section Transfer Page.

It will also be possible to have an overall detail of all metadata modified by an individual data processing, by selecting it in the audit window and clicking on **'Metadata Log Report'** from the context menu.

Audit

Contents

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Audit Concepts

What is it for

Audit facilitates checking and controls. It allows the tracking of all operations on CCH Tagetik, based on logins, and records who did what, when, where and for what reason.

Traceability

Operations can be traced by:

- entering the following information for the data and metadata of each table:
  - **Origin**: part of the application that generated the data (see "Data origin" below)
  - **Userupd**: user who performed the operation
  - **Dateupd**: date on which the user performed the operation
- control engine for data and relative changes
- Query Analyzer function

Data origin

Background

Data can be generated on original scenarios or on consolidation scenarios Sometimes, the origins on the consolidation scenarios result from transformations of the origins on the original scenarios.

Origin from original scenarios

Origin code	Description
COPY_PERIOD	Original - Carry forward from previous period
INPUT_WEB	Manual input - Web windows
INPUT_DEFORM	Manual input – reports
MAP_	ETL
QDL	Quick Data Loader
SUB_INPUT_WEB	Submitted manual input - Web windows
SUB_INPUT_DEFORM	Submitted manual input - Reports
SUB_MAP	Submitted ETL
ENV_SYNC	Update Standard tables
COPY_SEGMENT	Copy Segment
COPY_CONSOLIDATED	Copy consolidated amount on original amount

Origin code	Description
COPY_GENERAL	Copy amounts and journals
COPY_ORIG	Source origin
PROC_IC_SPREADING	IC spreading reallocation
PROC_IC_SPREADING_CB	IC spreading write off
PROC_IC_BALANCING	IC - automatic balancing
PROC_IC_SPREADING_WOFF	IC - spreading - write off
DEPLOYMENT	Deployments
SUB	SUB
PREL_RESTORE	Original – Carry forward
PREL_RESTORE_RED	Originals - Carry-forward - Rules Engine Designer
PREL_COPY_PREVIOUS	Original - Copy from the previous period
PREL_DECONSOLIDATION1	Original - Deconsolidation 1
PREL_DECONSOLIDATION2	Original - Deconsolidation 2
PROC_DECONSOLIDATION3	Original - Deconsolidation 3
PREL_MERGE1	Original - mergers
PREL_MERGE2	Original - mergers 2
PROC_MERGE3	Original - mergers 3
PREL_REVERSE1	Original - Reverse journal 1
PREL_REVERSE2	Original - Reverse journal 2
PREL_REVERSE3	Original - Reverse journal 3
PROC_FININV_DEPLOY	Original – Loading Financial investments
PROC_OSR_FI_LOADING	Original - Import financial investment data into ownership structure register
PROC_COUNTERPARTY	Original - Create automatic Ctps
PROC_DEPRECIATION	Original – Depreciation
PROC_DEFERRED_TAXES	Original - Deferred tax
PROC_ DEFERRED_ TAXES_ CALC_	Original - Deferred taxes - Calculation logics
BASIC	Basic
PROC_CTP_ELIMINATIONS	Original - Eliminations for Ctp
PROC_FI_ELIMINATIONS	Original - Eliminate Financial investments from Ownership structure register
PROC_IC_PROFIT	Original - Intragroup assets assignment
PROC_RED	Original - Rules Engine Designer
PROC_BALANCING	Original – Balancing
PROC_CALCULATED_BASIC	Original - Basic calculation logics
SUB_PROC_CALCULATED_BASIC	Origins - Submitted basic calculation logics
PROC_CALCULATED	Original - Calculation logics
SUB_PROC_CALCULATED	Origins - Submitted calculation logics
PROC_CALCULATED_MD	Original - MD calculation logics
PROC_FININV_RECL	Original – Financial investments reclassification
PROC_FININV_RECL_CALC	Original - Financial investments reclassification - Calculation logics
PROC_FININV_RECL_CALC_BASIC	Original - Financial investments reclassification - Basic calculation logics
PROC_NETEQUITY_RECL	Original – NER
PROC_CLOSING	Original – Closing
PROC_MONTHLY_SPLIT	Original - Monthly splitting
PROC_MONTHLY_SPLIT_NEW	Original - Monthly splitting
PROC_ MONTHLY_ SPLIT_ NEW_ CALC_ BASIC	Original - Monthly splitting - Basic calculation logics
PROC_ALG_OPERATIONS	Original - Algebraic operations
PROC_FORECAST	Original – Forecast
PROC_BDG_FN	Financial product
PROC_BDG_DEPRECIATION	Depreciation
PROC_RECLASSIFICATION	Preconsolidation reclassification
PROC_EPU	Original - Pick-up
PROC_EPU_CALC_BASIC	Original - Pick-up - Basic calculation logics
PROC_LOOK_THROUGH	Original - Look Through
EPU_DEPLOY	Pick-up loading

Origin code	Description
CFP_BUDGET	CFP - Copy from budget
CFP_LT_FINANCING	CFP - L/T financing
CFP_FIXED_ASSET	CFP – Assets
CFP_BOND	CFP – Titles
CFP_EXCHRATE_COVERAGE	CFP - FX hedging
CFP_PROJECT	CFP - Work orders
CFP_CASH_IN_OUT	CFP - Collection and Payments
CFP_PAYMENT_CALENDAR	CFP - Payment calendar
CFP_SPLITTING	CFP – Weights
CFP_PAYMENT_CALENDAR_WOFF	CFP - Payment calendar - Write off
CFP_SPLITTING_WOFF	CFP - Weights - Write off
CFP_VAT_DOWN_PAYMENT	CFP - VAT – Down payment
CFP_VAT_ACCOUNTING	CFP - VAT – Accounting
CFP_CORRISPETTIVI_WOFF	CFP - Consideration - Write off
CFP_VAT_PAYMENT	CFP - VAT – Payment
CFP_ VAT_ TRANS_ ACCOUNT_ WOFF	CFP - VAT - Transitory write off account
CFP_TAXES_PAYMENT	CFP - Taxes – Payment
CFP_TAXES_DOWN_PAYMENT	CFP - Taxes - Down payment
CFP_ TAXES_ DOWN_ PAYMENT_ WOFF	CFP - Taxes - Down payment write off account
CFP_FINAL_EXCHRATE_EVAL	CFP - Evaluation at final FX rate
CFP_TAXES_PL	CFP – Taxes
CFP_TAXES_ACC	CFP – Taxes
CFP_ TAXES_ DOWN_ PAYMENT_ DEBT	CFP - Taxes - Debit Down payment
CFP_TAXES_DEFERRED	CFP - Deferred taxes
CFP_TAXES_DEFERRED_WOFF	CFP - Deferred taxes - Write off down payment
CFP_TAX_CREDIT	CFP - Tax credit
CFP_CQNC	CFP – CNQC
CFP_FINANCIAL_RULE	CFP - Financial policy
CFP_INTEREST_PL	CFP - P&L interests
CFP_INTEREST_ACC	CFP - Interests – Accounting
CFP_ INTEREST_ DEFERRED_ PAYMENT	CFP - Interests - Collection / Deferred payment*
CFP_GROUP_CASH_POOLING	CFP - Cash Pooling
CFP_GROUP_VAT	CFP - Group VAT
CFP_DIVIDENDS	CFP - Dividends
CFP	Cash Flow Planning
RULE_DEPLOYMENT	CFP - Rules distribution
EVENT_RULE	CFP - Create Events by a rule
EVENT_RULE_MIXED	CFP - Create Events by hybrid rule
DIAGNOSTIC	Financial investments congruence
USER_LIM_WIZARD	Basic module: User rights - Simplified management
PROC_TAXES	Taxes
PC_COSTS	Production Control: Costs
PC_VALUES	Production Control: Values
PC_MOVEMENTS	Production Control: Stock transactions
BREAK_BACK	Form - Break Back
SUB_BREAK_BACK	Form - Submitted Break back
PROC_DIVIDENDS	Dividends
ELAB_DEADLINES	Generate Submission deadline dates

Origin from consolidation scenarios

Origin code	Description
INPUT	Input Derived from INPUT_WEB, INPUT_DEFORM, SUB_INPUT_WEB, SUB_INPUT_DEFORM, QDL on original scenarios
MAP	ETL Derived from SUB_ MAP or one of the possible origins starting with MAP_ on the original scenarios
PRCO_RESTORE	Consolidation – Carry forward
PRCO_RESTORE_MIN	Consolidation - Carry forward - Minorities
PRCO_RESTORE_RED	Consolidation - Carry forward - Rules Engine Designer - Consolidable
PRCO_RESTORE_RED_2_CONV	Consolidation - Carry forward - Rules Engine Designer - Converted
PRCO_RESTORE_RED_1_DECO	Consolidation - Carry forward - Rules Engine Designer - Proportional
PRCO_PERC_CHANGE	Consolidation - percentage change
CONS_BALANCING	Consolidation – Balancing
CONS_ GENEXCHRATE_DEPLOY	Consolidation - FX rate loading
CONS_CALCULATED	Consolidation - Calculation rule
CONS_ CALCULATED_MD	Consolidation - MD calculation logics
CONS_ CONVERSION_CONV	Consolidation – Conversion
CONS_ CONVERSION_DIFF	Consolidation - FX rate differences
CONS_ CONVERSION_DIFF_PROP_CHANGE	Consolidation - FX rate differences due to change in consolidation percentage
CONS_ CONVERSION_DIFF_RESTORE	Consolidation - FX rate differences calculated on restored values
CONS_ CONVERSION_HIST	Consolidation - Conversion at historical FX rate
CONS_ EQUITYEVAL_CALC	Consolidation - Evaluation with equity method
CONS_ EQUITYEVAL_CHANGE	Consolidation - Change from/to equity method
CONS_ NETEQUITY_RECL	Consolidation - Net equity accounts reclassification (NER)
CONS_MINORITY	Consolidation – Minorities
CONS_ MINORITY_COPY_PERIOD	Consolidation - Minorities interest - Copy previous period
CONS_ DECONSOLIDATION	Consolidation – Deconsolidation
CONS_ DECONSOLIDATION_2	Consolidation - Deconsolidation on converted amount
CONS_ DECONSOLIDATION_2_MIN	Consolidation - Deconsolidation on converted amount - Third parties
CONS_ DECONSOLIDATION_RED_2_CONV	Consolidation - Deconsolidation on converted amount - Rules Engine Designer
CONS_COPY_PREVIOUS	Consolidation - Copy from previous period
CONS_COPY_ PREVIOUS_MIN	Consolidation - Deconsolidation on converted amount - Copy from previous period - Minorities
CONS_COPY_ PREVIOUS_RED_2_ CONV	Consolidation - Deconsolidation on converted amount - Copy from previous period - Rules Engine Designer
CONS_ EXCHRATE_ DEPLOY	FX rates loading on Consolidation scenarios

Origin code	Description
CONS_ EXCHRATE_ DEPLOY	Consolidation - FX rate loading
CFP	Cash Flow Planning Derived from one of the possible origins starting with CFP_ on original scenarios
CONS_FORCE_1	Override Proportional amount type
CONS_FORCE_1_FINAL	Consolidation - Override Proportional amount type - Final
CONS_FORCE_1_PROP	Consolidation - Override Proportional amount type - Proportional calculation
CONS_FORCE_2	Override Converted amount type
CONS_FORCE_2_CONV	Consolidation - Override Converted amount type - Conversion
CONS_FORCE_2_FINAL	Consolidation - Override Converted amount type - Final
CONS_FORCE_2_MIN	Consolidation - Override Converted amount type - Minorities
CONS_FORCE_3	Override Consolidable amount type
CONS_ FORCE_ 3_ EQUITY	Consolidation - Override Consolidable amount type - Equity evaluation
CONS_FORCE_3_FINAL	Consolidation - Override Consolidable amount type - Final
CONS_PROP_CHANGE	Consolidation - Proportional calculation - Percentage change
CONS_PROP_ENTRY	Consolidation - Proportional calculation - Entities included in the fiscal year - Variations account change for the period before the first consolidation
CONS_ PROP_ NETEQUITY_VAR_RECL	Variations account change for Net equity reclassification
CONS_ LOOK_ THROUGH	Consolidation - Look Through
CONS_ EQUIVALENT_ SECTORAL_REVERSE	Consolidation - Look Through - Write off: Equivalent and sectoral
CONS_ EQUIVALENT_ SECTORAL_MINORITY	Consolidation - Look Through - Minorities: Equivalent and sectoral
CONS_ SECTORAL_ RECLASSIFICATION	Consolidation - Look Through - Account reclassification: sectoral
CONS_ RECLASSIFICATION	Consolidation - Consolidation reclassification
CONS_ IFRS5_ RECLASSIFICATION	Consolidation - IFRS5 Consolidation reclassification
CONS_RED	Consolidation - Rules Engine Designer - Consolidable
CONS_RED_2_CONV	Consolidation - Rules Engine Designer - Converted
CONS_RED_1_DECO	Consolidation - Rules Engine Designer - Proportional

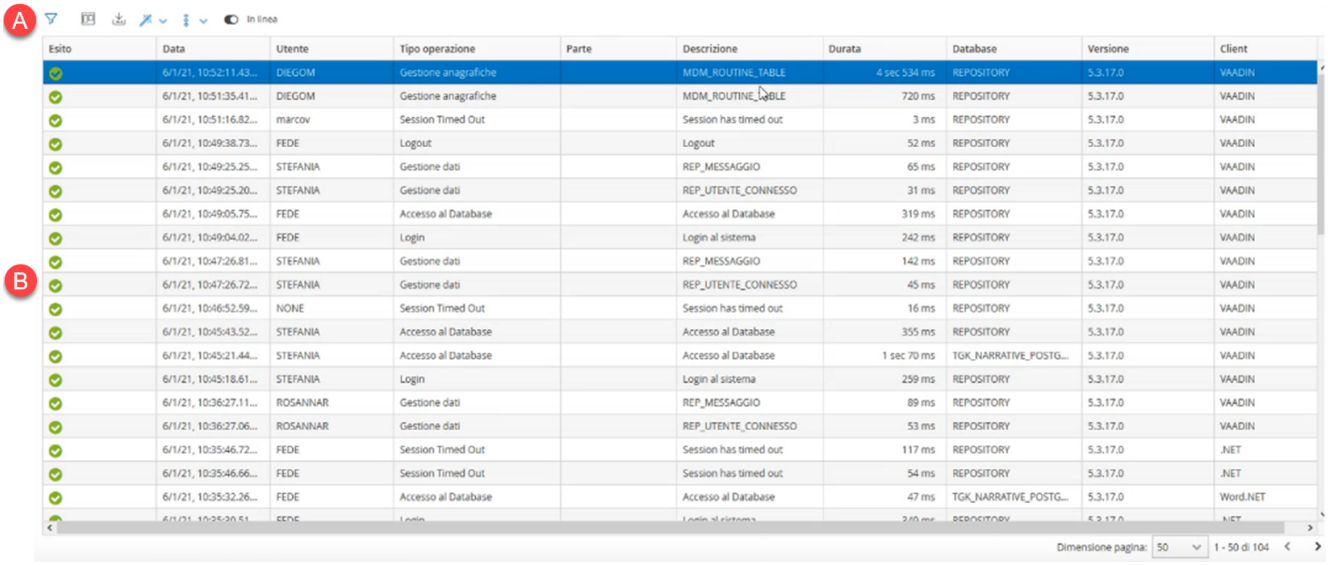
Audit page

Repository / Application database > Audit > Audit

Purpose of the page

This page allows access to the log report, see changes and identifyany errors.

What it looks like



Element	Description
To	Tool bar
B	Log report

Buttons and/or links on the page

Element	Description
	(only in the Repository) Allows you to: • store offline the operations present in the auditing trace • delete the operations present in the auditing trace • delete the operations present in the log of the auditing trace The user must specify : - the **criterion** for the definition of the date the system will use to filter the data to be deleted or archived. The possible values are: ◦ **Relative**. the reference date is calculated by the system starting from today and then going back the number of days specified by the user (**# Days up to today**); ◦ **Absolute**. The reference date is specified by the user (**Specific date**). In both cases, the system works on the information present in the audit log with the date before or equal to the reference date. - a **filter on the operation type** so as to partialize the data present in the audit log to be deleted based on the operation type. . These data processes can also be scheduled.  Offline data can be consulted via the functionality accessible from the web interface of the administrator user by following this path: *Repository > Audit > Audit* , then click on  Online
	• **Details**: opens the selected operation's details page. • **Details of all parts**: for operations split into several parts, it groups the activities into the relative parts and shows the details. • **Merged details**: for operations split into several parts, this shows all the activities inherent to an operation. • Metadata log report: opens the "Metadata log report" page relating to the selected data processing (currently only available for the"DB update: Element lists update" data processing). The report page will display a grid with details of deleted, inserted and updated element lists as a result of the data processing launch • (only if the licence includes Machine Learning module) **Download ML log details**: by selecting an operation with one of the following descriptions: *Train ML* processing, *Execute ML data processing*, *Execute a JOB*, *Financial Reporting: Run data processing by form*, *Data processing started*, you can download the .zip file containing the log files relating to the each Train or Execute ML data processing performed. In the event that no log file is available for the selected data processing or the service cannot be accessed at that time, an error message will be displayed.
	• **Online**: updates the data in real time • **Archived**: shows the archived data

Information on Logs

Element	Description
Result	Operation outcome: • : completed successfully • : completed successfully, but with warnings • : failed
Date	Date and time activity started
Operation type	Operation logged by the system.
Part	Part of the data process (if split).
Data processing / Managed data	Only for management operations. Indicates the table of the database involved.

Element	Description
JDBC statistics	Information on SQL queries run by the operation. • **n**: total queries run • **t**: total run time • **M**: longest query run time • **P**: database time as a percentage of total time
Mode	Operation start mode
Concurrency level	Maximum number of simultaneous data processes
OID	Unique ID of the operation
Correlation OID	Unique ID of a group of operations. Useful for data processes made up of several parts.

Audit Details page

Repository / Application database > Audit > Audit > ⋮

Purpose of the page

This page shows details on the selected operation, or part of it.

Details tab

Element	Description
Log	Details of activities related to the selected operation
Free parameters	Incoming parameters received from the operation, not linked to a dimension
Parameters	Incoming parameters received from the operation, linked to a dimension
Data	Edits to the data managed by the operations performed
Data - Mass updates	Mass updates to the managed data

Log tab: action buttons

Element	Description
Next issue	Shows the next issue.
	Shows/hides the duration in milliseconds
	Creates a .PDF file containing the following: • the operation execution tree • a list of the parameters and free parameters used by the operation • if present, details of each action related to the operation
	Exports the data in a zip folder with several files in Excel format. The files are as follows: • operation execution tree • list of the parameters and free parameters used by the operation • if present, details of each action related to the operation • in case of errors in the operation, a file for every stack trace These files cannot be reimported into CCH Tagetik.

Log tab: Page fields

Field	Description
Exceptions	Shows any serious issues found (red).
Statistics	Shows the statistics for the SQL query run by the selected activity.
SQL information	Shows all the SQL queries run by the selected activity if the related audit parameter of SQL queries has been selected.
Selected rows	If present, shows additional details relating to data worked on by the tracked activity..

Database Logins Report page

Repository > Audit > Report > Database Logins Report

Purpose of the page

This page gives access to the database login reports.

Buttons and/or links on the page

Element	Description
	**Historical**: shows the login history for the selected user.

Metadata log report page

Repository / Application database > Audit > Report > Metadata log report

Purpose of the page

This page allows the filtering of changes to tables from a given date.

Page fields

Field	Description
Database	Database in which to search for changes
Include offline tables	Includes data already archived.
Enables filter on the first date of the search (to reduce waiting time)	Limits the search to the specified start date.
Managed data	Edited database table
Data key	Code (physical key) of the edited data in the relevant database table
Field	Name of the edited field

Audit management procedures

Order log rows

Click a column header to order the rows in descending/ascending order.

Enable operation logging

Repository > General Settings > Audit

1. Select information to be tracked.
2. Click **Save**.

Filter changes

Repository / Application database > Audit > Report > Metadata log report

1. If present in the Repository, select the database in which to search for changes .
2. Set the desired period.
3. Select the desired options to limit search results.
4. Select **Run**: the search result will appear in the table.
5. For additional or different filtering of results, select , set the desired filters and select **Apply**.

Use the Query Analyzer function

This function can be used on the linked database, one query at a time.

<application database>

Click **Other links** and then **Query Analyzer**.

Endpoint

Contents

This section covers the following topics:

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Labels for Azure Information Protection 91
Configuring the Azure Information Protection label application 91
Configuring the endpoint for the Azure Information Protection labels 91
Configure an endpoint. 91

Endpoint page

Repository > Endpoint

Purpose of the page

This page allows you to configure the access to resources outside the application by defining an Endpoint. These Endpoints can be used when reading/writing from the Job ETL and other modules of Tagetik Software S.r.l..

Attributes

Page fields

Please find below the most important fields.

Field	Description
Code	It specifies the unique identifier by which the Endpoint will be referenced by the modules ofTagetik Software S.r.l. that use it.
Description	Shows the description in full.
Endpoint type	Indicates the mode of access to the resource, selectable from URL, Tagetik Data Loader, Tagetik Application, OData V2 service, OData V4 service, REST API, SharePoint, Google BigQuery, Snowflake, MS Fabric.

URL

By selecting the **URL** type, a resource can be referenced via certain protocols depending on the syntax entered in the **URL** field.

Default Endpoint

It instructs the system that files produced by CCH Tagetik tools (such as Application Import/Export) that require prolonged storage, must be saved to the remote URL reported by the Endpoint. If this option is not enabled for any Endpoint, files produced by CCH Tagetik tools will be saved to the directory set as *writable folder*.l This option can be enabled on one Endpoint only.

Managed protocols

Below is a list of managed protocols, with relevant examples.

Protocol	Windows URL examples	Unix URL examples
File	• file://localhost/c|/WINDOWS/ • file:///c|/WINDOWS/ • file://localhost/c:/WINDOWS/	• file://localhost/etc/ • file:///etc/
Ftp	ftp://host[:port]/url-path	
Ftp (secure)	ftps://host[:port]/url-path	
Http	http://webdav/files	
Https	https://webdav/files	
Sftp	sftp://host[:port]/url-path	
Smb	smb://host/shared-folder-path	

SFTP protocol

The '**Key Exchange**' (the process of negotiating the secret key used to encrypt data during the session) used by CCH Tagetik is the'*diffie-hellman-group-exchange-sha256*'.

The '**host key**' algorithms (algorithms used during key exchange to verify that the client is actually communicating with the correct server) managed by CCH Tagetik are the following:*ssh-ed25519,ecdsa-sha2-nistp256,ecdsa-sha2-nistp384,ecdsa-sha2-nistp521,rsa-sha2-512,rsa-sha2-256,ssh-rsa*

Note: *the ossh-rsa* algorithm is a weak algorithm and its use is not recommended (this algorithm has been deprecated by OpenSSH)

Tagetik Application

By selecting the **Tagetik Application** type, a Tagetik application can be referenced by its specific data source.

The configuration requires the following fields:

Field	Description
Tagetik Application	
URL	Tagetik application URL
Application datasource	Identifier of a datasource among those defined for the Tagetik application

Tagetik Data Loader

By selecting the **Tagetik Data Loader** type, it is possible to refer a Tagetik Data Loader previously registered on CCH Tagetik.

Configuration requires:

Field	Description
Tagetik Data Loader	
Tagetik Data Loader Name	to be selected from those defined.
Tagetik Data Loader source type	the type of source form which data must be extracted/uploaded, either *From database* or *From file*

Note: Once the source type from which data must be extracted/uploaded has been specified, clicking the **Update Tagetik Data Loader** data button will update the list of datasources or paths configured on the specified Tagetik Data Loader.

Database

Selecting the **From database** type tells Tagetik Data Loader to use the databases configured on it as a resource.

Configuration requires:

Field	Description
Tagetik Data Loader Datasource	the logical name of the datasource from which data must be extracted/uploaded

File

Selecting the **From file** type tells Tagetik Data Loader to use the repositories configured on it as a resource.

Configuration requires:

Field	Description
Tagetik Data Loader path	the logical name of the path on which the application can run read/write operations.

OData service

By selecting the **OData V4 service** or **OData V2 service** type, an OData service can be referenced for reading collections of entities (the EntitySets defined in the service root's Entity Data Model). For the **OData V4** service type, it is also possible to refer a service for reading collections of entities from functions (FunctionImport).

Configuration requires:

Field	Description
URL	the OData service root URL
Use direct connection (only for instances in the same region)	It allows the use of a direct connection for saastagetik services in the same cloud region. The flag is only present if: • it is a cloud environment • the system property '*it.grupposervizi.easy.api.internal.host*' is available • the url host ends with "*.saastagetik.com*" . Under the conditions described above, by default, the flag is enabled
Pagination size	Maximum number of records in each data packet to be loaded to CCH Tagetik. • > 0: packet-based reading enabled • 0: reads all data simultaneously

In the event that large amounts of data are to be transferred and a significant memory occupancy is observed on the CCH Tagetik application server side, it is possible to enable the transmission of packet data by setting the **pagination size** property to a value >0.

The system handles any server-side pagination of OData service (performed via the caption *@odata.nextLink*) even if the data packet transmission is enabled on CCH Tagetik.

REST API

By selecting the **REST API** type, it is possible to refer to a service that expose data via RESTfull API.

Configuration requires:

Field	Description
URL	the REST API root URL
Use direct connection (only for instances in the same region)	It allows the use of a direct connection for saastagetik services in the same cloud region. The flag is only present if: • it is a cloud environment • the system property '*it.grupposervizi.easy.api.internal.host*' is available • the url host ends with "*.saastagetik.com*" . Under the conditions described above, by default, the flag is enabled
Response type	Response format (JSON and CSV).
JSON:	
Path	The path to the object containing the list of data to read, using the JSONPath syntax.
Managed array level	The number of nested arrays to be included in the result.
CSV:	
Header	Indicates the presence of the header row

Field	Description
Columns separator	Indicates the separator character of the columns: "," - ";" - "|" - "Tab" - "Space"
Value delimiter mode	Indicates the value delimitation mode: None, Minimal, Non Numeric, All
Value delimiter	Specifies the values delimiter: None, ", '
Row separator	Specifies the row separator character: Carriage return + Line feed (\r\n), Carriage return (\r), Line feed (\n)
Pagination	
Usage	Indicates where the pagination property is to be specified: None, Header, Query param
Property name	Indicates the name of the property that manages the pagination size.
Size	Specifies the pagination size
Next page	
Position in response	Indicates where to retrieve the property defined in the response to access the next page: None, Header, Body attribute
Property name	Indicates the name of the property in the response that contains the information for retrieving the next page. If the selected position is *Body attribute*, the path must be indicated using the JSONPath syntax.
Position in request	Indicates where the property should be specified in the request to access the next page: None, Header, Query param, Complete URL
Property name	Indicates the name of the property to be specified in the request that contains the information for retrieving the next page.

Example - JSON

Endpoint configuration:

URL: *https://company.com/api/employees*

Response type: *JSON*

Path: *data.employees*

Managed array level: *0*

Pagination:

Usage: *Query param*

Property name: *maxSize*

Size: 3

Next page:

Position in response: *Body attribute*

Property name: *nextPage*

Position in request: *Query param*

Property name: *next*

The first request made by the system is: *https://company.com/api/employees?maxSize=3*
the returned body is:

```
{
"data":{
"employees":[
{"name":"Shyam", "email":"shyamjaiswal@gmail.com", "role":{"name":"Developer", "level":"Senior"},
"qualifications":[{"id": "AAA","details":{"id":"001", "list":["A01","A02"]}}]},
{"name":"Bob", "email":"bob32@gmail.com", "role":{"name":"Developer", "level":"Junior"},
"qualifications":[{"id": "BBB","details":{"id":"002", "list":["B01"]}}]},
{"name":"Jai", "email":"jai87@gmail.com", "role":{"name":"Manager", "level":"Senior"},
"qualifications":[{"id": "CCC","details":{"id":"003", "list":["C01","C02"]}}]}
]
},
"nextPage":"adWOdawe1"
}
```

La seconda richiesta effettuata dal sistema sarà:
https://company.com/api/employees?maxSize=3&next=adWOdawe1

The attributes of each employee will be read and the compound objects deployed:

- *"name"*
- *"email"*
- *"role.name"*
- *"role.level"*

whereas arrays (*"qualifications"*) will be ignored

If the **Managed Array Level** attribute is set to *1*, then the *qualifications. id* and qualifications.details.id fields will also be added

If the **Managed Array Level** attribute is set to *2*, then *qualifications. id, qualifications.details.id* and *qualifications.details.list* fields will also be added

Note: The handling of nested arrays results in the production of more records in the result rows, as they are obtained via Cartesian product.

Example - CSV

Endpoint configuration:

URL: *https://company.com/api/employees*
Response type: *CSV*
Header: *true*
Column separator: ,
Value delimiter mode: *All*
Value delimiter: '
Row separator: *Carriage return + Line feed (\r\n)*

Pagination:

Usage: *Header*
Property name: *maxSize*
Size: 3

Next page:

Position in response: *Header*
Property name: *nextPage*
Position in request: *Complete URL*
Property name:

The first request made by the system is: *https://company.com/api/employees* con Header: *maxSize: 3*
the returned body is:

```
"name","email","role.name","role.level"\r\n
"Shyam","shyamjaiswal@gmail.com","Developer","Senior"\r\n
"Bob","bob32@gmail.com","Developer","Junior"\r\n
"Jai","jai87@gmail.com","Manager","Senior"\r\n
```

The returned headers are:

```
Connection: keep-alive
Content-Type: text/csv
...
nextPage: https://company.com/api/employees?next=adWOdawe1
```

The second request made by the system will be: *https://company.com/api/employees?next=adWOdawe1* con Header: *maxSize: 3*

SharePoint

By selecting the SharePoint type, a SharePoint Site can be referenced.
Configuration requires:

Field	Description
Tenant id	SharePoint tenant id
Client id	SharePoint application client id

Site name	Name of the site on SharePoint. The name is case sensitive and must respect spaces, upper and lower case.
Tenant name	(optional) Name of the tenant retrieved from the SharePoint URL: *<tenantname>.sharepoint.com/sites/<sitename>* *Example*: "mytenant" from the URL: mytenant.sharepoint.com/sites/mysite

Note: The following requirements must be met in order to connect to SharePoint via CCH Tagetik:

- The minimum permissions (Microsoft Graph scope) associated with the configured client must be:
  - If the 'tenant name' field is filled: *Site.Selected*
  - If the 'tenant name' field is not filled: *Site.FullControl.All*
- The configured permissions must be of 'Application' type
- Authentication can be by secret or by certificate

For further details on SharePoint configuration, see Microsoft's official documentation

Google Big Query

By selecting the **Google Big Query** type, a Google Big Query project can be referenced.
Configuration requires:

Field	Description
Project Id	Resource project identifier on Google Cloud Platform

Snowflake

By selecting the **Snowflake** type, a Snowflake database can be referenced.
Configuration requires:

Field	Description
URL	The url to reach Snowflake's service. It must have the following format: https://<account locator>.<region>.snowflakecomputing.com For example: https://xx12345.eu-central-1.snowflakecomputing.com
User	User identifier Snowflake
Warehouse	Identifier of the Warehouse on which the resource is located
Database	Identifier of the Database on which the resource is located
Scheme	Identifier of the Scheme on which the resource is located
Role	Role with which the Snowflake user accesses resources

MS Fabric

By selecting the **MS Fabric** type, an MS Fabric database can be referenced
Configuration requires:

Field	Description
SQL Endpoint	The SQL Endpoint can be retrieved directly from the settings of the Warehouse Fabric to be managed. To be reported exactly as provided by Fabric.
Warehouse	The name of the Warehouse Fabric.
Tenant Id	Id of the tenant for which the Service Principal has been defined added to the Workspace Fabric to which the Warehouse to be managed belongs.

***Note**: In order to connect to MS Fabric via CCH Tagetik, a **Service Principal** meeting the following requirements is required:*

- Access to the **Workspace** where the **Warehouse** to be used is located, with authorisation level at least equal to **Contributor**
- **Secret**-only authentication.

For more details on MS Fabric configuration, please refer to the official Microsoft documentation.

Pretty Good Privacy (PGP) encryption

If the **URL** Endpoints and **Tagetik Data Loader** use **PGP** encryption, it will be possible to configure its features in the **Data Encryption** section.

The **Enable PGP** option allows enabling the **PGP** encryption so that files are decrypted when read and encrypted when written using this standard.

Requested information:

Field	Description
Public key	Public key of the key pair generated by the client
Secret key	Private key of the key pair generated by CCH Tagetik
Secret passphrase	Passphrase used to generate CCH Tagetik keys (Field value is hidden)

To successfully upload an encrypted file, it is necessary to comply with certain restrictions:

- no signature must be applied after encryption;
- the encrypted file must contain one recipient only.

If the endpoint is only used to read data, only the *secret key* and *secret passphrase* pair needs to be configured; if it is only used to write data, only the *public key* needs to be configured.

Authentication

If the service provides an authentication mode, it is possible to select a type from the **Available authentication types** field in the Authentication tab.

The following types of authentication are currently supported:

Field	Description
None	
Valid for all types of Endpoints, when the remote resource does not require any authentication type	
Basic Authentication	
Valid for **URL** type Endpoints (not with *"file"*), **OData service t**ype and **REST APIs**, when the remote resource requires requires an authentication consisting of user and password. For **URL-type** endpoints with the *'smb'* protocol, the username must contain the domain reference (e.g. <DOMAIN>\<username>)	
Username	identifies the user name that can access the requested resource
Password	identifies the password associated with the user name defined at the previous point
OAuth 2	
Valid for Endpoints of type **OData service**, **REST API**, and **Snowflake**, it allows you to configure, using the OAuth2 standard, the issuing of an access token by an authorization server to a third-party client, subject to the approval of the user who owns the resource being accessed. See OAuth website	
Grant type	It can be: Client credentials or Password credentials
Client Authentication	Indicates whether to send authentication in the header, body (with Client credentials or Password credentials) or authenticate with a signed JWT with a private key (Client credentials only)
Token Endpoint	The url to obtain the token
Client Id	OAuth2 Client Id
Client Secret	OAuth2 Client Secret
Username	Username for flow with password credentials
Password	Password for flow with password credentials

Field	Description
Kid	Represents the key identifier used if you have multiple keys on the authentication server to obtain a signed JWT with a private key
Lifetime	Indicates the time, in seconds, to be added to the current time for configuring the EXP field of the JWT token. Possible values are those in the range [1 - 3600].
Private key in PEM format	File containing the private key in *PEM* format to obtain a signed JWT with a private key. The private key must be generated with the RS256 algorithm in PEM format without passphrase and must have this format: -----BEGIN PRIVATE KEY----- MII... -----END PRIVATE KEY---- The claims included in the JWT token are: **iss**, **sub**, **aud**, **exp**, **iat**, **jti** (see RFC7523).
Scopes	OAuth2 Scopes
Resource	OAuth2 Resource
Use proxy configuration to obtain the token	Indicates whether the proxy configuration is required to reach the url for obtaining the token
Use configured certificates to obtain the token	Indicates whether authentication via certificates is required to reach the url to obtain the token
Key Pair	
Only valid for **Snowflake** endpoints. It allows you to configure authentication using your own private key.	
Account Identifier	Snowflake account identifier
Keyfile	Account's private key file
Passphrase	(optional) Passphrase for opening the private key
Fingerprint public key	Fingerprint of the associated public key on Snowflake. The fingerprint must be in the format SHA256:<hash key>.
Json Key	
Only valid for endpoints of type BigQuery. Allows you to configure the ADC key used for the connection	
ADC Json Key	ADC key in Json format of the service account used for connection (after saving, the private key field is obscured)
Underlying	
Valid only for endpoints of **BigQuery** type. It allows you to connect to Google BigQuery using the Google Cloud environment on which CCH Tagetik is installed.	

IMPORTANT: *beta functionality must be enabled to use this authentication mode.*

Authentication tab for endpoint of SharePoint type

Please find below the most important fields.

Field	Description
Scope	SharePoint Permissions Scope. If not specified, the default value is used: *https://graph.microsoft.com/.default*
Available authentication types	*Can be:* ***Via Secret*** *or* ***Via Certificate.*** ***Via*** **Secret - Secret** value: secret client value **Via Certificate,** it provides the following options: **Certificate Type:** can be: **PEM (.cer .crt .pem .key)** or **PKCS#12 (.pfx .p12)** **Certificate file**: contains the certificate in PEM or PKCS#12 format **Key File**: contains the key of the certificate (the field is only present if the certificate type is PEM) **Passphrase**: used to generate the certificate (the field is only present if the certificate type is PKCS#12)

Authentication tab for MS Fabric endpoints

Field	Description
Client Id	Client Attribute Id of the Azure Application defined for the Main Service
Secret	Client Secret value of Application Azure (Service Principal)

Proxy

If the Endpoint **OData service**, **REST API** and **Snowflake** are exposed via a Proxy server, this can be configured in the **Proxy** section.

Field	Description
Server	Proxy server identifier
Port	Proxy server port
User	Username for authentication on the proxy server
Password	Password for authentication on the proxy server

Custom headers

If the Endpoint **OData service, REST API** and **Snowflake** require additional headers to be sent in the request, these can be configured in the **Headers** section.

Field	Description
Key	Key of the custom header
Value	Value of the custom header

Certificates

If the **OData service** and **REST API** Endpoints require authentication via certificates, these can be configured in the **Certificates** section.
This section is only visible if the protocol used in the url is *https*.

Field	Description
Certificate Authority	
CA file	File containing the CA certificate in *PEM* format
Client	
Certificate Type	It can be: *PEM (.cer .crt .pem .key)* or *PKCS#12 (.pfx .p12)*
Certificate file	File containing the certificate in *PEM* or *PKCS#12* format
Key file	File containing the certificate key in *PEM* format (the field is only present if the certificate type is *PEM*)
Passphrase	Passphrase used to generate the certificate. (the value of the field is hidden)

Advanced

For some endpoint types, the following additional connection parameters can be set

Field	Description
Endpoint type: Snowflake	
Query timeout	Setting for Snowflake query timeout (in seconds). This option refers to the processing time of the query by Snowflake.
Type of Endpoint: Google Big Query	
Use target table	Use a target table for the query result.  ***IMPORTANT**: in case the compressed outcome of the query exceeds 10GB, this setting is mandatory)*
Use existing Project Id	For the definition of the target table, use the Project Id specified in the attributes of the Big Query endpoint

Field	Description
Project Id	Definition of the Project Id of the target table. If it has been set that the existing Project Id must be used, the field is not editable.
Dataset name	Definition of the name of the target table dataset
Table Name	Definition of the name of the target table
Write mode	Defines how data is written to the temporary table according to Google BigQuery : • **Write Truncate**: If the table already exists, BigQuery overwrites data, removes restrictions and uses the scheme given by the query. • **Write Truncate Data**: If the table already exists, BigQuery overwrites data, but keeps the restrictions and the scheme of the existing table. • **Write Append**: If the table already exists, BigQuery adds new data to the table and returns the results of the table. • **Write Empty**: If the table already exists and contains data, the data processing returns an error of 'duplicate' type.

Connection

For each Endpoint, in the **Connection** section, the **Test** button can be used to check the validity of the entered information and test the connection to the remote resource. At the end of the test, a message containing details on the test result is displayed in the lower area.

Endpoint management procedures

Adding an endpoint

Repository > Endpoint

1. Click : a new row will appear in the table.
2. Select the newly created row.
3. Set the fields as needed in the **Attributes** tab.
4. If the Endpoint requires it, it is possible to specify the necessary fields in the tabs **Authentication**, **Proxy**, **Custom headers** and **Certificates**
5. Click **Save**.

Clone an Endpoint

Repository > Endpoint

1. Click : a new row will appear in the table.
2. Select the newly created row.
3. In the tabs, the information of the source Endpoint is copied except the password type fields and the fields of the **Certificates** tab.
4. Manually set up the **Code** field and the required fields.
5. Click **Save**.

Connection test

This procedure allows the validity of the entered information to be verified and the connection to the endpoint to be tested.

Repository > Endpoint

1. Select the desired row in the table.
2. In the **Connection** tab, click on **Test**: The result of the tests performed appears.

Access to Endpoint functions via API

Introduction

CCH Tagetik also allows endpoints to be managed via APIs (RESTful APIs).

Certain conditions are required to use this consultation mode.

Access conditions

To be able to use these specific APIs, the user must be authenticated (Basic Auth, OAuth2).

APIs must also be enabled in the Repository. For reverse proxies, further settings are necessary. They can be used with both *http* and *https* protocols.

For more details on authentication and enabling, see REST API Overview.

API to read default Endpoints

Method

GET

URL

```
http://server:port/tagetikcpm/api/admintools/endpoint/v1/default
```

API to read all Endpoints

Method

GET

URL

```
http://server:port/tagetikcpm/api/admintools/import/v1/elements
```

API to read all Endpoints filtered by code

Method

GET

URL

```
http://server:port/tagetikcpm/api/admintools/import/v1/elements?filter=code_filter
```

API to read a single Endpoint

Method

GET

URL

```
http://server:port/tagetikcpm/api/admintools/import/v1/elements/endpoint_code
```

API to create a new Endpoint

Method

POST

URL

```
http://server:port/tagetikcpm/api/admintools/import/v1/elements
```

Property

In the body of the request it is possible to include the following properties in JSON format:

Property	Description	Options
code	Mandatory.	-
description	Mandatory.	-
endpointType	Mandatory.	It can take the following values: • U: URL • A: Tagetik Data Loader • P: Application • O: OData V4 • 2: OData V2 • R: REST API • S: SharePoint • B: Google Big Query • F: Snowflake

endpointType properties

Moreover, depending on the endpoint type it is possible to add the following properties:

URL: properties	Description	Options
url	Mandatory.	-
defaultEndpoint	-	true/false (default = false)
authenticationType	-	It can take the following values: • N: None (default) • B: Basic
basicUsername	Mandatory if authenticationType = B.	-
basicPassword	Defined if authenticationType = B.	-
pgpEnabled	-	true/false (default = false)
pgpPublicKey	Mandatory if pgpEnabled = true	-
pgpPrivateKey	Mandatory if pgpEnabled = true	-
pgpPassphrase	mandatory if pgpEnabled = true	-

Agent: properties	Description	Options
tdlAgentId	Mandatory.	-
tdlSourceType	Mandatory.	It can take the following values: • P: Path • D: Datasource
tdlPath	Mandatory if tdlSourceType = P	-
tdlDatasource	Mandatory if tdlSourceType = D	-
pgpEnabled	-	true/false (default = false)
pgpPublicKey	Mandatory if pgpEnabled = true	-
pgpPrivateKey	Mandatory if pgpEnabled = true	-
pgpPassphrase	Mandatory if pgpEnabled = true	-

Application: properties	Description	Options
url	Mandatory.	-
applicationDatasource	Mandatory.	-

Odata (V4, V2): properties	Description	Options
url	Mandatory.	-
odataPaginationSize	Mandatory.	>= 0 (default = 0)
authenticationType	Mandatory.	It can take the following values: • N: None (default) • B: Basic • O: OAuth2.0
basicUsername	Mandatory if authenticationType = B	-
basicPassword	Defined if authenticationType = B	-

Odata (V4, V2): properties	Description	Options
oauth2GrantType	Defined if authenticationType = O	It can take the following values: • CLIENT (default) • PASSWORD
oauth2TokenEndpoint	Mandatory if authenticationType = O	-
oauth2ClientId	Mandatory if authenticationType = O	-
oauth2ClientSecret	Mandatory if authenticationType = O	-
oauth2Username	Mandatory if authenticationType = O e oauth2GrantType = PASSWORD	-
oauth2Password	Mandatory if authenticationType = O e oauth2GrantType = PASSWORD	-
oauth2Scopes	Defined if authenticationType = O	-
oauth2Resources	Defined if authenticationType = O	-
oauth2ClientAuthentication	Defined if authenticationType = O	It can take the following values: • HEADER (default) • BODY • JWT
oauth2Kid	The Header parameter 'kid' (key ID) is a hint as to which key was used to protect JWS. Defined if authenticationType = O and oauth2GrantType = CLIENT and oauth2ClientAuthentication = JWT	-
oauth2Lifetime	The duration is the time in seconds added to the iat (issue at time) to create the JWT expiration time at or after which the JWT MUST NOT be accepted for data processing). Defined if authenticationType = O and oauth2GrantType = CLIENT and oauth2ClientAuthentication = JWT	>= 1 and <= 3600 (default = 300)
jwtCertificateFile	Generated private key code in PEM format. Mandatory if authenticationType = O and oauth2GrantType = CLIENT and oauth2ClientAuthentication = JWT	-
oauth2UseProxy	-	true/false (default = false)
oauth2UseCertificates	-	true/false (default = false)
proxyServer	-	-
proxyServerPort	-	-
proxyUsername	-	-
proxyPassword	-	-
customHeaders	-	-

REST API: properties	Description	Options
url	Mandatory.	-
restResponseType	Mandatory.	It can take the following values: • JSON (default) • CSV
restPath	Defined if restResponseType = JSON	-
restJsonArrayLevel	-	>= 0 (default 0)
restCsvHeader	Defined if restResponseType = CSV	true/false (default = false)

REST API: properties	Description	Options
restCsvColumnSeparator	Mandatory if restResponseType = CSV	It can take the following values: • "," (default) • ";" • "|" • "\t" • " "
restCsvValueDelimiterMode	Mandatory if restResponseType = CSV	It can take the following values: • NONE (default) • MINIMAL • NON_NUMERIC • ALL
restCsvValueDelimiter	Mandatory if restResponseType = CSV	It can take the following values: • N (default) • " • '
restCsvRowSeparator	Mandatory if restResponseType = CSV	It can take the following values: • CRLF (default) • CR • LF
restPaginationUsage	Mandatory.	It can take the following values: • N: None (default) • H: Header • Q: Query param
restPaginationPropertyName	Mandatory only if restPaginationUsage = H or Q	-
restPaginationSize	Mandatory only if restPaginationUsage = H o Q	Must be > 0
restNextPageResponsePosition	Mandatory.	It can take the following values: • N: None (default) • H: Header • B: Body attribute (only if restResponseType = JSON)
restNextPageResponsePropertyName	Mandatory only if restPaginationUsage = H or B	-
restNextPageRequestPosition	Mandatory.	It can take the following values: • N: None (default) • H: Header • Q: Query param • U: Complete URL
restNextPageRequestPropertyName	Mandatory only if restPaginationUsage = H o Q	-
authenticationType	Mandatory.	It can take the following values: • N: None (default) • B: Basic • O: OAuth2.0
basicUsername	Mandatory if authenticationType = B.	-

REST API: properties	Description	Options
basicPassword	Defined if authenticationType = B	-
oauth2GrantType	Defined if authenticationType = O	It can take the following values: • CLIENT (default) • PASSWORD
oauth2TokenEndpoint	Mandatory if authenticationType = O	-
oauth2ClientId	Mandatory if authenticationType = O	-
oauth2ClientSecret	Mandatory if authenticationType = O	-
oauth2Username	Mandatory if authenticationType = O e oauth2GrantType = PASSWORD	-
oauth2Password	Mandatory if authenticationType = O e oauth2GrantType = PASSWORD	-
oauth2Scopes	Defined if authenticationType = O	-
oauth2Resources	Defined if authenticationType = O	-
oauth2ClientAuthentication	Defined if authenticationType = O	It can take the following values: • HEADER (default) • BODY • JWT
oauth2Kid	The Header parameter 'kid' (key ID) is a hint as to which key was used to protect JWS. Defined if authenticationType = O and oauth2GrantType = CLIENT and oauth2ClientAuthentication = JWT	-
oauth2Lifetime	The duration is the time in seconds added to the iat (issue at time) to create the JWT expiration time at or after which the JWT MUST NOT be accepted for data processing). Defined if authenticationType = O and oauth2GrantType = CLIENT and oauth2ClientAuthentication = JWT	>= 1 and <= 3600 (default = 300)
jwtCertificateFile	Generated private key code in PEM format. Mandatory if authenticationType = O and oauth2GrantType = CLIENT and oauth2ClientAuthentication = JWT	-
oauth2UseProxy	-	true/false (default = false)
oauth2UseCertificates	-	true/false (default = false)
proxyServer	-	-
proxyServerPort	-	-
proxyUsername	-	-
proxyPassword	-	-
customHeaders	-	-

The certificates cannot be configured via API

Sharepoint: properties	Description	Options
sharepointTenantId	**Mandatory**	-
sharepointClientId	**Mandatory**	-
sharepointSiteName	**Mandatory**	-
sharepointScope	**-**	-
sharepointAuthenticationType	**Mandatory**	It can take the following values: • S: Secret Note: the field can take the read-only value C (Certificate)
sharepointSecretValue	**Mandatory**	-

MS Fabric: properties	Description	Options
fabricSqlEndpoint	**Mandatory**	SQL Warehouse Endpoint
fabricWarehouse	**Mandatory**	Warehouse Name
fabricTenantId	**Mandatory**	Tenant Id
fabricClientId	**Mandatory**	Apllication Azure client id
fabricSecret	**Mandatory**	Secret generated for Application Azure.

 Certificate authentication is not managed via API.

 Creation, replacement, update, and deletion operations for *Google BigQuery* and *Snowflake* endpoints are not allowed when using the REST API service. Executing these operations will return a '501 – Not Implemented' response.

API to replace an Endpoint

Method

PUT

URL

```
http://server:port/tagetikcpm/api/admintools/import/v1/elements/endpoint_code
```

Property

In the body of the request, in JSON format, it is possible to include the same properties as in the creation according to the type of endpoint, with the exception of 'code', which is already specified in the url.

For an endpoint with oauth2ClientAuthentication = JWT, this operation will have no effect (400 - Bad request: The following endpoint cannot be replaced: <ENDPOINT_CODE>)

API to update an Endpoint

Method

PATCH

URL

```
http://server:port/tagetikcpm/api/admintools/import/v1/elements/endpoint_code
```

Property

In the body of the request, in JSON format, it is possible to include, as properties to be updated, the same as the creation according to the type of endpoint, with the exception of the following fields which cannot be changed:

- code
- endpointType
- tdlAgentId
- tdlSourceType
- sharepointAuthenticationType

Other non-editable fields belong to the case of an endpoint modification with oauth2ClientAuthentication = JWT:

- authenticationType
- oauth2GrantType
- oauth2ClientAuthentication
- jwtCertificateFile

API to delete an Endpoint

Method

DELETE

URL

```
http://server:port/tagetikcpm/api/admintools/import/v1/elements/endpoint_code
```

Labels for Azure Information Protection

Purpose of the page

This page provides information on creating an application and its endpoint to access the Azure Information Protection label list.

Configuring the Azure Information Protection label application

Registering a new application

From the Microsoft Entra administration centre, section 'Applications -> Enterprise applications', create a new application for label access. Select create your own application. Enter a name for the application. Click on the Create button. An application will be created and you will navigate to the application overview page.

Take note of the Application (Client) ID of the newly created application.

References: https://entra.microsoft.com/

API permissions

Assign the following permissions, see section 'Applications -> App registration -> API Permissions'.

APIs	API / Permission Name	Type
Microsoft Graph	InformationProtectionPolicy.Read.All	Application
Microsoft Information Protection Sync Service	UnifiedPolicy.Tenant.Read	Application

Once the permissions have been entered, give consent to the grant for Microsoft (see the relevant button).

Secret

From the 'Certificates and Secrets' section, create a new secret for the application and note the value of the secret.

Caution: The secret value will only be accessible for a short time.

Configuring the endpoint for the Azure Information Protection labels

Repository > Endpoint

Required information

The following information will be required for configuration:

- Directory (tenant) ID.
  Example: 51ce633b-266a-4f37-0001-b9d0d77e2d90
- The Application (client) ID of an application configured to access label information.
- A secret related to the authentication application
  Example: fwH8Q~kxWMe6bmnWwZjVpW4Ujn6ayzUXhItnacuF

Be careful to use the secret and not the secret id.

Configure an endpoint.

Add a new endpoint by means of the add button.

Attributes Section

Field	Value
Code	Identified endpoint code for labels, Example: LABELS
Description	Shows the description in full.
Endpoint type	Select API REST.

REST API subsection

Field	Value
URL	Enter the following URL: https://graph.microsoft.com/beta/dataClassification/sensitivityLabels
Type of response	Select: JSON
Path	enter: value
Managed levels	Set value 2

Leave the rest of the subsections untouched.

Authentication section

Field	Value
Authentication Type	Select: OAuth 2.0
Grant type	Select: Client credentials
Token Endpoint	Enter the following text replacing the text <Tenant ID> with the Tenant ID of your organisation. https://login.microsoftonline.com/<Tenant ID>/oauth2/v2.0/token Example: https://login.microsoftonline.com/73727f63-b5c4-4b0d-905d-e9c951753cd3/oauth2/v2.0/token
Client Id	Insert Cliend ID.
Client Secret	Enter the Secret of the application configured to access the labels.
Scopes	Enter the following text: https://graph.microsoft.com/.default
Resource	Leave blank.
Client Authentication	Select: Send as basic authentication header.

Save the configuration.

It will now be possible to set the labels to be used for Azure Information Protection. See: Endpoint Label Setting

TGK Service

Contents

This section covers the following topics:

TGK Service page 93
Generate keys window 94
Manage the TGK Service 95

TGK Service page

Repository > TGK Service > TGK Service

***IMPORTANT**: this page is only visible if, for the user's role, function 0003130 has been enabled in the repository and if the **TGK Service** tile was enabled via .*

Purpose of the page

This page makes it possible to manage Windows servers on which TGKMSService (CCH Tagetik's .NET webservice) has been installed.

Attributes tab

Please find below the most important fields.

Field	Description
Code	Unique ID of the TGKMSService instance being managed
URL	URL at which the instance of the TGKMSService is located
Client code	ID code of the client invoking TGKMSService. Generated by the **Generate Keys** window.
Available	Indicates the node status (**false**: off-line, **true**: active).

SSAS tab

This tab makes it possible to manage Tagetik Analytics models on Microsoft Analysis Services (SSAS).

Please find below the most important fields.

Field	Description
Enabled	Makes it possible to generate an SSAS resource.
Number of instances	Instances that can be run simultaneously. Maximum value: 100.
Polling interval (seconds)	CCH TagetikProcess waiting time between polling requests to the service while a command is running. Maximum value: 99999
Template	Name of a database on the Analysis Services server to use as a template
Type	Type of instance used in Analysis Service model management
Host	Name or URL of the Analysis Services instance used by TGKMSService to generate standard CCH Tagetik OLAP cubes.

SAP tab

This tab makes it possible to run ETL on SAPs generated by Tagetik Connector Enabled for SAP.

Please find below the most important fields.

Field	Description
Enabled	Makes it possible to run ETL on SAP.
Number of instances	Instances that can be run simultaneously. Maximum value: 1.
Polling interval (seconds)	CCH TagetikProcess waiting time between polling requests to the service while a command is running. Maximum value: 99999
Template	Name of a database on the Analysis Services server to use as a template
URL Logs	URL containing the Reporting Services reports generated by extracting data from SAP. Default value: http://localhost/ReportServer/ReportViewer.aspx?/OneConnect/LoggingOverview
Master Package	Name, including the path, of the SSIS packet that must be run by TGKMSService during the extraction of data from SAP. This extraction is performed by the SAP connecter to populate the staging database used by the ETL routine. Default value: .\\SSISPackages\\EDW_STA_TagetikMasterPackage.dtsx
Connection database	Connection string towards the OneConnect configuration database. Default value: TagetikLog0.
Execute SSIS	Allows ETL to call back the TGKMSService to run tasks developed with SQL Server Integration Services.

Excel tab

This tab makes it possible to generate Excel forms linked to CCH Tagetik Excel .NET forms.

Field	Description
Enabled	Makes it possible to generate the Excel form.
Number of instances	Instances that can be run simultaneously. Maximum value: 1.
Polling interval (seconds)	CCH TagetikProcess waiting time between polling requests to the service while a command is running. Maximum value: 99999
Timeout	Sets the maximum waiting for the database's response.
Callback URL	Parameter used by the Excel client to find out which CCH Tagetik server to connect to in order to run the form.

Word tab

This tab makes it possible to generate Word documents linked to Collaborative Office.

Field	Description
Enabled	Makes it possible to generate the Word document.
Number of instances	Instances that can be run simultaneously. Maximum value: 1.
Polling interval (seconds)	CCH TagetikProcess waiting time between polling requests to the service while a command is running. Maximum value: 99999
Timeout	Sets the maximum waiting for the database's response.
Callback URL	Parameter used by the Word client to find out which CCH Tagetik server to connect to in order to run the form.

Generate keys window

Repository > TGK Service > Generate keys

***IMPORTANT**: this page is only visible if, for the user's role, function 0002962 is enabled in the repository and if the **TGK Service** tile was enabled via .*

Purpose of window

This window makes it possible to generate the pair of keys to ensure that communications between the CCH Tagetik application and TGKMSService (CCH Tagetik's .NET webservice) are secure. RSA-standard encryption is used.

Fields in window

Field	Description
Public Key	Public key generated by the system in Base64 format and encoded in accordance with the *SubjectPublicKeyInfo X.509* standard.

Buttons in the window

Element	Description
Generate Key	Generates and shows the asymmetrical pair of keys (public and private). If the pair of keys is re-generated, the services using the public key must be updated.
Export public key	• Exports the public key in standard ASN.1 format. • Creates and downloads a file with .cer extension.

Manage the TGK Service

Generate an SSAS/SAP/Excel/Word resource

Repository > TGK Service > TGK Service

1. Click : a new row will appear in the table.
2. In the **Attributes** tab, set the parameters as desired.
3. Select the desired tab and set the desired parameters.
4. Click **Save**.

Introduction to REST API

CCH Tagetik provides several sets of APIs that allow you to read or write data and perform numerous operations without requiring access to the graphical interface.

This enables numerous integrations with third-party software and allows the automation of various processes.

Access to audit data via API

Introduction

Audit logs can be accessed by specially authorised and trained users, including through APIs (RESTful APIs) developed according to the OData v4.0 standard.
By default, the elements are ordered with respect to the dateTi*meUtc property in descending order, from most recent to least recent.*

Note The service returns up to a maximum of 10,000 items.

Certain conditions are required to use this consultation mode.

Authorisation

To be able to use this specific API, the user must have the 0001286 function enabled at least in read mode.

APIs for data processing

STATUS	SWAGGER
ACTIVE	-
FORMAT	ROOT ENDPOINT
OData	**<api root>/audit/v1/dataprocessing**

The URLs for obtaining the audit data on data processing are provided below:

URL	Description
<api-root>/audit/v1/dataprocessing/	Service displayed
<api-root>/audit/v1/dataprocessing/$metadata	Returns the following metadata: • Scheme • EntityType • EntityContainer • EntitySet In particular, data types are specified for all entity properties and, in the case of changes, the additionalFields property is present
<api-root>/audit/v1/dataprocessing/	Returns the name and description of all the application's data processing.
<api-root>/audit/v1/dataprocessing/<ELABORATION_CODE>	Returns all occurrences of the selected data process (ELABORATION_CODE).

The *additionalFields* property shows the data changed during data processing; it lists a collection of values, each of which has three properties:

- field: shows the name of changed data
- old_<value> = value before editing
- new_<value> = value after editing

Contents of the data returned

By invoking the second endpoint, the following data are returned:

Element	Description
oid	Elaboration code
dateTimeUtc	Date and time performed (format: ISO 8601, UTC)
status	Data processing outcome
parts	Processing step, if divided into parts (0 = pre-run)
user	Executing user
database	Execution database
duration	Data processing duration (millisec)
client	Data processing client
ipClient	IP address of the device originating the data processing

These data are followed by the dimensional parameters, if present, together with the relative values.

APIs for managing tables and data

The URLs for obtaining audit data on the management of tables and data are provided below:

URL	Description
`<api-root>/audit/v2/datamanagement/`	Service displayed
`<api-root>/audit/v2/datamanagement/$metadata`	Returns the following metadata: • Scheme • EntityType • EntityContainer • EntitySet In particular, the data types of the columns of each table are specified, and the *additionalFields* property is present in case of changes.
`<api-root>/audit/v2/datamanagement/`	Returns the code and description of the edited tables.
`<api-root>/audit/v2/datamanagement/TABLE_CODE`	Returns all occurrences of the changes in the indicated table (TABLE_CODE)

Contents of the data returned

By invoking the second endpoint, the following data are returned:

Element	Description
oid	Operation code
dateTimeUtc	Date and time edited (format: ISO 8601, UTC)
status	Edit outcome
user	Executing user
database	Execution database
duration	Data processing duration (ms)
client	Data processing client
ipClient	IP address of the device originating the data processing
editType	Type of edit (entry, edit or deletion)
factTable	Type of table edited: • true = facts table • false = master table
key	Physical key of the edited data

This data may be followed by *additionalFields* properties for each column of the selected table if changes are made:

- field: shows the name of changed data
- old_<value> = value before editing
- new_<value> = value after editing

Multi-value properties are aggregated on the same row and separated by a comma.

APIs for database logins

STATUS	SWAGGER
ACTIVE	-
FORMAT	ROOT ENDPOINT
OData	**<api root>/audit/v1/database**

The URLs for obtaining audit data on database logins are provided below:

URL	Description
<api-root>/audit/v1/database/	Service displayed
<api-root>/audit/v1/database/$metadata	Returns the following metadata: • Scheme • EntityType • EntityContainer • EntitySet
<api-root>/audit/v1/database/CONNECT_TO_DATABASE	Returns a list of logins to the various databases.

Contents of the data returned

By invoking the second endpoint, the following data are returned:

Element	Description
oid	Operation code
dateTimeUtc	Date and time of login format: ISO 8601, UTC)
status	Operation outcome
user	Executing user
database	Execution database
duration	Data processing duration (ms)
client	Data processing client
ipClient	IP address of the device that ran the data processing

API for user logins

The URLs for obtaining audit data on user logins are provided below:

URL	Description
<api-root>/audit/v1/user/	Service displayed
<api-root>/api/audit/v1/user/$metadata	Returns the following metadata: • Scheme • EntityType • EntityContainer • EntitySet
<api-root>/audit/v1/user/LOGIN	Returns a list of user logins.

Contents of the data returned

By invoking the second endpoint, the following data are returned:

Element	Description
oid	Operation code
dateTimeUtc	Date and time of login (format: ISO 8601, UTC)
status	Operation outcome
user	Executing user
client	Client used by the user
ipClient	IP address of the device that ran the data processing

Element	Description
sessionDuration	Session duration (ms), terminated by logout or timed out
timedOut	Type of session interruption: • true = session timed out • false = user logout

Filter audit data through API

Introduction

To reduce the quantity of data retrieved, several types of filter can be applied:

- to header properties
- to dimensional parameters

Query parameter

$filter=()

Filter heading properties: filter by date

Property

dateTimeUtc (format: ISO 8601, UTC)

Description

Filter data based on operation date.

Operators

Operator	Net result
gt	Shows records from dates after the selected date.
lt	Shows records from dates prior to the selected date.
ge	Shows records from or after the selected date.
le	Shows records from or prior to the selected date.

Example

http://127.0.0.1:8080/tagetikcpm/api/audit/v1/dataprocessing/SOLLECITATORE_ CERTIFICAZIONI?$filter=(dateTimeUtc lt 2020-08-21T08:01:26.758Z)

http://127.0.0.1:8080/tagetikcpm/api/audit/v1/datamanagement/CONTO?$filter=(dateTimeUtc gt 2020- 12-21T**08**:00:00.000Z) and (dateTimeUtc lt 2020- 12-21T**12**:00:00.000Z)

Particular features of the dateTimeUtc property

The value stored in the database is more precise than can be specified in the endpoint. For this reason, the search results will be based on the level of precision of the endpoint.

Filter header properties: string-type data

Property

oid, **status**, **user**, **database**, **client**, **ipClient**, **editType** and **key**.

Description

Filter data based on header properties.

Operators

Operator	Net result
eq	Shows records for which the parameter is the same as the one reported in quotes.
ne	Shows records for which the parameter is different from the one reported in quotes.

Example

http://127.0.0.1:8080/tagetikcpm/api/audit/v1/datamanagement/UTENTE?$filter=(clientIp eq '10.10.10.10')

http://127.0.0.1:8080/tagetikcpm/api/audit/v1/datamanagement/UTENTE?$filter=(user eq 'UTENTEA') or (user eq 'UTENTEB')

http://127.0.0.1:8080/tagetikcpm/api/audit/v1/datamanagement/UTENTE?$filter=(key eq '%7BTEST, GRAFFE%7D')

***Note**: if the **key** property contains braces {}, encoding must be performed as in the example.*

Filter dimensional parameters

Description

This filter is applied to the data processing APIs. It filters data by applying dimensional parameters, invoking the endpoint which returns the metadata.

Operators

Operator	Net result
eq	Shows records whose header string is the same as the one reported in quotes.
ne	Shows records with a different header string from the one reported in quotes.

Example

http://127.0.0.1:8080/tagetikcpm/api/audit/v1/dataprocessing/ESPLOSIONE(periodo eq '10')

http://127.0.0.1:8080/tagetikcpm/api/audit/v1/datamanagement/ESPLOSIONE(raccolta ne 'ACT')

In the following two examples, the records shown are those where the “currency” parameter has ‘no value’:

http://127.0.0.1:8080/tagetikcpm/api/audit/v1/datamanagement/AGGIORNA_PARAM_BANCARIE(valuta eq 'no value')

http://127.0.0.1:8080/tagetikcpm/api/audit/v1/datamanagement/AGGIORNA_PARAM_BANCARIE(valuta eq null)

***Note**: it is not possible to filter for a parameter that has not been specified (e.g. “currency” : null).*

Pagination of results

Description

Makes it possible to subdivide the data returned on pages with specific characteristics. This type of query allows retrieval of a maximum of 10,000 elements.

Parameters

Parameter	Net result
$top=N1	(N1 positive integer) Reports N results.
$skip=N2	(N2 positive integer) Omits the first N results.
?$top=N1&skip=N2	Combines the results of the two parameters.

***Note**: It is possible to use several query parameters by concatenating them with the & symbol.*

Example

http://127.0.0.1:8080/tagetikcpm/api/audit/v1/dataprocessing/SOLLECITATORE_CERTIFICAZIONI?$top=2&$skip=5

http://127.0.0.1:8080/tagetikcpm/api/audit/v1/datamanagement/CONTO?$top=5

Results sorting

Data can be sorted by using:

Parameter	Net result
$orderby=P1 ORD,P2 ORD, ...	(P1, P2, ...) These represent the set of properties you want to sort on (you can sort by one or more properties) (ORD) This represents the *asc or* desc sorting type: for ascending and descending sorting respectively.

***Note**: By default, if no sort type is specified, the ascending sort will be applied.*

The properties on which the sorting can be performed are the header properties: oid, dateTimeUtc, status, user, database (not present for the user login api), client, ipClient.

Moreover, sorting by *key* is available for the data management APIs

Several properties can be combined by following **?$orderby=*P1 ORD,P2 ORD,P3 ORD*.**

Example

http://127.0.0.1:8080/tagetikcpm/api/audit/v2/datamanagement/CONTO?$orderby=dateTimeUtc desc

http://127.0.0.1:8080/tagetikcpm/api/audit/v2/datamanagement/CONTO?$orderby=user asc,dateTimeUtc

Caching system

To optimise the performance of the query through APIs, a caching system has been introduced through the system property **it.grupposervizi.easy.audit.odata.cache.cron**.

The cron expression contains six asterisks (*) which indicate the following fields:

- seconds
- minutes
- hour
- day
- month
- year

***Note**: without that system property, the default value is 0 30 * * * *, and therefore a refresh is performed every 30 minutes.*

The following data are saved in the cache:

- dimensional characteristics, for data processing
- all the properties of the edited table, for the management of the tables and data

Changes made after the cache has been built will only appear after the next cache update. For data processing, the header properties and parameters present in the cache are always displayed.

Audit Data Migration Tool

In Service Pack 18, the audit persistence mode in CCH Tagetik has changed.
To enable the visualisation of data previously stored within the audit, a data migration tool was released in Service Pack 18, which can be accessed from the legacy UI by clicking on the red banner.

Starting with Service Pack 34, it is only possible to launch the audit data migration tool via API.

The following APIs enable the execution of the tool:

URL	Description
<api-root>/api/auditDataMigration/run?amount=n	Allows you to launch the migration tool for n records. n must be between 1 and 2^31 - 1, it is recommended to launch the tool with an n = 10000 records.
<api-root>/api/auditDataMigration/run	Allows you to run the migration tool for all records that have not yet been migrated.
<api-root>/api/auditDataMigration/amount	It lets you know how many records still need to be migrated.

Application Job exposure with RESTful API

CCH Tagetik exposes the execution of application jobs using Open RESTful APIs.

STATUS	SWAGGER
ACTIVE	-
FORMAT	ROOT ENDPOINT
OData	**<api root>/dataprocessing/job/v1**

Application job

To run the Application job, you can call the following URL through a **POST** method :

http://*server*:*port*/*tagetikcpm/api/dataprocessing/job/v1/*database_id**/job_code***

the application job *job_code* will be run on the environment specified via *database_id*.

 An error will be returned if the application job code does not exist.

Currently managed tasks are:

- ETL
- Loading to Analytical Workspace
- Update Financial Workspace from Analytical Workspace
- DTP
- Stored procedures
- Calculation logics
- Preliminary data processing on Original scenarios
- Preliminary on Consolidation area
- Data processing for Legal Entities
- Consolidation

In the response, the following will be returned in JSON format:

dataprocessingId: data processing ID

E.g.: { "dataprocessingId":"FF8081817A4264E5017A42D08E3C21EB" }

To check the execution status of the application job, a GET request can be used to invoke the URL:

http://*server*:*port*/*tagetikcpm/api/dataprocessing/job/v1/*dataprocessingId**

In the response, the following will be returned in JSON format:

status: *RUNNING*/*FAILED*/*COMPLETED* - depending on whether the data processing is running, failed or successfully completed.

E.g.: {"requestDateTime": "2021-06-25T10:56:42.850+0000", "status": "COMPLETED"}

To interrupt the execution of the application job, you can call the following URL via **DELETE** method:

http://*server*:*port*/*tagetikcpm/api/dataprocessing/job/v1/*dataprocessingId**

In the response, the following will be returned in JSON format:

message: "Stop on dataprocessingOid *dataprocessingId*" / "Notify on dataprocessingOid *dataprocessingId*" - depending on whether or not the data processing is complete.

E.g.: {"message": "Stop on dataprocessingOid FF8081817A4264E5017A42D08E3C21EB"}

 Warning: For the job to be executed via API, all tasks must use fixed parameters set to 'From elements list'. Cases where a parameter is set to 'From Parameter' are not supported.

Access to Master Data Management functions via APIs

Introduction

CCH Tagetik makes it possible to run Master Data Management data processing via APIs (RESTful APIs) developed according to the OData V4.0 standard.

STATUS	SWAGGER
ACTIVE	-
FORMAT	ROOT ENDPOINT
REST / OpenAPI	**<api root>/mdm**

Certain conditions must be met to use this consultation method.

Access conditions

To be able to use these specific APIs, the user must be authenticated (Basic Auth, OAuth2).

APIs must also be enabled in the Repository. For reverse proxies, further settings are necessary. They can be used with both *http* and *https* protocols.

For more details on authentication and enabling, see REST API Overview.

API for running the export (DB to File)

Method

POST

URL

```
<api-root>/mdm/export/v1/routine_code
```

Parameters

routine_code = routine MDM to be exported

***IMPORTANT**: If the indicated Routine MDM is not DB to File, an error message is returned.*

Data returned

In the response, the following ID will be returned in JSON format:

dataprocessingId = data processing ID.

E.g.: { "dataprocessingId":"FF8081817A4264E5017A42D08E3C21EB" }

API for checking export status (DB to File)

Method

GET

URL

```
<api-root>/mdm/export/v1/dataprocessingId
```

Data returned

In the response, the data processing status will be returned in JSON format:

status: RUNNING/FAILED/COMPLETED

Status	Description
RUNNING	Running
FAILED	Failed
COMPLETED	Completed successfully

E.g.: {"requestDateTime": "2021-06-25T10:56:42.850+0000", "status": "COMPLETED"}

fileName*_[routine_name]*: name of the produced file

E.g.: {"requestDateTime": "2021-06-25T10:56:42.850+0000", "status": "COMPLETED", "parameters": {"fileName_001": "tcpm_mdmTables_USER_DB_xxx_part_1_of_1.zip"}}

API for stopping the export (DB to File)

Method

DELETE

URL

```
<api-root>/mdm/export/v1/dataprocessingId
```

Data returned

In the response, the following ID will be returned in JSON format:

message = Stop on dataprocessingOid *dataprocessingId*" / "Notify on dataprocessingOid *dataprocessingId*"

Message	Description
Stop on dataprocessingOid	Requests the interruption of data processing still running
Notify on dataprocessingOid	Requests the interruption of data processing that has already ended. The system responds with the end of processing message.

E.g.: {"message": "Stop on dataprocessingOid FF8081817A4264E5017A42D08E3C21EB"}

API for running the import (File to DB)

Method

POST

URL

```
<api-root>/mdm/import/v1/routine_code
```

Parameters

routine_code = routine MDM to be imported

***IMPORTANT**: If the indicated Routine MDM is not File to DB, an error message is returned.*

Property

The following properties can be included in JSON format in the body of the request.

Property	Description	Options
filename	Mandatory. Files to import.	-
endpoint	Endpoint to be used if the file is located in a remote directory. See "Endpoint page" on page 75.	-
dir	Directory in which the files are located	If an endpoint is specified, it allows the sub-directory to be specified based on the endpoint's root. If an endpoint has not been specified, the following can be specified: • empty field = directory configured as a CCH Tagetik writable folder • relative path = writable folder sub-directory • absolute path = generic directory indicated in the Allowed Directories

Examples

{ "filename":"export.zip", "endpoint":"", "dir":""} - import of the "export.zip" file from the *Writable folder*

{ "filename":"export.zip", "endpoint":"", "dir":"tmp"} - import of the "export.zip" file from the "tmp" sub-directory of the *Writable folder*

{ "filename":"export.zip", "endpoint":"", "dir":"C:\impexp"} - import of the "export.zip" file from the "C:\impexp" directory

{ "filename":"export.zip", "endpoint":"SFTP", "dir":""} - import of the "export.zip" file via the SFTP endpoint

{ "filename":"export.zip", "endpoint":"SFTP", "dir":"tmp"} - import of the "export.zip" file present in the "tmp" sub-directory of the SFTP endpoint

Data returned

In the response, the following ID will be returned in JSON format:

dataprocessingId: data processing ID.

E.g.: { "dataprocessingId":"FF8081817A4264E5017A42D08E3C21EB" }

API for checking import status (File to DB)

Method

GET

URL

```
<api-root>/mdm/import/v1/dataprocessingId
```

Data returned

In the response, the data processing status will be returned in JSON format:

status: RUNNING/FAILED/COMPLETED

Status	Description
RUNNING	Running
FAILED	Failed
COMPLETED	Completed successfully

E.g.: {"requestDateTime": "2021-06-25T10:56:42.850+0000", "status": "COMPLETED"}

API for stopping the import (File to DB)

Method

DELETE

URL

```
<api-root>/mdm/import/v1/dataprocessingId
```

Data returned

In the response, the following ID will be returned in JSON format:

message = Stop on dataprocessingOid *dataprocessingId*" / "Notify on dataprocessingOid *dataprocessingId*"

Message	Description
Stop on dataprocessingOid	Requests the interruption of data processing still running
Notify on dataprocessingOid	Requests the interruption of data processing that has already ended. The system responds with the end of processing message.

E.g.: {"message": "Stop on dataprocessingOid FF8081817A4264E5017A42D08E3C21EB"}

API for running the import (DB to DB)

Method

POST

URL

```
<api-root>/mdm/import/v1/routine_code
```

Parameters

routine_code = routine MDM to be imported

IMPORTANT: *If the indicated Routine MDM is not DB to DB, an error message is returned.*

Data returned

In the response, the following ID will be returned in JSON format:

dataprocessingId = data processing ID

E.g.: { "dataprocessingId":"FF8081817A4264E5017A42D08E3C21EB" }

API for checking import status (DB to DB)

Method

GET

URL

```
<api-root>/mdm/import/v1/dataprocessingId
```

Data returned

In the response, the data processing status will be returned in JSON format:

status: RUNNING/FAILED/COMPLETED

Status	Description
RUNNING	Running
FAILED	Failed
COMPLETED	Completed successfully

E.g.: {"requestDateTime": "2021-06-25T10:56:42.850+0000", "status": "COMPLETED"}

API for stopping the import (DB to DB)

Method

DELETE

URL

```
<api-root>/mdm/import/v1/dataprocessingId
```

Data returned

In the response, the following ID will be returned in JSON format:

message = Stop on dataprocessingOid *dataprocessingId*" / "Notify on dataprocessingOid *dataprocessingId*"

Message	Description
Stop on dataprocessingOid	Requests the interruption of data processing still running
Notify on dataprocessingOid	Requests the interruption of data processing that has already ended. The system responds with the end of processing message.

E.g.: {"message": "Stop on dataprocessingOid FF8081817A4264E5017A42D08E3C21EB"}

User provisioning SCIM API

Tagetik exposes user provisioning REST API following the System for Cross-domain Identity Management (SCIM) 2.0 API. SCIM specification is designed to make managing user identities in cloud-based applications and services easier.

In this manual, we will describe what part of SCIM API has been implemented following the specification detailed in the RFC 7644.

STATUS	SWAGGER
ACTIVE	-
FORMAT	ROOT ENDPOINT
SCIM v2	**<api root>/scim/v2**

Authorisation

Users accessing SCIM APIs should also have access to the application repository database as an Admin. The default $SYSADMIN role or any roles derived from this one will satisfy this requirement.

User endpoints

This endpoint allows users to be created, modified or deleted and is visible from the path:

```
<api-root>/scim/v2/Users
```

By default, users for whom the value of the FLAG_ELIMINATED ('deleted flag') column within the REP_USER table is equal to DELETED('1'), RECODED('2') or FORGOTTEN('5') are filtered out and cannot be extracted by the SCIM API.

Operations

Operation	Http Method	Body of the request	Path variable	Query string parameters	Body of the answer	HTTP response body	Response Status (SCIM Exception)
Create	POST	User	-	-	User	201 (Created)	405 (Method not permitted) 409 (Conflict)
Read	GET	-	userId	-	User	200 (OK)	404 (Not found)
Delete	DELETE	-	userId	-	-	200 (OK)	404 (Not found)
Replace	PUT	User	userId	-	User	200 (OK)	404 (Not found)
Search	GET	-	-	filter sortBy sortOrder startIndex count (all parameters are optional)	User List	200 (OK)	400 (Invalid request) 405 (Method not permitted)
Update	PATCH	PatchRequest	userId	-	User	200 (OK)	400 (Invalid request) 404 (Not found)

SCIM exceptions

Examples of invocation

HTTP Status	Descriptions
400 (Invalid request)	It occurs when the request does not match the service specification, e.g. by invoking the 'search' method when the filter is not well defined (wrong parameters/syntax errors) or by invoking the 'update' method when the Patch request is invalid.
404 (Not found)	It occurs when the requested resource is non-existent; for example, if an attempt is made to replace, update, search or delete a user but there is no user matching the identification code used.
405 (Method not permitted)	It occurs when the HTTM method is not permitted for the requested endpoint or if the serialised object supplied does not correspond to the object expected by that method. For instance, when an incorrect user object is provided in the body of the request.
409 (Conflict)	This occurs when, for example, a request is made to create a new user using a user ID code already in use in the system.

Supported standard SCIM fields

The following table shows the standard SCIM user fields supported.

SCIM User Field	Supported cardinality (collection only)	Supported Values
userName	-	-
name.givenName	-	-
name.familyName	-	-
emails	only 1	*String *Formula for establishing type work
phoneNumbers	only 1	*String *Formula for establishing type work
meta.lastModified		
password		
userType		L(Local)/E(External)
groups		
preferredLanguage	only 1	*LanguageCaption - CountryCaption
active		True/False

*Note: the preferredLanguage field only supports values in the format LanguageCaption - CountryCaption (example: en-US) and only values managed by the Tagetik application are allowed

If values are used for the preferredLanguage field that do not comply with the required format or are not managed by the application during creation or update, a Status Code '400 Bad request' is sent with an error message 'Invalid Language Code' or 'Language Code Not Managed'.

Custom fields

The following table shows the custom user fields of CCH Tagetik.

SCIM User Field	Description	Supported Values
userReferences	Indicates the existing user from whom the new user will inherit roles	-
locked	user lock status	True/False (immutable, equal to the opposite of the 'active' field)
mustChangePwdAtNextLogin	Forces user to change password at next login	True/False
interactionType	Client interaction type	0 (interactive)/1 (Machine to Machine)
lastLogin	Last access in application	-
validityStartDate	Validity start date	**
validityEndDate	Validity end date	**
descriptionLanguage	Descriptions Language	-
userRoles	The list of databases the user can access and the corresponding role	-
dbAllowed	List of databases associated with the user, with ID, description and lock status	-

**Note: when creating or updating, it is possible to specify the time for the fields validityStartDate and validityEndDate. Nevertheless, any time entered is ignored and replaced with '00:00:00' for both fields.

SCIM Enterprise User Schema Extension fields

The following table shows the supported Enterprise scheme user fields.

SCIM User Field	Description	Supported Values
department	Specifies the user's department	-

Examples

Show list of all users	
Method	GET
URL	<api-root>/scim/v2/Users
Body	-
Note	

Get single user	
Method	GET
URL	<api-root>/scim/v2/Users/MYUSER
Body	-
Example answer	{ "schemas": ["urn:ietf:params:scim:schemas:core:2.0:User", "urn:tgk:params:scim:schemas:extension:2.0:User", "urn:ietf:params:scim:schemas:extension:enterprise:2.0:User"], "id": "MYUSER", "externalId": "MYUSER", 'meta': { "resourceType': 'User', "lastModified": 1700149220782 }, "userName": "MYUSER", "name': { "familyName": "My", "givenName": "User". }, "userType': 'E', "preferredLanguage': 'en-IT', "active": true, "emails": [{ "value": "myuser@example.com" }], "partiallyBlocked': false, "urn:tgk:params:scim:schemas:extension:2.0:User": { "userReferences": ["REF_USER_CODE"], "locked": false, "mustChangePwdAtNextLogin": false, "interactionType": "INTERACTIVE", "lastLogin": "2024-01-01T10:10:10.000Z", "validityStartDate": "2020-01-01T00:00:00Z", "validityEndDate": "2030-01-01T00:00:00Z", "descriptionLanguage": "es", "userRoles": [{ "dbId": "[!]", "role": "$ALL" }, { "dbId": "TCPM_DEMO_DEV", "role": "$ALL" }], "dbAllowed": [{"id": "TCPM_DEMO_DEV", "description": "TCPM_DEMO_DEV", "locked": false}] }, "urn:ietf:params:scim:schemas:extension:enterprise:2.0:User": { "department": "User Department" }
Note	With this request we obtain the attributes of an existing user MYUSER with the preferredLanguage 'en-IT'.

Create new user	
Method	POST
URL	<api-root>/scim/v2/Users
Body	{ "schemas": ["urn:ietf:params:scim:schemas:core:2.0:User"], "externalId": "MYUSER", "userName": "MYUSER", "userType': 'E', "active": true, "preferredLanguage': 'en-IT', "emails":[{ "value":"myuser@example.com", "type":"work", "primary": true }], "name": { "familyName": "User", "givenName": "My" }, "urn:tgk:params:scim:schemas:extension:2.0:User": { "userReferences": ["REF_USER_CODE"], "descriptionLanguage": "it" } }
Note	This request adds an external user MYUSER with email myuser@example.com and with preferredLanguage 'en-IT', and sets the reference user to 'REF_USER_CODE'.

Apply changes to existing users	
Method	PATCH
URL	`<api-root>/scim/v2/Users/MYUSER`
Body	`{ "schemas":["urn:ietf:params:scim:api:messages:2.0:PatchOp", "urn:tgk:params:scim:schemas:extension:2.0"], "Operations":[{ "op":"add", "path":"urn:tgk:params:scim:schemas:extension:2.0:User:userReferences", "value":["REF_USER_CODE"]}] }`
Note	This request adds the reference user 'REF_USER_CODE' to the existing user 'MYUSER'. To replace the existing reference user, use " `"op":"replace"` instead of `"op":"add"`.

Apply changes to user roles	
Method	PATCH
URL	`<api-root>/scim/v2/Users/MYUSER`
Body	`{ "schemas":["urn:ietf:params:scim:api:messages:2.0:PatchOp", "urn:tgk:params:scim:schemas:extension:2.0"], "Operations":[{"op":"add", "path":"urn:tgk:params:scim:schemas:extension:2.0:User:userRoles", "value":[{"dbId":"DB_ID", "role":"COD_RUOLO"}]}] }`
Note	This request adds the role 'COD_RUOLO' to the existing user 'MYUSER' for the database 'DB_ID'. To replace the role, use ' `"op":"replace"` instead of `"op":"add"`. Please note that, starting from SP34, for the add and replace operations involving the user–application DB association, the "Application DB Restrictions" field will be set to "Restricted" by default, in order to ensure the same behaviour as the CCH Tagetik web interface.

Assign type 'work' to email	
Method	PATCH
URL	`<api-root>/scim/v2/Users/MYUSER`

Body	{
	
	"schemas":["urn:ietf:params:scim:api:messages:2.0:PatchOp",
	"urn:tgk:params:scim:schemas:extension:2.0",
	"urn:ietf:params:scim:schemas:extensions:enterprise:2.0:User"],
	"Operations":[{"op":"add", "path":"emails[type eq \"work\"].value", "value":"test@test.it"}]}]
	
	}

Note	This request modifies the 'email' field to set its type to 'work'.

Delete user	
Method	DELETE
URL	`<api-root>/scim/v2/Users/MYUSER`
Body	-
Note	This request deletes the MYUSER user.

Schemas endpoint

This endpoint allows you to retrieve the SCIM JSON schemas used by Tagetik user provisioning functionalities and it is published at the following path:

`<api-root>/scim/v2/Schemas`

Operations

The endpoint provides the following operations:

Operation	Http Method	Path Variable	Response Body	HTTP Response Status	Response Status (SCIM Exception)
Lists of all supported schemas	GET	-	Schema List	200 (OK)	-
Get schema by urn	GET	-	Scheme	200 (OK)	404 (Not Found)

SCIM Exceptions

Invocation examples

HTTP Status	Descriptions
404 (Not Found)	It occurs when trying to read a schema, but no schema with the specified URN is available

Examples

Read a list of all supported schemas	
Method	GET
URL	`<api-root>/scim/v2/Schemas`
Body	-
Notes	

Read a specific schema	
Method	GET
URL	`<api-root>/scim/v2/Schemas/urn:tgk:params:scim:schemas:extension:2.0:User`
Body	-
Notes	Retrieves the "User" schema extension provided by CCH Tagetik.

Bulk endpoint

This endpoint allows you to create, edit or delete multiple users within a single bulk request and it is exposed in the path:

`<api-root>/scim/v2/Bulk`

Operations

Operation	Http Method	Request Body	Path Variable	Query String Parameters	Response Body	HTTP Response Status	Response Status (SCIM Exception)
Bulk	POST	BulkRequest	-	-	BulkResponse	200 (OK)	400 (Bad Request)

SCIM Exceptions

Invocation examples

HTTP Status	Descriptions
400 (Bad Request)	It occurs when the bulk request does not match the service specification, for example invoking it providing the same bulk id to more than one operation inside the bulk request or missing some required attribute.

Supported BulkRequest fields

The supported standard SCIM User fields are listed in the following table.

SCIM User field	Supported cardinality (for collection only)	Supported values
Operations	1..*	-

Operations fields

The supported standard SCIM Operations fields are listed in the following table. Not all the SCIM Operations standard fields are currently supported by CCH Tagetik.

SCIM User field	Supported cardinality (for collection only)	Supported values
method	1..*	-
bulkId	-	-
path	-	-
data	-	-
location	-	-
response	-	-
status	-	-

Examples

List all users	
Method	POST
URL	`<api-root>/scim/v2/Bulk`

List all users

Body

```
{
"schemas": ["urn:ietf:params:scim:api:messages:2.0:BulkRequest"],
"Operations": [{
    "method": "POST",
    "path": "/Users",
    "bulkId": "1",
    "data": {
        "schemas": ["urn:ietf:params:scim:schemas:core:2.0:User"],
        "externalId": "testUser1",
        "userName": "testUser1",
        "name": {
            "familyName": "testUser1",
            "givenName": "testUser1"
        },
        "urn:tgk:params:scim:schemas:extension:2.0:User": {
            "userReferences": ["TAGETIK"]
        }
    }
},{
    "method": "POST",
    "path": "/Users",
    "bulkId": "2",
    "data": {
        "schemas": ["urn:ietf:params:scim:schemas:core:2.0:User"],
        "externalId": "testUser2",
        "userName": "testUser2",
```

List all users

```
            "name": {
                "familyName": "testUser1",
                "givenName": "testUser1"
            },
            "urn:tgk:params:scim:schemas:extension:2.0:User": {
                "userReferences": ["TAGETIK"]
            }
        }
    },{
        "method": "DELETE",
        "path": "/Users/testUser3"
    }]}
```

List all users

Sample Response

```
{
    "schemas": ["urn:ietf:params:scim:api:messages:2.0:BulkResponse"],
    "Operations": [
        {
            "method": "POST",
            "bulkId": "1",
            "location": "/Users/testUser1",
            "status": "201"
        },{
            "method": "POST",
            "bulkId": "2",
            "location": "/Users/testUser2",
            "status": "201"
        },{
            "method": "DELETE",
            "location": "/Users/testUser3",
            "status": "200"
        }
    ]
}
```

Notes This call combines two user creations (testUser1, testUser2) and one deletion (testUser3).

Configuration and Reporting languages - API

Introduction

STATUS	SWAGGER
ACTIVE	–

FORMAT	ROOT ENDPOINT
REST / OpenAPI	**<api root>/settings/**

CCH Tagetik allows to receive Configuration and Reporting languages via APIs (RESTful APIs).

Certain conditions are required to use this consultation mode.

Access conditions

To be able to use these specific APIs, the user must be authenticated with the authentication mode chosen for CCH Tagetik (Authentication - Cloud).

APIs must also be enabled in the Repository.

API for reading languages

Method

GET

URL

```
http://server:port/tagetikcpm/api/settings/languages/v1
```

Data returned

The response, in JSON format, returns the list, in alphabetical order, of the languages (FIELD_STRINGA) contained in the REP_CONFIGURATION table.

status: 200 OK

Ex:{"applicationLanguages": ["aa","ar","de","en","es","it","zh","zu"]}

Database Lock API

Tagetik exposes REST APIs to orchestrate database maintenance. Through administrative APIs it's possible to either lock/unlock appication databases (endpoint) or the entire application, eg. for overall or repository maintenance activities.

Functionalities supported by such APIs are described in "Database list page" on page 11.

STATUS	SWAGGER
ACTIVE	**<api root>/admintools/swagger-ui/index.html**
FORMAT	ROOT ENDPOINT
Open API	**<api root>/admintools**

Authorisation

Users accessing SCIM APIs should also have access to the application repository database as an Admin. The default $SYSADMIN role or any roles derived from this one will satisfy this requirement.

Access to Endpoint functions via API

Introduction

CCH Tagetik also allows endpoints to be managed via APIs (RESTful APIs).

Certain conditions are required to use this consultation mode.

Access conditions

To be able to use these specific APIs, the user must be authenticated (Basic Auth, OAuth2).

APIs must also be enabled in the Repository. For reverse proxies, further settings are necessary. They can be used with both *http* and *https* protocols.

For more details on authentication and enabling, see REST API Overview.

API to read default Endpoints

Method

GET

URL

```
http://server:port/tagetikcpm/api/admintools/endpoint/v1/default
```

API to read all Endpoints

Method

GET

URL

```
http://server:port/tagetikcpm/api/admintools/import/v1/elements
```

API to read all Endpoints filtered by code

Method

GET

URL

```
http://server:port/tagetikcpm/api/admintools/import/v1/elements?filter=code_filter
```

API to read a single Endpoint

Method

GET

URL

```
http://server:port/tagetikcpm/api/admintools/import/v1/elements/endpoint_code
```

API to create a new Endpoint

Method

POST

URL

```
http://server:port/tagetikcpm/api/admintools/import/v1/elements
```

Property

In the body of the request it is possible to include the following properties in JSON format:

Property	Description	Options
code	Mandatory.	-
description	Mandatory.	-

Property	Description	Options
endpointType	Mandatory.	It can take the following values: • U: URL • A: Tagetik Data Loader • P: Application • O: OData V4 • 2: OData V2 • R: REST API • S: SharePoint • B: Google Big Query • F: Snowflake

endpointType properties

Moreover, depending on the endpoint type it is possible to add the following properties:

URL: properties	Description	Options
url	Mandatory.	-
defaultEndpoint	-	true/false (default = false)
authenticationType	-	It can take the following values: • N: None (default) • B: Basic
basicUsername	Mandatory if authenticationType = B.	-
basicPassword	Defined if authenticationType = B.	-
pgpEnabled	-	true/false (default = false)
pgpPublicKey	Mandatory if pgpEnabled = true	-
pgpPrivateKey	Mandatory if pgpEnabled = true	-
pgpPassphrase	mandatory if pgpEnabled = true	-

Agent: properties	Description	Options
tdlAgentId	Mandatory.	-
tdlSourceType	Mandatory.	It can take the following values: • P: Path • D: Datasource
tdlPath	Mandatory if tdlSourceType = P	-
tdlDatasource	Mandatory if tdlSourceType = D	-
pgpEnabled	-	true/false (default = false)
pgpPublicKey	Mandatory if pgpEnabled = true	-
pgpPrivateKey	Mandatory if pgpEnabled = true	-
pgpPassphrase	Mandatory if pgpEnabled = true	-

Application: properties	Description	Options
url	Mandatory.	-
applicationDatasource	Mandatory.	-

Odata (V4, V2): properties	Description	Options
url	Mandatory.	-
odataPaginationSize	Mandatory.	>= 0 (default = 0)
authenticationType	Mandatory.	It can take the following values: • N: None (default) • B: Basic • O: OAuth2.0
basicUsername	Mandatory if authenticationType = B	-
basicPassword	Defined if authenticationType = B	-

Odata (V4, V2): properties	Description	Options
oauth2GrantType	Defined if authenticationType = O	It can take the following values: • CLIENT (default) • PASSWORD
oauth2TokenEndpoint	Mandatory if authenticationType = O	-
oauth2ClientId	Mandatory if authenticationType = O	-
oauth2ClientSecret	Mandatory if authenticationType = O	-
oauth2Username	Mandatory if authenticationType = O e oauth2GrantType = PASSWORD	-
oauth2Password	Mandatory if authenticationType = O e oauth2GrantType = PASSWORD	-
oauth2Scopes	Defined if authenticationType = O	-
oauth2Resources	Defined if authenticationType = O	-
oauth2ClientAuthentication	Defined if authenticationType = O	It can take the following values: • HEADER (default) • BODY • JWT
oauth2Kid	The Header parameter 'kid' (key ID) is a hint as to which key was used to protect JWS. Defined if authenticationType = O and oauth2GrantType = CLIENT and oauth2ClientAuthentication = JWT	-
oauth2Lifetime	The duration is the time in seconds added to the iat (issue at time) to create the JWT expiration time at or after which the JWT MUST NOT be accepted for data processing). Defined if authenticationType = O and oauth2GrantType = CLIENT and oauth2ClientAuthentication = JWT	>= 1 and <= 3600 (default = 300)
jwtCertificateFile	Generated private key code in PEM format. Mandatory if authenticationType = O and oauth2GrantType = CLIENT and oauth2ClientAuthentication = JWT	-
oauth2UseProxy	-	true/false (default = false)
oauth2UseCertificates	-	true/false (default = false)
proxyServer	-	-
proxyServerPort	-	-
proxyUsername	-	-
proxyPassword	-	-
customHeaders	-	-

REST API: properties	Description	Options
url	Mandatory.	-
restResponseType	Mandatory.	It can take the following values: • JSON (default) • CSV
restPath	Defined if restResponseType = JSON	-
restJsonArrayLevel	-	>= 0 (default 0)
restCsvHeader	Defined if restResponseType = CSV	true/false (default = false)

REST API: properties	Description	Options
restCsvColumnSeparator	Mandatory if restResponseType = CSV	It can take the following values: • "," (default) • ";" • "|" • "\t" • " "
restCsvValueDelimiterMode	Mandatory if restResponseType = CSV	It can take the following values: • NONE (default) • MINIMAL • NON_NUMERIC • ALL
restCsvValueDelimiter	Mandatory if restResponseType = CSV	It can take the following values: • N (default) • " • '
restCsvRowSeparator	Mandatory if restResponseType = CSV	It can take the following values: • CRLF (default) • CR • LF
restPaginationUsage	Mandatory.	It can take the following values: • N: None (default) • H: Header • Q: Query param
restPaginationPropertyName	Mandatory only if restPaginationUsage = H or Q	-
restPaginationSize	Mandatory only if restPaginationUsage = H o Q	Must be > 0
restNextPageResponsePosition	Mandatory.	It can take the following values: • N: None (default) • H: Header • B: Body attribute (only if restResponseType = JSON)
restNextPageResponsePropertyName	Mandatory only if restPaginationUsage = H or B	-
restNextPageRequestPosition	Mandatory.	It can take the following values: • N: None (default) • H: Header • Q: Query param • U: Complete URL
restNextPageRequestPropertyName	Mandatory only if restPaginationUsage = H o Q	-
authenticationType	Mandatory.	It can take the following values: • N: None (default) • B: Basic • O: OAuth2.0
basicUsername	Mandatory if authenticationType = B.	-

REST API: properties	Description	Options
basicPassword	Defined if authenticationType = B	-
oauth2GrantType	Defined if authenticationType = O	It can take the following values: • CLIENT (default) • PASSWORD
oauth2TokenEndpoint	Mandatory if authenticationType = O	-
oauth2ClientId	Mandatory if authenticationType = O	-
oauth2ClientSecret	Mandatory if authenticationType = O	-
oauth2Username	Mandatory if authenticationType = O e oauth2GrantType = PASSWORD	-
oauth2Password	Mandatory if authenticationType = O e oauth2GrantType = PASSWORD	-
oauth2Scopes	Defined if authenticationType = O	-
oauth2Resources	Defined if authenticationType = O	-
oauth2ClientAuthentication	Defined if authenticationType = O	It can take the following values: • HEADER (default) • BODY • JWT
oauth2Kid	The Header parameter 'kid' (key ID) is a hint as to which key was used to protect JWS. Defined if authenticationType = O and oauth2GrantType = CLIENT and oauth2ClientAuthentication = JWT	-
oauth2Lifetime	The duration is the time in seconds added to the iat (issue at time) to create the JWT expiration time at or after which the JWT MUST NOT be accepted for data processing). Defined if authenticationType = O and oauth2GrantType = CLIENT and oauth2ClientAuthentication = JWT	>= 1 and <= 3600 (default = 300)
jwtCertificateFile	Generated private key code in PEM format. Mandatory if authenticationType = O and oauth2GrantType = CLIENT and oauth2ClientAuthentication = JWT	-
oauth2UseProxy	-	true/false (default = false)
oauth2UseCertificates	-	true/false (default = false)
proxyServer	-	-
proxyServerPort	-	-
proxyUsername	-	-
proxyPassword	-	-
customHeaders	-	-

The certificates cannot be configured via API

Sharepoint: properties	Description	Options
sharepointTenantId	**Mandatory**	-
sharepointClientId	**Mandatory**	-
sharepointSiteName	**Mandatory**	-
sharepointScope	**-**	-
sharepointAuthenticationType	**Mandatory**	It can take the following values: • S: Secret Note: the field can take the read-only value C (Certificate)
sharepointSecretValue	**Mandatory**	-

MS Fabric: properties	Description	Options
fabricSqlEndpoint	**Mandatory**	SQL Warehouse Endpoint
fabricWarehouse	**Mandatory**	Warehouse Name
fabricTenantId	**Mandatory**	Tenant Id
fabricClientId	**Mandatory**	Apllication Azure client id
fabricSecret	**Mandatory**	Secret generated for Application Azure.

 Certificate authentication is not managed via API.

 Creation, replacement, update, and deletion operations for *Google BigQuery* and *Snowflake* endpoints are not allowed when using the REST API service. Executing these operations will return a '501 – Not Implemented' response.

API to replace an Endpoint

Method

PUT

URL

```
http://server:port/tagetikcpm/api/admintools/import/v1/elements/endpoint_code
```

Property

In the body of the request, in JSON format, it is possible to include the same properties as in the creation according to the type of endpoint, with the exception of 'code', which is already specified in the url.

For an endpoint with oauth2ClientAuthentication = JWT, this operation will have no effect (400 - Bad request: The following endpoint cannot be replaced: <ENDPOINT_CODE>)

API to update an Endpoint

Method

PATCH

URL

```
http://server:port/tagetikcpm/api/admintools/import/v1/elements/endpoint_code
```

Property

In the body of the request, in JSON format, it is possible to include, as properties to be updated, the same as the creation according to the type of endpoint, with the exception of the following fields which cannot be changed:

- code
- endpointType
- tdlAgentId
- tdlSourceType
- sharepointAuthenticationType

Other non-editable fields belong to the case of an endpoint modification with oauth2ClientAuthentication = JWT:

- authenticationType
- oauth2GrantType
- oauth2ClientAuthentication
- jwtCertificateFile

API to delete an Endpoint

Method

DELETE

URL

```
http://server:port/tagetikcpm/api/admintools/import/v1/elements/endpoint_code
```

Access to the database import and export functions via APIs

Introduction

STATUS	SWAGGER
ACTIVE	-
FORMAT	ROOT ENDPOINT
REST/OpenAPI	**<api root>/admintools/**

CCH Tagetik CCH Tagetik makes it possible to run database import and export functions via APIs (RESTful APIs) developed according to the OData v.4.0 standard.

Certain conditions are required to use this consultation mode.

Access conditions

To be able to use these specific APIs, the user must be authenticated (Basic Auth, OAuth2).

APIs must also be enabled in the Repository. For reverse proxies, further settings are necessary. They can be used with both *http* and *https* protocols.

For more details on authentication and enabling, see REST API Overview.

API for running the export

Method

POST

URL

```
http://server:port/tagetikcpm/api/admintools/export/v1/database_id
```

Parameters

database_id = ID of db to export

***Note**: to export the Repository, database_id = REPOSITORY*

Property

In the body of the request it is possible to include the following properties in JSON format:

zipMaxSize = maximum size in MB of files to be produced (default = 2048)

E.g.: { "zipMaxSize":"2048" }

Returned data

The response returns the following ID in JSON format:

dataprocessingId = data processing ID.

E.g.: { "dataprocessingId":"FF8081817A4264E5017A42D08E3C21EB" }

API for checking export status

Method

GET

URL

```
http://server:port/tagetikcpm/api/admintools/export/v1/dataprocessingId
```

Returned data

The response returns the data processing status in JSON format:

status: RUNNING/FAILED/COMPLETED

Status	Description
RUNNING	Running
FAILED	Failed
COMPLETED	Successfully completed

E.g.: {"requestDateTime": "2021-06-25T10:56:42.850+0000", "status": "COMPLETED"}

impexp.zipFiles.produced: names of produced files

Es: {"requestDateTime": "2021-06-25T10:56:42.850+0000", "status": "COMPLETED", "parameters": {"impexp.zipFiles.produced": "tcpm_USER_DB_xxx_part_1_of_1.zip"}}

API for stopping the export

Method

DELETE

URL

```
http://server:port/tagetikcpm/api/admintools/export/v1/dataprocessingId
```

Returned data

The response returns the following ID in JSON format:

message = Stop on dataprocessingOid *dataprocessingId*" / "Notify on dataprocessingOid *dataprocessingId*"

Message	Description
Stop on dataprocessingOid	Data Processing was stopped
Notify on dataprocessingOid	Data processing in progress

E.g.: {"message": "Stop on dataprocessingOid FF8081817A4264E5017A42D08E3C21EB"}

API for running the import

Method

POST

URL

```
http://server:port/tagetikcpm/api/admintools/import/v1/database_id
```

Parameters

database_id = ID of database to which data is exported

***Note**: to import the Repository, database_id = REPOSITORY*

Property

In the body of the request it is possible to include the following properties in JSON format:

Property	Description	Options
filename	Mandatory. File to import.	It is possible to specify: • an individual file name • a list of file names (separator = **;**) • all files with **.zip** extension (filename = *****)
endpoint	Endpoint to use when the file is located in a remote directory See "Endpoint page" on page 75.	-

Property	Description	Options
dir	Directory on which files reside	If an endpoint is specified, it allows a sub directory to be specified based on the root of the endpoint. If no endpoint has been specified, it allows the following to be specified: • empty field = directory set up as writable folder of CCH Tagetik • relative path = sub-directory of the writable folder • absolute path = generic directory
emptyDb	Indicates whether the data in the target database must be completely deleted before importing.	true/false (default = false)
deletePartitions	Indicate whether all present AIH partitions (tables with prefix AW_) are to be deleted. Not to be used if the target database is the Repository.	true/false (default = true)

Note: The dynamic:Multi File syntax can be used for the *filename* parameter. In this case, since the files to be processed are ZIP archives, the syntax will not evaluate the content of the archives, but only the ZIP files themselves as entities

Examples

{ "filename":"export.awex", "endpoint":"", "dir":"","emptyDb":"false","deletePartitions":"true"} - import of the "export.awex" file from the *Writable folder*

{ "filename":"export.awex", "endpoint":"", "dir":"tmp","emptyDb":"false","deletePartitions":"true"} - import of the "export.awex" file from the "tmp" della *Writable folder* subdirectory

{ "filename":"export.awex", "endpoint":"", "dir":"C:\impexp","emptyDb":"false","deletePartitions":"true"} - import of the "export.awex" file from the "C:\impexp" directory

{ "filename":"export.awex", "endpoint":"SFTP", "dir":"","emptyDb":"false","deletePartitions":"true"} - import of the "export.awex" file through the SFTP endpoint

{ "filename":"export.awex", "endpoint":"SFTP", "dir":"tmp","emptyDb":"false","deletePartitions":"true"} - import of the "export.awex" file lpresent in the "tmp" subdirectory of the SFTP endpoint

{ "filename":"export_1_of_2.awex;export_2_of_2.awex", "endpoint":"SFTP", "dir":"","emptyDb":"false","deletePartitions":"true"} - import of the "export_1_of_2.awex" and "export_2_of_2.awex" files present on the SFTP endpoint

{ "filename":"*", "endpoint":"SFTP", "dir":"","emptyDb":"false","deletePartitions":"true"} - import of all files with the ".awex" extension present on the SFTP endpoint

Returned data

The response returns the following ID in JSON format:

dataprocessingId = data processing ID.

E.g.: { "dataprocessingId":"FF8081817A4264E5017A42D08E3C21EB" }

API for checking import status

Method

GET

URL

```
http://server:port/tagetikcpm/api/admintools/import/v1/dataprocessingId
```

Returned data

The response returns the data processing status in JSON format:

status: RUNNING/FAILED/COMPLETED

Status	Description
RUNNING	Running
FAILED	Failed
COMPLETED	Successfully completed

E.g.: {"requestDateTime": "2021-06-25T10:56:42.850+0000", "status": "COMPLETED"}

API for stopping the import

Method

DELETE

URL

```
http://server:port/tagetikcpm/api/admintools/import/v1/dataprocessingId
```

Returned data

The response returns the following ID in JSON format:

message = Stop on dataprocessingOid *dataprocessingId*" / "Notify on dataprocessingOid *dataprocessingId*"

Message	Description
Stop on dataprocessingOid	Data Processing was stopped
Notify on dataprocessingOid	Data processing in progress

E.g.: {"message": "Stop on dataprocessingOid FF8081817A4264E5017A42D08E3C21EB"}

User Rights

What they are for

User rights allow the configuration of security criteria by controlling the elements and functionalities the user can access in each application database.
The configuration on the Data Model determines which elements of a dimension can be displayed or edited by the user.

***IMPORTANT**: By default, every new user has management rights on all items. Such rights can be controlled through restrictions. Exceptions are those on category and consolidation scenario, which are restricted by default and must be set by the administrator.*

Types of User Rights

For all functionalities the user can access, the type of rights can be specified:

Element	Description
Visibility Rights	allow consultation of data only
Editability Rights	enable the updating (insertion/modification) and viewing of data
Submission rights	allow data submission

For further details, consult the Users section.

User Rights

Admin

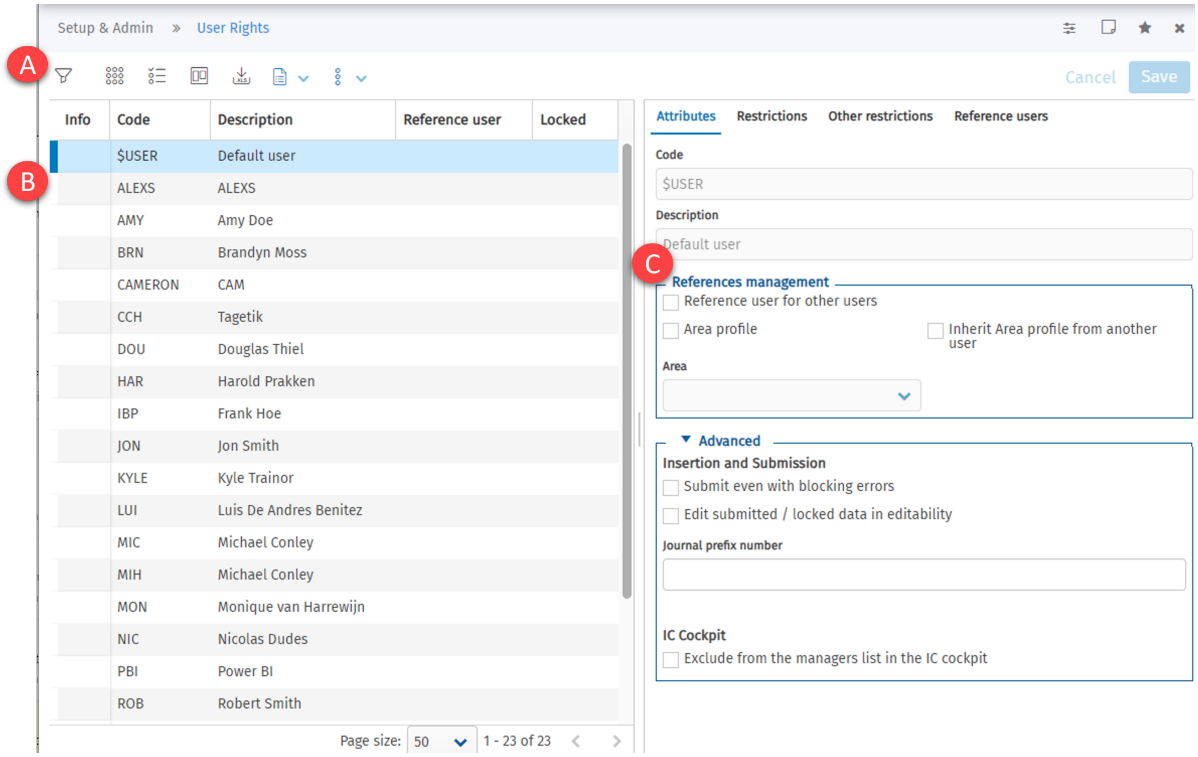
application database > User Rights

Purpose of the page

This page allows you to configure user rights related to the relevant environment.

***IMPORTANT**: By default, each new user has management rights over all existing elements (except category and consolidation scenario). These rights may be controlled through restrictions.*

What it looks like



Element	Description
A	Tool bar with action buttons on the page
B	List of defined users associated with individual databases
C	Configuration tab

Buttons on the page

Element	Description
	This filters the data displayed on the page.
	Activates/de-activates the edit function directly in the table and hides the configuration tabs.
	Enables multi-row selection to allow mass changes.
	Selects the columns to be shown/hidden in the table and those to be locked when scrolling the table.
	Exports data in Excel format. This file cannot be re-imported into CCH Tagetik.
	Shows the launch window of the User Rights Report.
	• **Copy user restrictions and settings from another user**: copies selected restrictions and settings associated with another existing user. • **Refresh users**: copies the Repository users who are authorised to access the application database where the utility is launched. The data processing deletes users no longer present in the Repository and prevents access to users **temporarily blocked** on the Repository • **Copy Entity Rights**: clones the restrictions for the selected user from one Entity hierarchy to another.

User Rights Report

Application Database > User Rights >  *> User Rights Report*

Purpose of window

This window allows you to generate analysis reports on user rights.

Fields in window

Section	Description
Users	In this section, you can specify the user list and the restriction type to be exported: • **All**: includes all users in the environment allows the export of all users in the elements list. If disabled, you must manually populate the list with the relevant users. • **Entity rights and Tasks** : indicates that the report shall include Entity rights and enables the **Hierarchy** section to open the page for selecting Entity hierarchies • **Restrictions on dimensions:** tells the report to export restrictions on system dimensions, custom dimensions, workflow step and analytical dimensions • **Other restrictions**: instructs the report to export and enable restrictions on forms, Collaborative Office documents, ETL domains, ETL routines, ETL jobs, closing routines, stored procedures, analytical workspace, and analytical workspace datasets
Report Type	Allows you to choose between **Analytical** or **Synthetic** reports. The analytical report lists the deployed restrictions for all their elements

User Rights: Attribute Setup

application database > User Rights> Attributes tab

Attributes tab

Please find below the most important fields.

Field	Description
Code	User's unique ID
Description	User description In multilingual projects, a description field is shown for each language enabled in the environment.
References management	
Reference user for other users	User available as a reference model for defining other users' restrictions.
Area Profile	The selected user is a dummy user with an area profile function. This user will be used by the administrator to manage other users by grouping them (by area), and allows users to inherit the restrictions associated with that profile. **Note**: *The administrator can set the **maximum number of users belonging to an area** from the repository on the **Area Profile** user : if this number is zero, the number of users that can belong to an area is unlimited* (see User Settings)
Inherit Area profile from another user	The user is associated with an area profile which limits their authorisations.

Field	Description
Area	User group. If the user represents an area profile, this field contains the user code of that user.
Advanced: Insertion and Submission	
Submit even with blocking errors	The user can submit data even in the presence of blocking errors in the diagnostics.
Edit submitted / locked data in editability	The user can change any submitted/locked data in editability. The option will actually be active only for scenarios/periods where the option with the same name is enabled.
Journal prefix number	To be used when generating the adjustment number for manual entries, if the adjustment prefix generation mode is set to **Equal to user journal number prefix** or **Equal to the concatenation of the user's journal number prefix and Entity/node'.**
Advanced: IC cockpit	
Exclude from the managers list in the IC cockpit	Excludes the user from the list of Managers in the IC cockpit.

Users rights: Restrictions on dimensions

application database > User Rights > Restrictions tab

Purpose of the page

Restrictions on dimensions allow you to define, for each user, which elements they can view and enter data for.

***IMPORTANT**: Elements are both visible and editable by default. To prevent modifications, you need to assign read-only authorization.*

Restrictions Tab

Please find below the most important fields.

Field	Description
Entity and task	Restricts Entities the user can see. For more information see Entity and Task rights.
Accounts	You can configure restrictions either by individual accounts or by nodes of a hierarchy defined on the account dimension. **Note**: Individual accounts and/or lower-level nodes can be added or deleted.
Custom Dimensions	Restrictions can be defined only if the custom dimension is managed. It can be done by individual elements of the dimension or by nodes of a non-time-dependent hierarchy defined on the custom dimension; you can add or remove individual elements of the custom dimension and/or lower-level nodes to or from the nodes See Custom Dimensions
Categories	You can configure restrictions either by individual categories or by nodes of the category grouping. ***Note**: Individual accounts and/or lower-level nodes can be added or deleted.*
Original Scenarios	You can configure restrictions either by individual scenarios or by nodes of the scenario grouping. ***Note**: Individual accounts and/or lower-level nodes can be added or deleted.*

Field	Description
Consolidation Scenarios	You can configure restrictions either by individual consolidation scenarios or by nodes of the scenario grouping. **Note**: Individual accounts and/or lower-level nodes can be added or deleted. ***IMPORTANT**: It is necessary to specify whether Entities belonging to the consolidation scope must include visibility rights for the user. This option is useful when the structure of the consolidation scope does not match the organisational structures used for data collection.*
WF step	Specifies any restrictions to the workflow steps the user can manage.
Analytical Dimensions	Defining restrictions on analytical dimensions allows to define, for each user, which elements they are allowed to view and enter data for. For more information see Restrictions on Analytical Dimensions.

Restrictions can be set for each dimension with the following values.

Option	Description
Unrestricted	Default value (except for category dimensions and consolidation scenarios): the user manages all elements of the dimension in the elements list.
Restricted	The user only acquires rights to the specified dimension elements, which can be configured via the Define link. The result is displayed in the Definition Result window. ***IMPORTANT**: default value for the Consolidation Scenario. A new user has no editability and visibility rights on any consolidation scenario.*
By Area profile	A user belonging to an area profile inherits as editable/visible all elements associated with that profile. **Note**: *Option not available for analytical dimensions.*
Restricted to the contributor categories	Default value for the category dimension. A new user has editability and visibility rights over all contributor categories.

***IMPORTANT**: When saving, the system automatically deploys the definition of restrictions on dimensions: for each dimension it generates a detailed list of editable and/or visible elements.*
Exceptions are Entity and analytical dimensions for which the deployment has to be launched manually with the appropriate button

User Rights on Entity dimension

application database > User Rights > Restrictions

Entity and task setup

Rights on the Entity dimension are defined by organisational hierarchy. Users may have different rights for different data collection processes affecting the following :

- on **Entities** and **nodes**: they allow the administrator to define, for each user, the Entities/nodes for which data can be viewed and/or entered and submitted.
- on **tasks**: they allow the user, for the Entities they are enabled on, to perform only the assigned tasks (e.g. entering data in a specific form, executing a certain data processing etc.). For more details on Entity-specific workflow, see Business Workflow concepts.

Entity dimensions permission are defined according to the logic that any editable/submittable Entity/node is automatically also visible. So:

- whatever activity a user may perform on an Entity, its visibility is automatically implied.
- whatever activity a user may perform on a node, visibility of the node and all underlying Entities is automatically implied.

***IMPORTANT**: By default, a new user is **restricted** on Entities (i.e., they have no viewing rights on any Entity) and **unrestricted** on tasks. The administrator can later assign specific task-level restrictions within Workflow Models, in accordance with the limitations defined at the Entity level.*

Rights definition

Entity rights can take the following values:

Value	Description
Unrestricted in insertion, submission and display	The user can enter data and/or submit each Entity and node that is part of the current process. They also have visibility rights for all Entities in the elements list. ***Note** With this setting, it is not necessary to specify restrictions for each organizational hierarchy involved in a data collection process.*
Restricted	Default value. The user only has viewing and possibly data entry and submission rights on the Entities and nodes indicated in the *Entity Rights* window, which can be reached via the *Define* link. It is also possible to define submission and/or operational tasks that the user can manage. Additionally, workflows and steps the user is allowed to work on can be defined.
By Area profile	The user's restrictions match those of the set Area Profile.
Restricted in insertion and submission, but not restricted in display	The user has data entry and submission rights only on the Entities and nodes specified in the *Entity rights* window. These restrictions do not affect data visibility.
Restricted in Insertion, Submission or Display by Area profile	The user has data entry and submission rights only on the Entities and nodes specified in the *Entity rights* window. The visibility rights on the user's Entities match those of the Area Profile. *Note It is possible to further restrict the visibility rights of a user who inherits the area profile. In this case, the system will intersect the Entities of the Area profile with those defined in the window accessible by the **Define** link*

Entity and task restrictions setup

application database > User Rights > Restrictions > Entity and Task > Define link

***IMPORTANT**: the link is only active if the user has **restricted** rights*

Purpose of window

In this window, you can configure detailed restrictions on Entities and their corresponding groupings (nodes).

Entity rights

Entity rights can be set on the page, divided into the following tabs:

Option	Description
Entity rights	The following attributes can be specified on each Entity: • **Action** type: include or exclude • **Organization** organization type Entity hierarchy nodes • **Entity** entity code • **Insert data**: enables data entry • **Run submission tasks:** ◦ *Generic list*: Definition is done from the **Generic Task List** tab ◦ *Specific list* ◦ *All* ◦ *None* • ***Run operational tasks***: ◦ *Generic list*: Definition is done from the **Generic Task List** tab ◦ *Specific list* ◦ *All* ◦ *None*
Visibility	Specifies the Entities/nodes on which the user has view-only rights.
Contributor Node Rights and Consolidator Node Rights	Node **type** you can specify nodes and the level up to which the selected tasks should be applied ◦ *The current node* ◦ *The node and all child nodes* ◦ *Lowest children level* ◦ *X-level children* (levels 1 to 9) ◦ *All lowest elements* ◦ *All child nodes* ◦ *The node, all child nodes and all lowest elements* **Run operational tasks**: ◦ *All* users can perform all submission/approval tasks ◦ *Generic list* In this case, the definition is made in the **Generic Task list** tab of the Entity and Task Rights definition window and is valid for all nodes with the generic list option active ◦ *Specific list* in this case the definition is done via the **Define** link that appears next to the selection. This option allows you to restrict the user to specific tasks based on the node and/or the Entity ◦ *None* none of the submission /approval tasks can be performed by the user. **Run Submission tasks** specifies whether the user is enabled to run submission tasks. Possible values are: ◦ *All* users can perform all submission/approval tasks ◦ *Generic list* In this case, the definition is made in the **Generic Task list** tab of the Entity and Task Rights definition window and is valid for all nodes with the generic list option active ◦ *Specific list* in this case the definition is done via the **Define** link that appears next to the selection. This option allows you to restrict the user to specific tasks based on the node and/or the Entity ◦ *None* none of the submission /approval tasks can be performed by the user

***IMPORTANT**: The system processes records in the same order in which they are entered. It is therefore important to pay attention to the sequence of entries and nodes—what is added later could invalidate what was entered earlier*

Task rights: generic list and specific list

In Entity rights, task restrictions define which specific tasks within a business workflow a user is allowed to perform on the entities/nodes they are authorized for.

Task restrictions can be set as follows:

Option	Description
Run Submission tasks - All	The user can perform all submission and approval tasks
Run submission tasks - Generic list	The user can only perform certain submission and approval tasks defined in the **generic task list** tab.
Run submission tasks - Specific list	The user can only perform specific submission and approval tasks defined by the **Define** link, which appears next to the selection and allows the **specific task list** to be populated.  *This option allows you to restrict the user to specific tasks based on the node and/or the item.*
Run submission tasks - None	The user cannot perform any submission and approval tasks.
Run operational tasks - All	The user can perform all operational, i.e. non-submission and approval tasks
Run operational tasks - Generic list	The user can only perform certain operational tasks defined in the **generic task list** tab , which can be configured via the ***Define*** **link.**
Run operational tasks - Specific list	The user can only perform specific operational tasks defined by the **Define** link that appears next to the selection and allows the **specific task list** to be populated.  *This option allows you to restrict the user to specific tasks based on the node and/or the item.*
Run operational tasks - None	The user cannot perform any operational tasks.

Generic list definition methods

Two generic list definition methods are available:

Method	Description
By step	For each available Business Workflow, you can select the step, the entire task workflow, or the individual task the user is allowed to access.
By label	Different tasks, even across the different steps, can be grouped under a label. Thus, It is possible to select a specific label and restrict the user to the tasks associated with it.

Action buttons on this page

Method	Description
Deploy	Populates all tables in the database related to Entity rights as set in the window
Definition results	This window shows the list of configured restrictions. See Definition result

User Rights: Other Restrictions

application database > User Rights > Other Restrictions

Other Restrictions Tab

Please find below the most important fields.

Field	Description
Administration Restrictions	
ETL: Domain setup	The definition can be done: • for individual elements; • by grouping nodes, ETL environments or Job ETL or Routine ETL; individual elements and/or lower-level nodes can be added or removed to nodes
Collaborative Office: setup	Specifies whether the user can view, manage and edit documents in the Collaborative Office module. The definition is by **Document Type**.
Analytical Workspaces	The restriction will affect the list of analtyical workspaces visible to the user. The definition occurs only for individual elements, as there is no aggregation structure for these dimensions. the restrictions on the Analytical Worskpace also affect the Admin Homepage, so the user cannot see the Analytical Worskpace even from the Admin Page **IMPORTANT**: In order to display user-accessible workspaces on the homepage, it is still necessary to add the respective tiles with the edit button on the page .
User Restrictions	
Form	The definition can be done: • by individual forms or • by form grouping nodes; individual forms and/or lower-level nodes can be added or removed to the nodes.
ETL: Routine execution	It is possible to define routines that the user can display and run. The definition can be done: • by individual routines • by routine grouping nodes; individual routines and/or lower-level nodes can be added or removed to the nodes.
ETL: Run Job	It is possible to define jobs that the user can display and run. The definition can be done: • by individual jobs • by job grouping nodes; individual jobs and/or lower-level nodes can be added or removed to the nodes.
Closing: Routine	It is possible to define closing routines that the user can display and run. The definition occurs only for individual elements, as there is no aggregation structure for these dimensions.
Analytical Workspace Dataset	The definition of user-visible datasets can be performed: • by individual datasets; • by tags; individual datasets can be added or removed to tags restrictions on datasets only affect the user homepage

Field	Description
Stored procedure	The definition occurs only for individual elements, as there is no aggregation structure for these dimensions.
Collaborative Office: Editing and Submission	Specifies whether the user can view, enter and submit data. the definition is made by: • **Document type** • **Document activity** • **Invoice** • specifying whether to **include child parts** Access **type** It is possible to restrict access to view-only or to allow editing of everything or only comments. **Run operational tasks** **Run submission tasks** The user canrun only operational tasks or also submission tasks **Labels** identification labels linked to each workflow task **Edit submitted data** specifies whether the user is allowed to edit the certified but not yet approved or closed parts **Edit attachments** specifies whether the user is enabled to manage document attachments

***IMPORTANT**: By default, every new user has management rights on all items. Such rights can be controlled through restrictions. Exceptions are those on category and consolidation scenario which, by default, are restricted and must be set by the administrator user.*

Restrictions on Analytical Dimensions

Restrictions on analytical dimensions allow you to define, for each user, which elements they can view and enter data for.

Restrictions can be defined:

- for individual elements of the analytical dimension
- for nodes of a hierarchy defined on the analytical dimension. Individual elements of the analytical dimension and/or lower-level nodes can be added or removed from the nodes.

***IMPORTANT**: By default, each new user has management rights over all existing elements (except category and consolidation scenario).*

Elements (except category and consolidation scenario) are by default both visible and editable. To prevent modification, it is necessary to specify read-only.

Rights Definition

application database > User Rights > Restrictions > Define link

***IMPORTANT**: the link is only active if the user has **restricted** rights*

Purpose of the page

On this page you can configure restrictions

Attributes tab

Below are the most significant fields for system, custom and analytical dimensions.

Field	Description
Action	Allows you to specify the type of action for each detail row: **Include**: inserts a dimension element/node in the definition **Exclude**: excludes a dimension element/node from the definition
Dimension	The dimension to be configured. You can select Individual dimensions or groupings
Read-only	Enables consultation only.
Entity dimension specificities	
Additional fields by tab	it is possible to specify input/submission restrictions for each organisational hierarchy involved in an application process. If the rights on the various entities are always the same, it is possible to use the 'simplified management' feature to define the data entry and submission rights.

Below are the available fields for the WorkFlow step:

Field	Description
Business Workflows	Allows you to specify the Bw for which the task list is to be defined
Step	Allows you to specify the step.

Definition results

application database > User Rights > Restrictions > Entity and Task > Definition results link

***IMPORTANT**: the link is only active if the user has **restricted** rights*

Purpose of window

This window shows the list of set restrictions Entity rights can be deployed by the **Deploy** button in the top left corner 

Reference users

application database > User Rights > Reference Users

***IMPORTANT**: The field can only be edited if there is at least one reference user, i.e. with the **Reference user for other users** option enabled. For further details see User Settings.*

Reference Users Tab

Element	Description
	Allows you to add registered users to be used as reference users. It contains the list of the user's reference users.

Note: *If more than one reference user is indicated, the restrictions are intersected in order to derive the greatest common denominator.*

Self-Service portal

What it is

The Self-Service portal, accessible via any browser, displays key performance indicators (KPIs) related to the Cloud environments.

Advantages

It allows real-time monitoring of Cloud Availability and Licence Consumption for Cloud Environments.

Activation and access

CCH Tagetik Cloud customers can request activation from the CCH® Tagetik Support portal by submitting a 'General Request'. Once activated, the Self-Service Portal can be used by all Cloud customers, using the same login credentials as for the CCH® Tagetik Support Portal. For activation and access see How to Activate the Self-Service Portal.

***IMPORTANT**: The Self-Service Portal is compatible with CCH® Tagetik from version 5.3SP19.*

How to Activate the Self-Service Portal

Self - Service Portal Activation Procedure

1. "Create a General Request on CCH® Tagetik Support Portal" below
2. "Log in to the Self-Service Portal" below

Create a General Request on CCH® Tagetik Support Portal

Access the CCH® Tagetik Support Portal link

1. Select "**Technical request** "
2. Select "**Other Technical**" as the request type
3. Fill the form with the required information.

Log in to the Self-Service Portal

1. Check your inbox and open the e-mail received from the CCH® Tagetik support team at the end of activation.
2. Access the **web link** contained in the e-mail
3. Click the "**Login with SERVICE CLOUD**" button
4. Click the **Allow** button
5. Enter the credentials used to access the Support portal of CCH Tagetikto log in.

***IMPORTANT**: The self-service portal works with Single Sign-On entry on the CCH® Tagetik support portal. Use the credentials of the CCH® Tagetik support portal to log in, after allowing access with Salesforce, if required.*

Self-Service Portal: Main Tool Bar

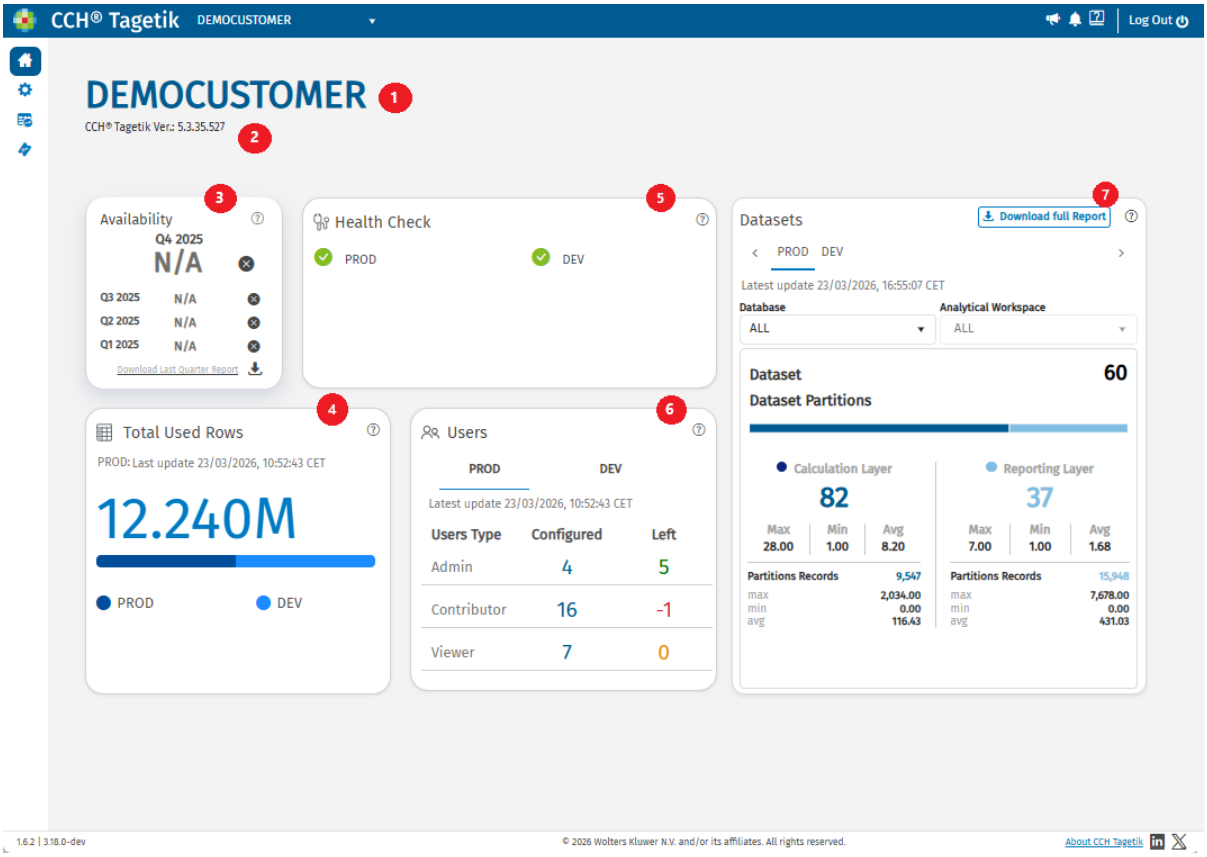
In the Self-Service Portal, the main toolbar is always visible, regardless of the application page being used.

Element	Description
CCH® Tagetik	Tagetik logo, not modifiable.
	**What's new** Show the latest updates of the Self-Service Portal.
	**Alerts & Notifications** It allows the user to view events tracked by the Self-Service Portal on the PROD environment. The Self-Service Portal tracks three types of events that may occur on the PROD environment: • **CPM Status** • **Performance** • **SAP HANA** **CPM Status** events include global outage, partial outage and back online events. An outage event indicates that some servers are not operational. These events do not include interruptions due to connectivity or database problems. In the case of a **partial outage**, only part of the servers are down; consequently, only some users may experience disruptions and, by logging out and logging in, will be automatically directed to a functioning server. A **global outage** affects all servers. In such cases, the cloud activates a self-restoring system that autonomously restarts the servers involved, removing the need to open a ticket or report the emergency. Finally, **back online** events indicate that one or more servers are back up and running after an interruption. **Performance** type events indicate that a data processing has exceeded the expected time, which can be configured in the Notifications section of the Self-Service Portal. **SAP HANA** type events are available only to customers using SAP HANA as the database vendor and indicate the occurrence of a 'memory exceeded' event. In particular, these events are triggered when the memory used by the database server exceeds 80 % of the available memory for a continuous period of at least 15 minutes.
	**Log Off** Allows the user to log out of the application.

Self-Service Portal: Overview

Summary

What it looks like



Element	Description
1	Environment name
2	Version of CCH® Tagetik in production
3	Card Cloud Availability
4	Card Total rows used
5	Card Integrity check
6	Card Users
7	Card Dataset

Cloud Availability card

Overview > Cloud Availability

This card shows the cloud availaibility (%) for the last 4 calendar quarters, with the option to download a detailed report for the last quarter.

Cloud Availability: details

Element	Description
Display values	The value is displayed in green if higher than 99.5 per cent; otherwise in red. If the cloud avalaibility value is not available for a given quarter, the displayed value is greyed out.
Update frequency	The card is updated four times a year (01 January, 01 April, 01 July, 01 October).
Calculation rules	For the definition and calculation of cloud avalaibility, see the Cloud Specification document on CCH Tagetik community

Total Used Rows card

Overview > Total Used Rows

This card shows the total number of rows used and the size of each environment (e.g. Prod, Dev, etc.).

Total Used Rows: Details

Element	Description
Display value	It allows users to monitor the number of rows used against the number of licensed rows.
Update frequency	The card is updated every 8 hours.
Calculation rules	The total number of rows used is calculated as the sum of all data tables (AIH partitions and Financial Workspace fact tables) of all existing environments. Specifically, • the Financial Workspace tables included in the calculation are all tables starting with DATI_ excluding tables starting with DATI_COMMENTI and DATI_NOTE. For example, the following are included: DATI_SALDI_LORDI, DATI_SALDI_IC, DATI_RETT_RIGA, etc... • aIH partitions included in the calculation are all tables starting with AW_ excluding Transaction Matching tables starting with AW_TM1. The archived data tables of the Financial Workspace (i.e. tables beginning with H_DATI_) and those of the ETL (i.e. tables beginning with MAP_) are not used in the calculation.

Health Check card

Overview > Health Check

This card shows the health status of the infrastructure for each environment (PROD, DEV or UAT).

Health Check: details

Element	Description
Display values	The green icon indicates that the environment is operational. The red icon means that the environment is not available. The yellow icon indicates that the environment is operational, but it is possible that one or more servers are not connected to the environment or are experiencing issues (e.g. reduced performance).
Update frequency	Every 60 seconds.

Users card

Overview > Users

This card shows the number of user licences purchased and configured for each environment.

Users: details

Element	Description
Display values	For each environment and each type of licence purchased, the licences configured and those yet to be configured are shown.
Update frequency	Every 8 hours.

Dataset card

Overview > Dataset

This card shows the dataset metrics for each environment.

Dataset: details

Element	Description
Display values	In the card you can filter by Database and Analytical Workspace to see specific metrics. If no filter is applied, the metrics of the datasets are relative to the internal environment. The card includes the number of datasets, the partition count per calculation and reporting layer based on the chosen filter. For each layer you can see: • maximum, minimum and average partitions • total records in partitions • maximum, minimum and average of partition records
Download full report	Allows you to download a file containing the number of records in each partition of the selected environment.
Update frequency	Every 24 hours.

Self-Service Portal: Notification Settings

Notification Settings

IMPORTANT:

- *The SAP HANA Status Check Notifications page is only visible if SAP HANA is used as the vendor database.*
- *The Consumed Credits Notifications page is visible only if the Intelligent Platform license has been installed.*

Performance Monitoring

Notification Settings > Performance Monitoring

Performance Monitoring Notifications

In this section, thresholds in seconds can be configured for the following types of data processing:

- DTP
- ETL – Job
- ETL – Routine
- Report

If a data processing exceeds the threshold set on the PROD environment, the Self-Service Portal creates a notification in the Alerts & Notifications section.

If a data processing exceeds the threshold set on one of the client's environments (PROD, DEV, UAT) and email sending is configured for that environment, the Self-Service Portal sends an email notification of the event that has occurred.

Environments & Email Notifications

In this section, it is possible to enable email notifications to be sent if one of the events tracked for performance monitoring in the Self-Service Portal occurs on one of the customer's environments. This can be configured for each environment available on the client side (PROD, DEV, UAT).

In addition, in this section it is possible to define the email addresses that will receive notifications for tracked events for performance monitoring in the Self-Service Portal.

SAP HANA Healthcheck

Notification Settings > SAP HANA Healthcheck

SAP HANA Healthcheck Notifications

This section displays the events related to SAP HANA status checks, for which email notifications can be subscribed to.

Environments & Email Notifications

In this section, it is possible to activate the transmission of email notifications if one of the events monitored during the SAP HANA status check is detected on one of the environments. Activation can be configured individually for each available environment.

In addition, it is possible to specify the email addresses that will receive notifications of monitored events for the SAP HANA status check via the Self-Service Portal.

Credit Overview

Notification Settings > Credit Overview Notifications

Credit Overview Notifications

In this section, it is possible to add or remove email addresses to which certain events will be notified:

- running out of credits on the Intelligent Platform STANDARD package resulting in blocked functionality
- running out of credits on the Intelligent Platform PREMIUM package and thus starting overage consumption for those functionalities

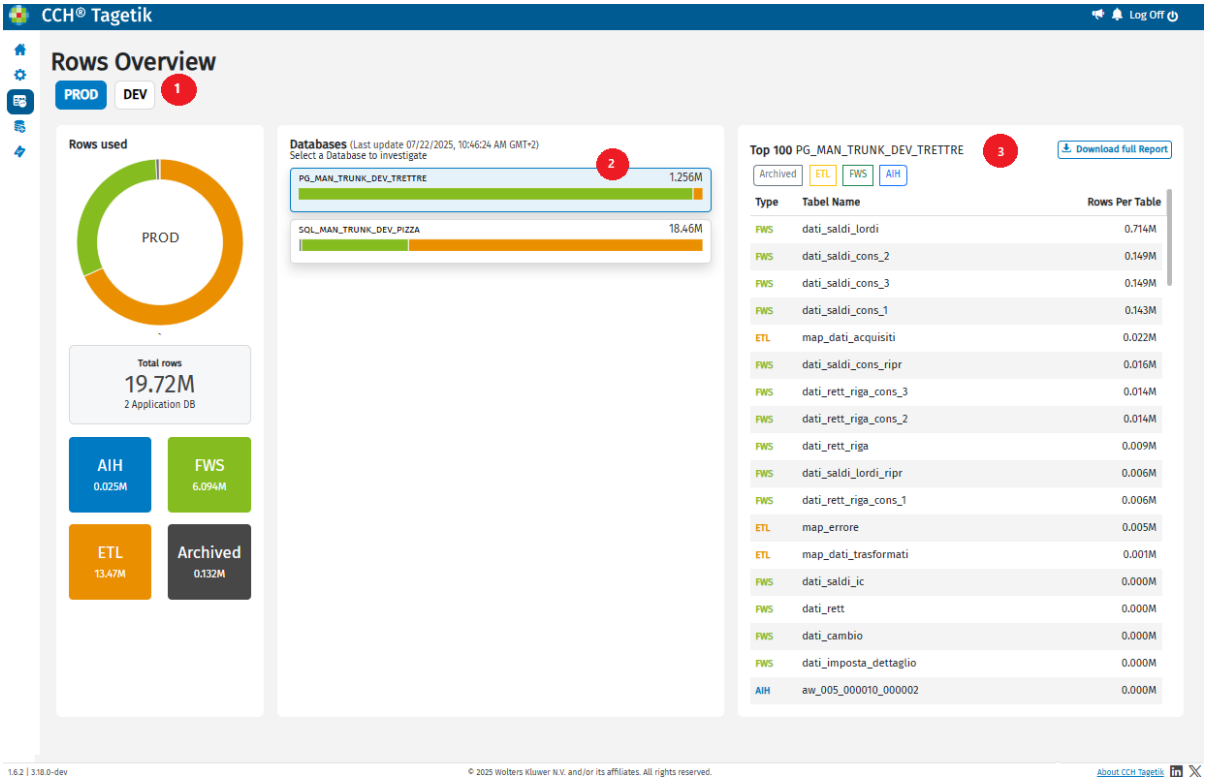
It is also possible to enable the sending of a warning email upon reaching the thresholds related to the percentage of usage of Standard or Premium license credits via specific toggles.

The available thresholds are 50%, 75%, 85% and 95%. Upon reaching 100% of the consumed credits, the email will still bee sent, and it cannot be disabled.

Self-Service Portal: Detail Rows

Detail Rows

What it looks like



Element	Description
1	Environment name
2	Rows by type and Application Database.
3	Top 100 Tables

Rows by Type and Application Database card

Detail Rows > Rows by Type and Application Database

This card shows the total number of rows for the selected environment/database.

Rows by Type and Application Database: Details

Element	Description
Value	This tab presents a different view than the Total Used Rows tab. ***IMPORTANT:*** *All data tables are included, even those that do not determine the value of the licence.*
Update frequency	The card is updated every 8 hours.
Calculation rules	For each database, the rows are grouped into: • AIH: sum of the rows of the AIH partitions, i.e. tables starting with AW_ excluding Transaction Matching tables starting with AW_TM1. • FWS: sum of the rows of the Financial Workspace tables including tables starting with DATI_ excluding tables starting with DATI_COMMENTI e DATI_NOTE. For example, the following are included: DATI_SALDI_LORDI, DATI_SALDI_IC, DATI_RETT_RIGA, etc... • ARCHIVED: sum of the rows of the archived data tables of the Financial Workspace, i.e. the tables beginning with H_DATI_. • ETL: sum of rows of ETL tables that include tables starting with MAP_. To reconcile this view with the Total Used Rows card on the Overview page, the grouped rows with the label AIH and the grouped rows with the label FWS of all databases and all environments (PROD, DEV, UAT) must be added up. The rows in the ETL and ARCHIVED groupings are shown to encourage cleaning and maintenance of the database, but do not determine the value of the licence.

of Rows – Top 100 Tables card

Detail Rows> Top 100 Tables

This card shows the top hundred tables in the database, ranked by number of rows, for the selected environment/database.

of Rows - Top 100 Tables: details

Element	Description
Display values	This card has a different display than the Total used Rows card. ***IMPORTANT:*** *All data tables are included, even those that do not determine the value of the licence.*
Update frequency	The card is updated every 8 hours.
Calculation rules	N.A.

Self-Service Portal: HANA Trend page

HANA Trend

Performance Chart

HANA Trend > Performance Chart

This card provides a chart showing the HANA memory usage in the selected environment. By default, it displays the trend of the last day (1 Day); however, data from the last 7 days (7 Days) and the last 30 days (30 Days) can also be viewed. The chart consistently displays the maximum available memory along with the 80% usage threshold

If the memory used in the PROD environment stays above 80% for at least 15 consecutive minutes, a “memory exceeded” event will be reported in the Alerts & Notifications section.

Self-Service Portal: Consumption

Consumption

***IMPORTANT**: The Consumption page within the Self-Service Portal is only visible if the Intelligent Platform licence has been installed.*

Credits used up by area

Consumptions > Objective AI, API and Intelligent Analytics cards

The Objective AI card

This card shows the number of credits and the number of execution hours consumed to perform one of the Objective AI functionalities: AI-Driver based analysis, AI-Anomaly detection, AI-Automapping.

The API card

This card shows the number of credits and the number of GB of consumed traffic spent to invoke the APIs for data display and processing.

The Intelligent Analytics card

This card shows the number of GB of data hosting consumed for the Intelligent Analytics functionalities.

Ask AI card

This card shows the number of requests made for Ask AI functionalities.

Consumed credits table

This table contains a row for each functionality that generated a consumption of credits. You can request to receive the data in the table by email by clicking the “Email Consumption” button. In this case, the user will receive an email within a few minutes containing either an attachment or a link to download a compressed file with the export of the consumption table

Consumed credits table: details

Element	Description
Plan	Package from which credit was deducted: Standard or Premium
Area	Area to which the functionality belongs: Objective AI, API, Intelligent Analytics or Ask AI
Tenant	Tenant, i.e. the environment where the functionality was executed
DB	Database where the functionality was performed
Workspaces	Workspace where the functionality was performed. Optional.
Date	Functionality execution date
Consumption	Consumption in terms of time or bytes or number of requests for the functionality
Credits	Credit consumption of the functionality
Description	Description of the functionality. Optional.
Operation Detail	Functionality detail. Optional.
Operation Detail Type	Type of functionality detail. This column allows the aggregation of functionalities. For example, in this column for APIs we can see whether it is a 'Data Exposure', a 'Data Management' or an 'Application orchestration'. Optional.
Operation Additional Detail	Further details. This column for APIs contains the invoked URL. Optional.
User Code	User responsible for the consumption. This column for APIs contains the code of the user who invoked the API. Only customers and partners can view the value of this column in plain text, whereas internal users—such as support users—see the information in an encrypted form Optional.

My Package

Consumptions > My Package

This section shows the packages purchased:

- Intelligent Platform - STANDARD
- Intelligent Platform - PREMIUM

The STANDARD package consumes credits on AI Automapping, AI Anomaly detection, AI driver-based analysis and APIs for data exposure and processing.

The PREMIUM package consumes credits on the same functionality as the STANDARD package, and in addition allows you to use the Intelligent Analytics and Ask AI features.

For each package, the number of credits consumed and the number of credits purchased are displayed.

The Overage Credits card

The Overage Credits card shows the credits consumed as well as the credits purchased in the PREMIUM package. The card is only shown if there are over-consumed credits.

The You ran out of credits card

The You ran out of credits card indicates that it is no longer possible to use the Intelligent Platform functionality because the STANDARD licence credits have run out.